

8

7

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1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV

ECN

DESCRIPTION OF REVISION

B

0000876898

PRODUCTION RELEASED

FX

CHINA

FDX

K84 MLB_DYN SCHEMATIC

03/19/2010

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K24_MLB

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-8568	1	SCHEM,MLB_DYN,K84	SCH	CRITICAL	
820-2883	1	PCBF,MLB_DYN,K84	PCB	CRITICAL	

SCHEM,MLB_DYN,K84

Apple Inc.

051-8568

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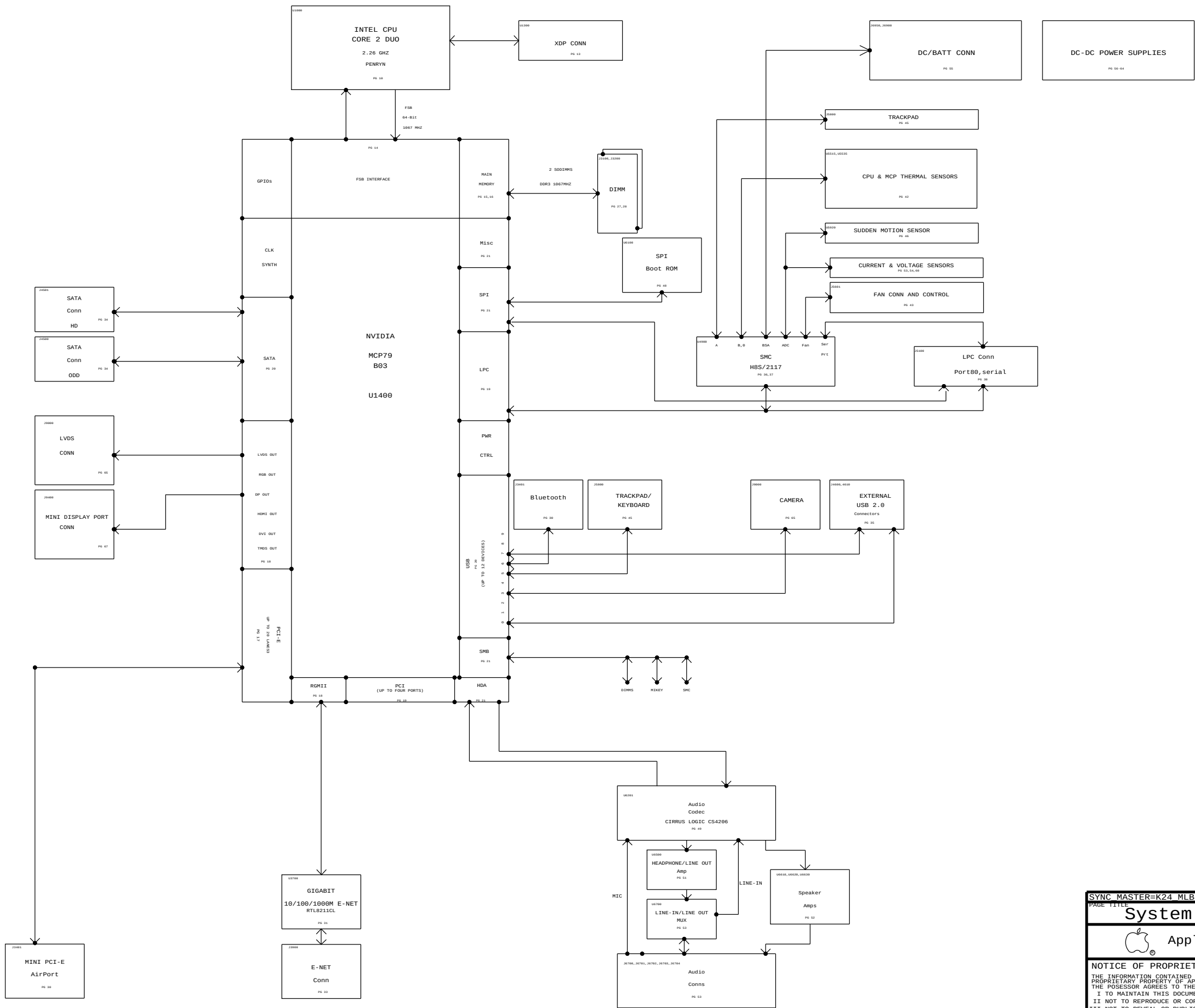
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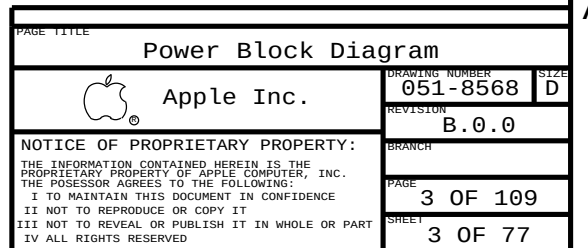
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System Block Diagram			
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BOM Variants		
BOM NUMBER	BOM NAME	BOM OPTIONS
639-1144	PCBA, MLB_DYN, FOX DDR CONN, KB4	KB4_COMMON_PVT, CPU: 2.26GHZ, FOX_DDR_CONN, EEEE_DD2X
639-1145	PCBA, MLB_DYN, MLX DDR CONN, KB4	KB4_COMMON_PVT, CPU: 2.26GHZ, MLX_DDR_CONN, EEEE_DD2Y
985-1651	KB4 MLB_DYN DEVELOPMENT PVT	KB4_DEVEL_PVT

BAR CODE LABELS / EEEE #'S

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
826-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DD2X]	CRITICAL	EEEE_DD2X
926-4393	1	LBL, P/N LABEL, PCB, 28MM X 6 MM	[EEEE_DD2Y]	CRITICAL	EEEE_DD2Y

BOM GROUP	BOM OPTIONS
KB4_COMMON	COMMON, ALTERNATE, KB4_MCP, KB4_MISC, KB4_DEBUG_ENG, KB4_PROGPARTS
KB4_COMMON_PVT	COMMON, ALTERNATE, KB4_MCP, KB4_MISC, KB4_DEBUG_PROD, KB4_PROGPARTS
KB4_MCP	MCP_B03, BOOT_MODE_USER, MCPSEQ_SMC
KB4_MISC	ONEWIRE_PU, DP_ESD, MIKEY, LDO_NO_MEM_SENSE, 1P05_HIGH_SIDE_SENSE, MCP_T_DIODE_SENSOR, MCPSMC_DIGITEMP_YES
KB4_PROGPARTS	BOOTROM_PROG, SMC_PROG, WELLSPRING_PROG
KB4_DEBUG_ENG	SMC_DEBUG_YES, XDP
KB4_DEBUG_PVT	DEVEL_BOM_PVT, SMC_DEBUG_YES, XDP, NO_VREFMRGN
KB4_DEBUG_PROD	SMC_DEBUG_YES, XDP, LPCPLUS_NOT, NO_VREFMRGN
KB4_DEVEL_ENG	DEBUG_ADC, XDP_CONN, LPCPLUS, VREFMRGN
KB4_DEVEL_PVT	XDP_CONN, LPCPLUS, VREFMRGN

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
33753769	1	POC, SL6VT, 2.26, 25W, 1866, R9, 3H, BGA, P7550	U1900	CRITICAL	CPU: 2.26GHZ
33859710	1	IC, GACP, MCP70, 35X35MM, BGA1437, B03	U1400	CRITICAL	MCP_B03
51659706	1	CONN, 284P, 500IHM, SOCKET, D0R3, RAM, BGA	J3200	CRITICAL	FOX_DDR_CONN
516-0291	1	CONN, 284P, 500IHM, P=0.6MM	J3100	CRITICAL	FOX_DDR_CONN
51659790	1	CONN, 284P, 500IHM, SOCKET, D0R3, RAM, NGA/SC	J3200	CRITICAL	MLX_DDR_CONN
516-0213	1	CONN, 284P, 500IHM, P=0.6MM, HF	J3100	CRITICAL	MLX_DDR_CONN
452-1788	4	SCR, M1.6X0.35X6.0, D4, NO.3, BLK, M07	SCREW1, SCREW2, SCREW3, SCREW4	CRITICAL	
514-0704	1	CONN, RCPT, R345, PLASTIC, HF, K83/KB4	J3900	CRITICAL	
514-0705	2	CONN, RCPT, USB, 4P, PLASTIC, HF, K83/KB4	J4600, J4610	CRITICAL	
514-0706	1	CONN, RCPT, MCP, 28P, PLASTIC, HF, K83/KB4	J9400	CRITICAL	
35352718	1	IC, 15L88842, 4X V MONTR, 2.78/2.86V, TDFN8	U7870	CRITICAL	
870-1885	4	POGO PIN, MED, NOISE-IMPROVED, KB4	Z50900, Z50901, Z50902, Z50903	CRITICAL	
870-1885	3	POGO PIN, MED, NOISE-IMPROVED, KB4	Z50908, Z50909, Z50911	CRITICAL	
870-1886	5	POGO PIN, TALL, NOISE-IMPROVED, KB4	Z50904, Z50905, Z50906, Z50907, Z50910	CRITICAL	
870-1886	5	POGO PIN, TALL, NOISE-IMPROVED, KB4	Z50912, Z50913, Z50914, Z50915, Z50919	CRITICAL	
870-1887	3	POGO PIN, THIN, NOISE-IMPROVED, KB4	Z50917, Z50918, Z50916	CRITICAL	
51859774	1	CONN, RCPT, 60P, P=0.4, STK HT 1.0	J1300	CRITICAL	XDP_CONN
33753680	1	POC, L602, PRQ, 2.40, 25W, 1866, R9, 3H, BGA	U1900	CRITICAL	CPU: 2.4GHZ

35352718 IS NEW INTERSIL PART FOR FIXING B4 DONGLE ISSUE
514-0704 IS CLOUD GREY 4/LB3 PLASTIC W/PDNI PLATING VERSION OF 514-0692 PART FOR RJ45 CONNECTOR
514-0705 IS CLOUD GREY 4/LB3 PLASTIC W/PDNI PLATING VERSION OF 514-0689 PART FOR USB CONNECTORS
514-0706 IS CLOUD GREY 4/LB3 PLASTIC W/PDNI PLATING VERSION OF 514-0691 PART FOR MINI DP CONNECTOR
514-0718 IS CLOUD GREY 4/LB3 PLASTIC W/PDNI PLATING VERSION OF 514-0694 PART FOR AUDIO CONNECTOR

DEVELOPMENT BOM

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
085-1651	1	KB4 MLB_DYN DEVELOPMENT PVT	DEVEL_PVT	CRITICAL	DEVEL_BOM_PVT

Programmable Parts

33859663	1	IC, SMC, HSB/2117, 9X9MM, TLP, HF	U4900	CRITICAL	SMC_BLANK
34152485	1	IC, SMC, KB4	U4900	CRITICAL	SMC_PROG
33550610	1	IC, FLASH, SPI, 32MBIT, 3.3V, 80MHZ, 8-SOP	U6100	CRITICAL	BOOTROM_BLANK
34170281	1	IC, PROGRAM, EP3 BOOTROM, KB4A	U6100	CRITICAL	BOOTROM_PROG
33752983	1	IC, P50C+ W/ USB, 56 PIN, MLF, CY8C24794	U5701	CRITICAL	WELLSPRING_BLANK
34152491	1	IC, WELLSPRING CONTROLLER, KB4	U5701	CRITICAL	WELLSPRING_PROG

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
15250693	15250778		ALL	DALE/VISHAY, MAGLAYERS AS ALTERNATE
15250796	15250685		ALL	CYNTEC AS ALTERNATE
15750050	15750055		ALL	DELTA AS ALTERNATE
13850603	13850602		ALL	MURATA AS ALTERNATE
12850093	12850218		ALL	KEMET AS ALTERNATE
15250874	15250516		ALL	MAGLAYERS AS ALTERNATE
15250847	15250586		ALL	MAGLAYERS AS ALTERNATE
10450010	10450023		ALL	DALE/VISHAY AS ALTERNATE
35352989	35352811		ALL	ISL625BAHRTZ-TR5304A AS ALTERNATE

K84 BOARD STACK-UP

Top	SIGNAL
2	GROUND
3	SIGNAL(High Speed)
4	SIGNAL(High Speed)
5	GROUND
6	POWER
7	POWER
8	GROUND
9	SIGNAL(High Speed)
10	SIGNAL(High Speed)
11	GROUND
BOTTOM	SIGNAL

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Functional Test Points

D

D

FAN CONNECTORS FUNC_TEST

PP5VRT S0	7 8
FAN_RT_PWM	43
FAN_RT_TACH	43

(NEED TO ADD 1 GND TP)

MIC FUNC_TEST

BI_MIC_LO	52 53
BI_MIC_HI	52 53
BI_MIC_SHIELD	52 53

SPEAKER FUNC_TEST

SPKRAMP_L_N_OUT	52
SPKRAMP_L_P_OUT	52
SPKRAMP_R_N_OUT	52
SPKRAMP_R_P_OUT	52
SPKRAMP_SUB_N_OUT	52
SPKRAMP_SUB_P_OUT	52

C

C

LVDS FUNC_TEST

PP3V3_LCDVDD_SW_F	7 64 (NEED 2 TP)
PP3V3_S0_LCD_F	64
PPVOUT_S0_LCDBKLT	7 47 64 67 (NEED 2 TP)
LVDS_IG_DDC_CLK	18 64
LVDS_IG_DDC_DATA	18 64
LVDS_IG_A_DATA_N<0>	18 64 71
LVDS_IG_A_DATA_P<0>	18 64 71
LVDS_IG_A_DATA_N<1>	18 64 71
LVDS_IG_A_DATA_P<1>	18 64 71
LVDS_IG_A_DATA_N<2>	18 64 71
LVDS_IG_A_DATA_P<2>	18 64 71
LVDS_IG_A_CLK_F_N	64 71
LVDS_IG_A_CLK_F_P	64 71
LED_RETURN_1	64 67
LED_RETURN_2	64 67
LED_RETURN_3	64 67
LED_RETURN_4	64 67
LED_RETURN_5	64 67
LED_RETURN_6	64 67
PP5V_S3_CAMERA_F	7 64
USB_CAMERA_CONN_P	64 72
USB_CAMERA_CONN_N	64 72

(NEED TO ADD 5 GND TP)

SATA ODD CONN FUNC_TEST

PP5V_SW_ODD	(NEED 2 TP) 7 34 47
SMC_ODD_DETECT	34 36
SATA_ODD_D2R_C_P	34 71
SATA_ODD_D2R_C_N	34 71
SATA_ODD_R2D_P	34 71
SATA_ODD_R2D_N	34 71

(NEED TO ADD 2 GND TP)

SATA HDD/SIL FUNC_TEST

PP5V_S0_HDD_FLT	(NEED 2 TP) 7 34
SATA_HDD_R2D_P	34 71
SATA_HDD_R2D_N	34 71
SATA_HDD_D2R_C_P	34 71
SATA_HDD_D2R_C_N	34 71
SYS_LED_ANODE_R	34

(NEED TO ADD 3 GND TP)

BATT POWER CONN FUNC_TEST

SMBUS_SMC_BSA_SCL	39 74
SMBUS_SMC_BSA_SDA	39 74
SYS_DETECT_L	54
BATT_POS_F	54 55

(NEED TO ADD 2 GND TP)

(NEED 2 TP)

HALL EFFECT CONNECTOR FUNC_TEST

PP3V42_G3H	7 8
SMC_LID_R	54

X16 WIRELESS CONN FUNC_TEST

PP3V3_S3_BT_F	30
CONN_PCIE_MINI_D2R_P	30 71
CONN_PCIE_MINI_D2R_N	30 71
CONN_PCIE_MINI_R2D_P	30 71
CONN_PCIE_MINI_R2D_N	30 71
PCIE_CLK100M_MINI_CONN_P	30 71
PCIE_CLK100M_MINI_CONN_N	30 71
PP3V3_WLAN	7 30 (NEED 2 TP)
PCIE_WAKE_L	17 30
CONN_USB2_BT_P	30 72
CONN_USB2_BT_N	30 72
MINI_CLKREQ_0_L	30
MINI_RESET_CONN_L	30

(NEED TO ADD 2 GND TP)

IPD_FLEX_CONN FUNC_TEST

PP3V3_S3_LDO	7 45
PP18V5_S3	7 45
Z2_CS_L	44 45
Z2_DEBUG3	44 45
Z2_M0SI	44 45
Z2_MISO	44 45
Z2_SCLK	44 45
Z2_BOOST_EN	45
Z2_HOST_INTN	44 45
Z2_CLKIN	44 45
Z2_KEY_ACT_L	44 45
Z2_RESET	44 45
PSOC_MISO	44 45
PSOC_MOSI	44 45
PSOC_SCLK	44 45
SMBUS_SMC_A_S3_SDA	39 74
SMBUS_SMC_A_S3_SCL	39 74
PSOC_F_CS_L	44 45
PICKB_L	44 45

(NEED TO ADD 2 GND TP)

KEYBOARD CONN FUNC_TEST

PP3V3_S3	7 8
PP3V42_G3H	7 8
WS_KBD1	44
WS_KBD2	44
WS_KBD3	44
WS_KBD4	44
WS_KBD5	44
WS_KBD6	44
WS_KBD7	44
WS_KBD8	44
WS_KBD9	44
WS_KBD10	44
WS_KBD11	44
WS_KBD12	44
WS_KBD13	44
WS_KBD14	44
WS_KBD15_CAP	44
WS_KBD16_NUM	44
WS_KBD17	44
WS_KBD18	44
WS_KBD19	44
WS_KBD20	44
WS_KBD21	44
WS_KBD22	44
WS_KBD23	44
WS_KBD_ONOFF_L	44
WS_LEFT_SHIFT_KBD	44
WS_LEFT_OPTION_KBD	44
WS_CONTROL_KBD	44

(NEED TO ADD 1 GND TP)

POWER NETS FUNC_TEST

PPVCORE_S0_CPU	8
PPVCORE_S0_MCP	8
PP0V75_S0	8
PP1V05_S0	8
PP1V5_S0	8
PP1V8_S0	8
PP5VLT_S0	8
PP5VRT_S0	7 8
PP3V3_S0	8
PP1V5_S3	8
PP3V3_S3	7 8
PP5V_S3	8
PP1V1R1V05_S5	8
PP3V3_S5	8
PP3V42_G3H	7 8
PPBUS_G3H	8
PP3V3_ENET_PHY	8
PP1V2R1V05_ENET	8
PP3V3_G3_RTC	21 22 25
PP3V3_WLAN	7 30
PP5V_SW_ODD	7 34 47
PP5V_S0_HDD_FLT	7 34
PP3V3_S5_AVREF_SMC	36 37
PP18V5_S3	7 45
PP3V3_S3_LDO	7 45
PP3V3_LCDVDD_SW_F	7 64
PPVOUT_S0_LCDBKLT	7 47 64 67
PP4V5_AUDIO_ANALOG	49
SMC_PM_G2_EN	36 56 62
PM_SLP_S4_L	21 36 37 62
PM_SLP_S3_L	21 32 36 62 66
PP5V_S3_CAMERA_F	7 64

(NEED TO ADD 1 GND TP)

DC POWER CONN FUNC_TEST

PP18V5_DCIN_FUSE	(NEED 2 TP) 54
ADAPTER_SENSE	54

(NEED TO ADD 2 GND TP)

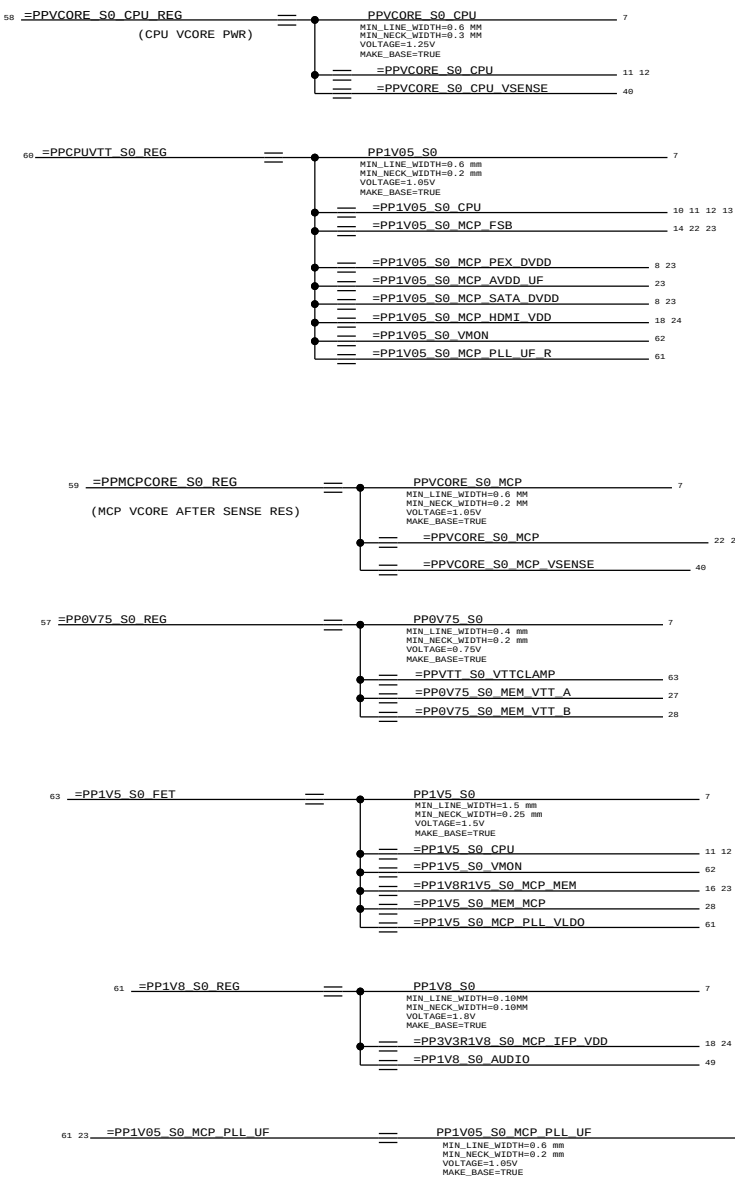
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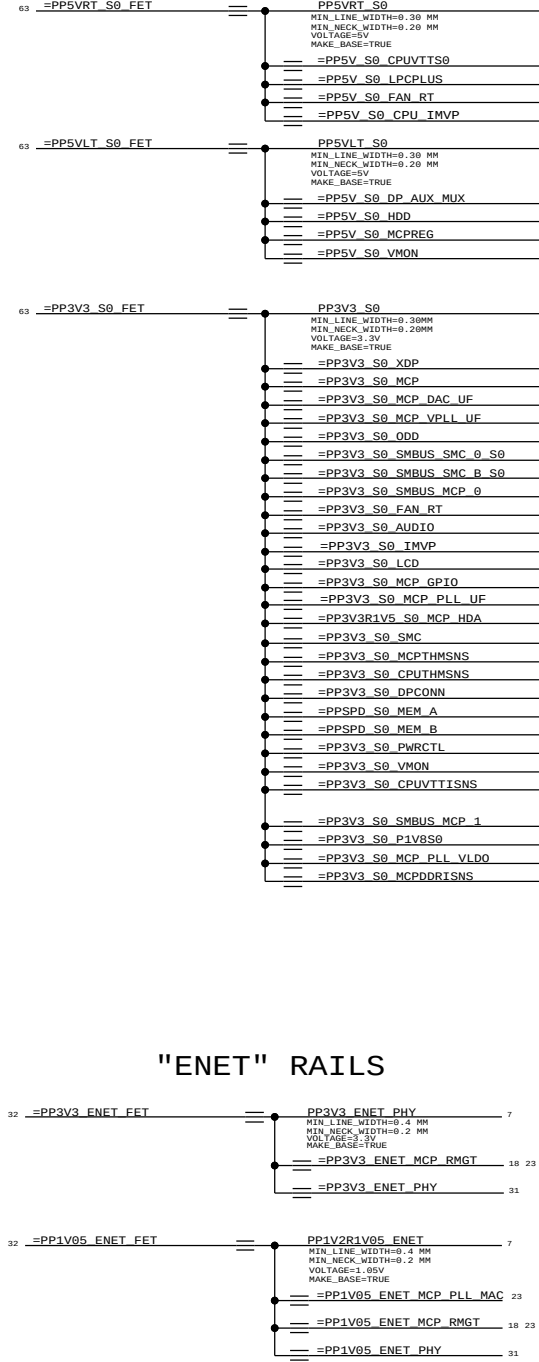
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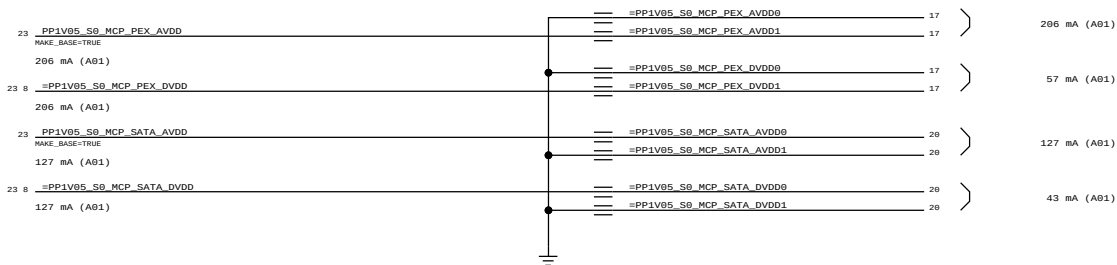
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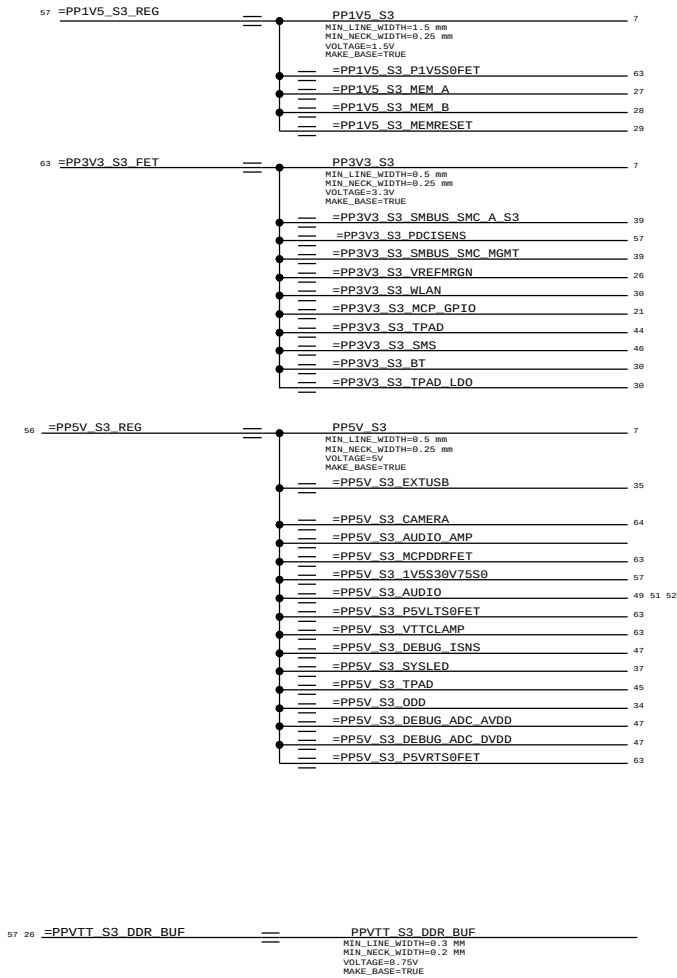
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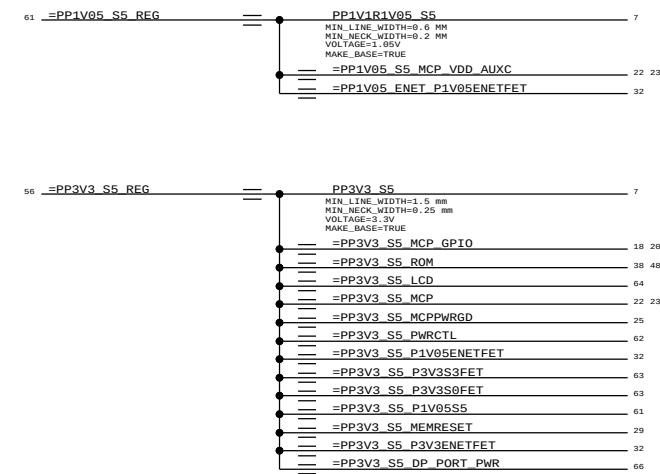
PEX & SATA AVDD/DVDD aliases



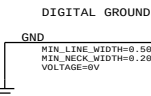
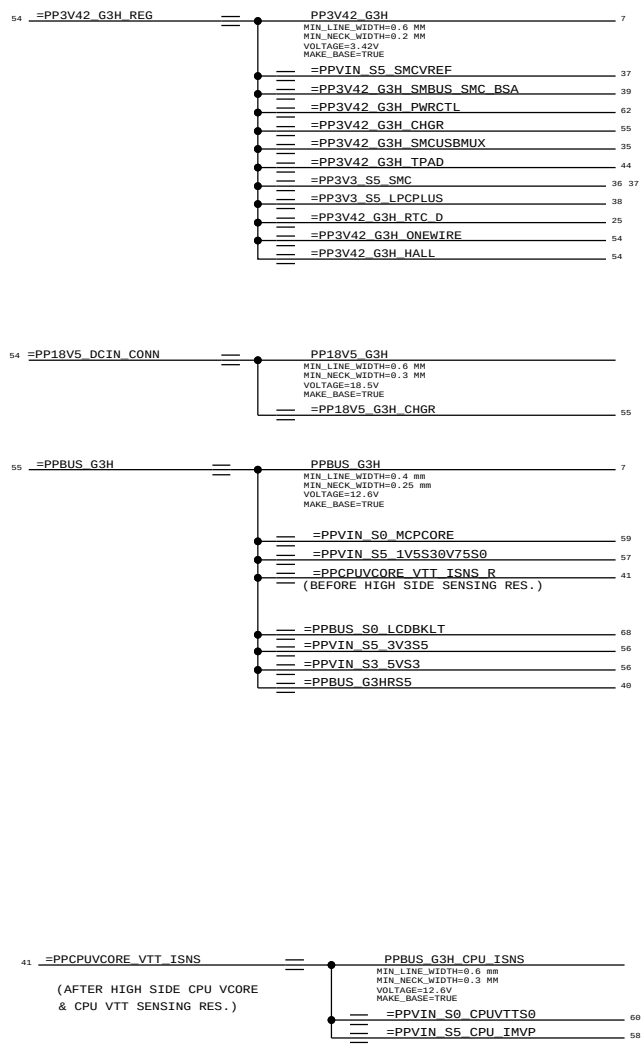
"S3" RAILS



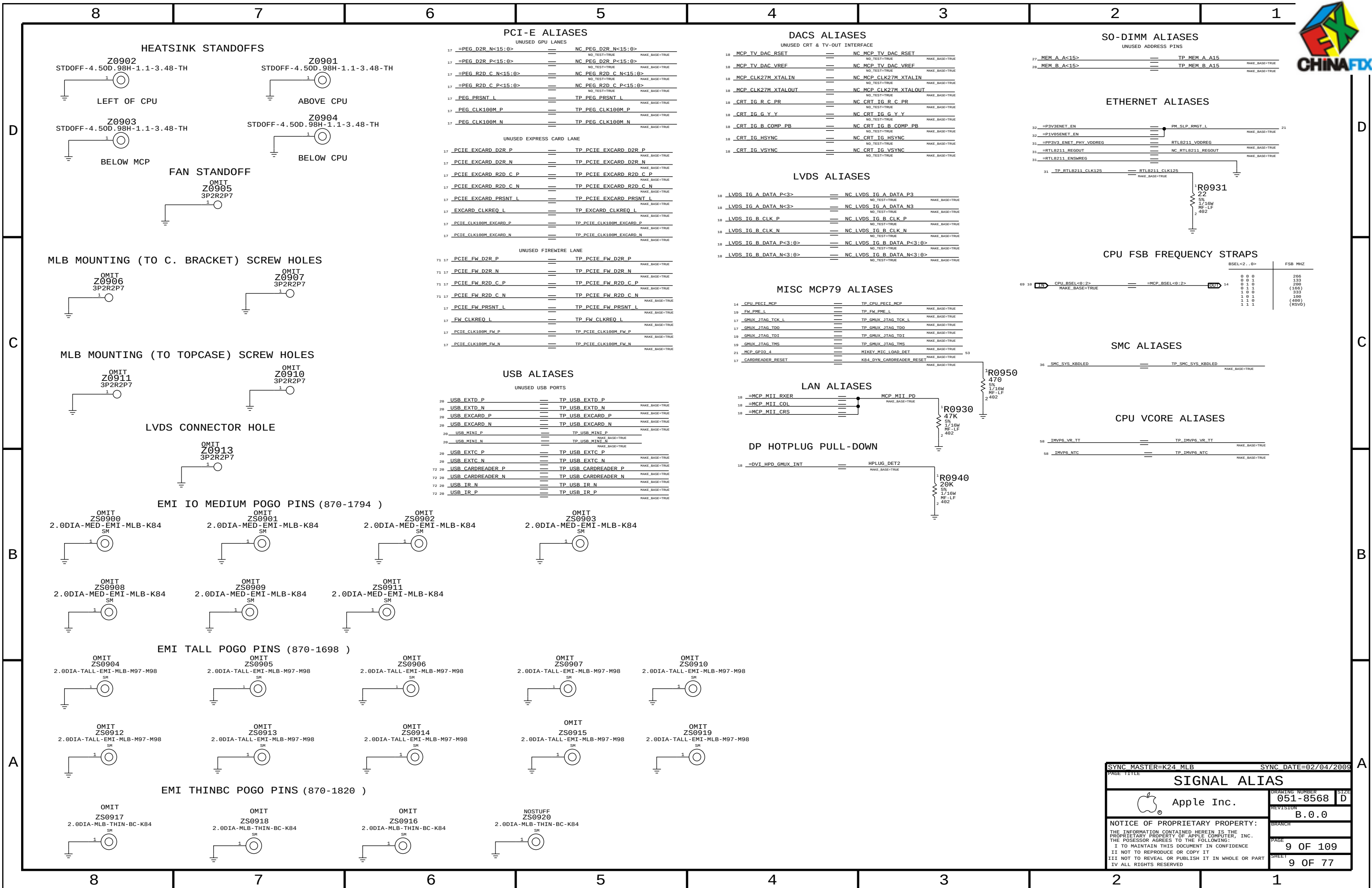
"S5" RAILS



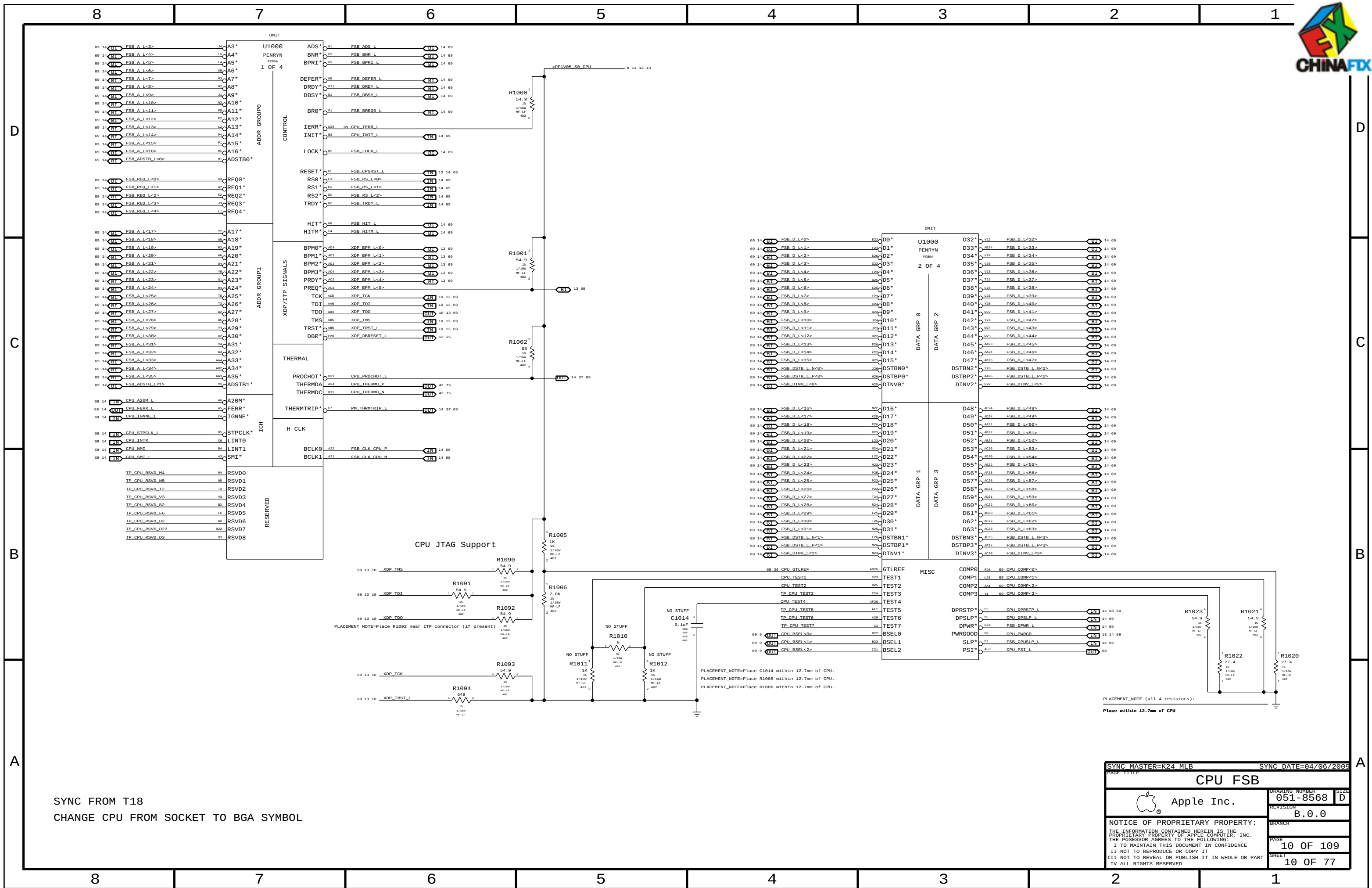
"G3H" RAILS



SYNC MASTER=K24 MLB		SYNC DATE=02/04/2009	
PAGE TITLE		Power Aliases	
Apple Inc.		DRAWING NUMBER	051-8568
		REVISION	B.0.0
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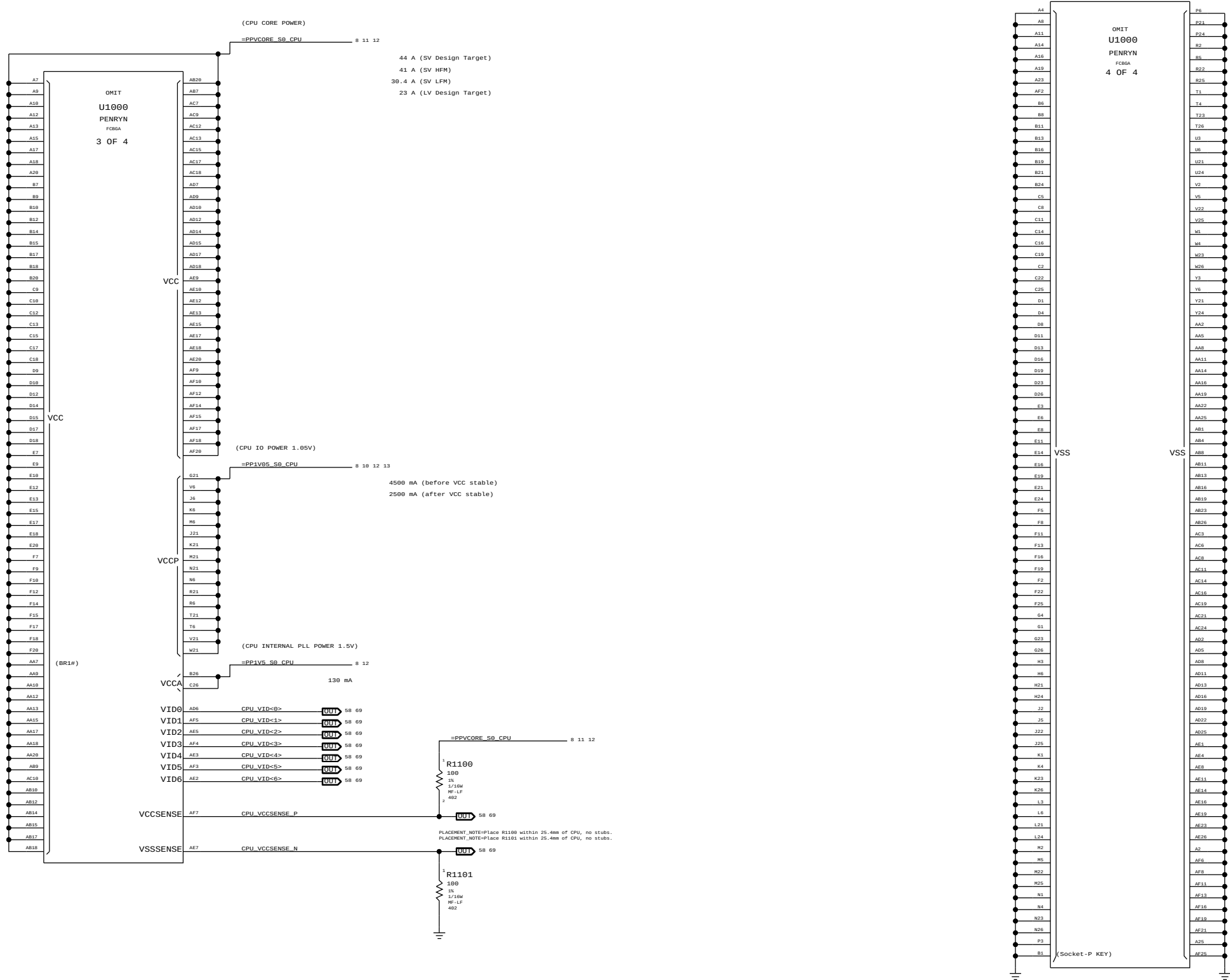


SYNC MASTER=K24 MLB		SYNC DATE=02/04/2009	
PAGE TITLE		REVISION	
SIGNAL ALIAS		051-8568	
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SYNC FROM T18
CHANGE CPU FROM SOCKET TO BGA SYMBOL

SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE			
CPU FSB			
	Apple Inc.		DRAWING NUMBER
			051-8568
		REVISION	D
		B.0.0	
BRANCH			
PAGE			
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SYNC FROM T18
CHANGE CPU FROM SOCKET TO BGA SYMBOL

SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE			
CPU Power & Ground			
 Apple Inc.		DRAWING NUMBER	051-8568
		REVISION	B.0.0
		BRANCH	
		PAGE	11 OF 109
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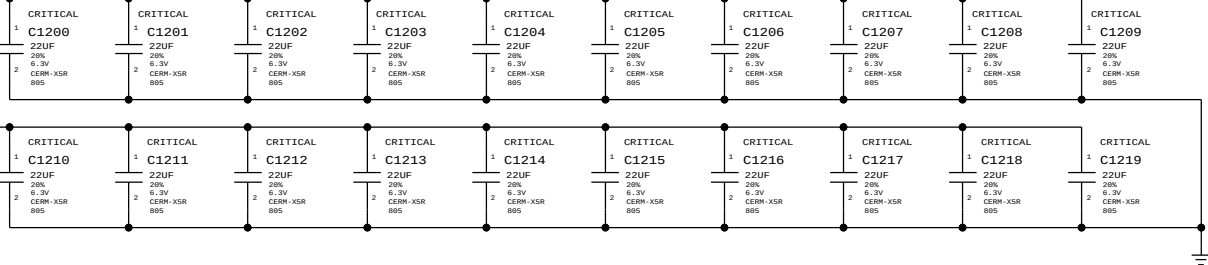


CPU VCore HF and Bulk Decoupling

4X 330UF. 20X 22UF 0805

PLACEMENT_NOTE (C1200-C1219):

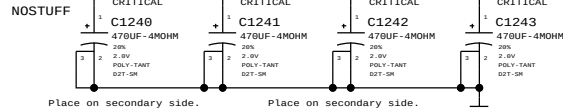
Place inside socket cavity on secondary side.



PLACEMENT_NOTE (C1240-C1243):

Place on secondary side.

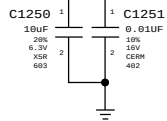
Place on secondary side.



VCCA (CPU Avdd) DECOUPLING

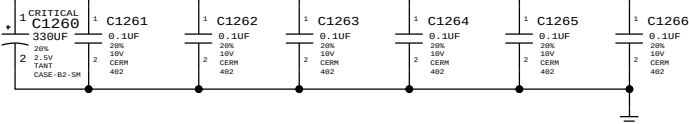
1x 10uF, 1x 0.01uF

PLACEMENT_NOTE=Place C1251 near CPU pin B26.



VCCP (CPU I/O) DECOUPLING

1x 330uF, 6x 0.1uF 0402



SYNC FROM T18
REMOVE NO STUFF CAPS C1220 TO C1231
REMOVE C1244 & C1245
CHANGE C1240-C1243 AND C1260 FROM 128S0241(9 MILLI-OHM) TO 128S0231(6 MILLI-OHM)

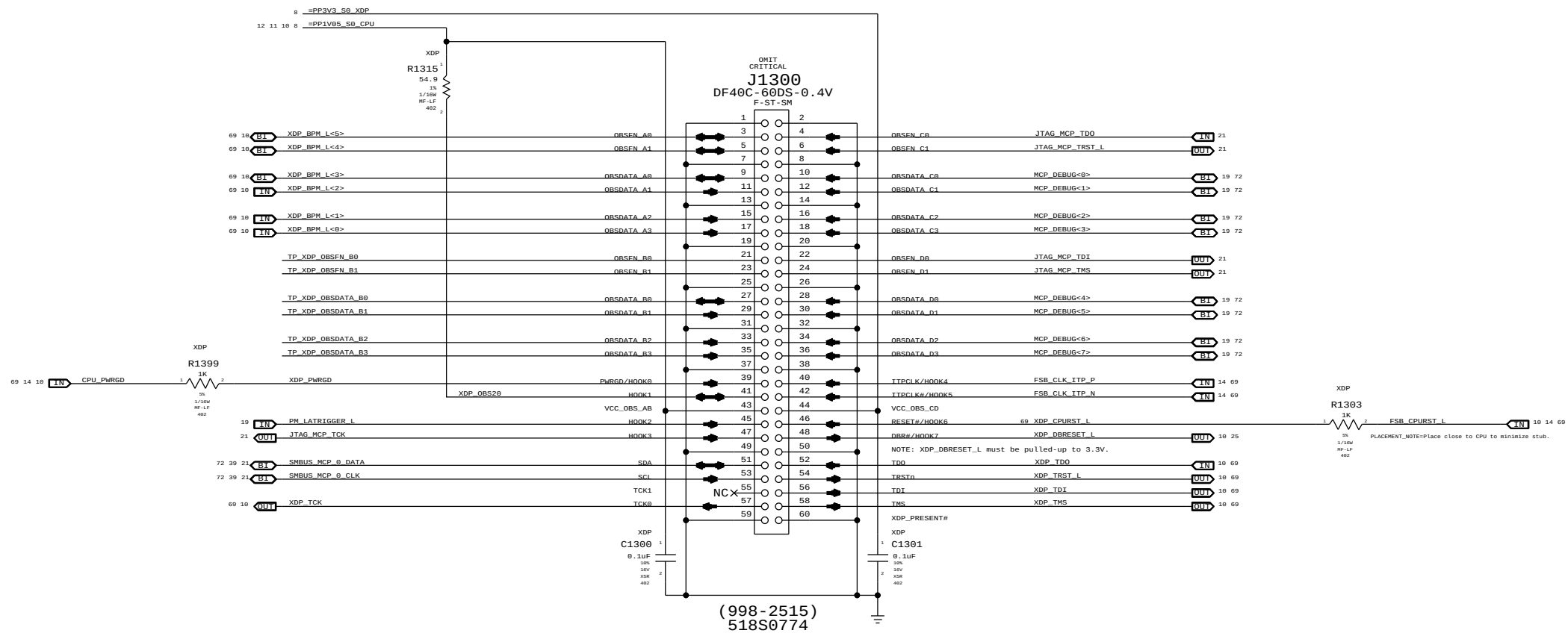
SYNC MASTER=K24 MLB		SYNC DATE=03/30/2009	
PAGE TITLE		CPU Decoupling	
DRAWING NUMBER		051-8568	SIZE D
REVISION		B.0.0	BRANCH
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Mini-XDP Connector

NOTE: This is not the standard XDP pinout.
USE WITH 929-9782 ADAPTER FLEX TO SUPPORT CPU, MCP DEBUGGING.

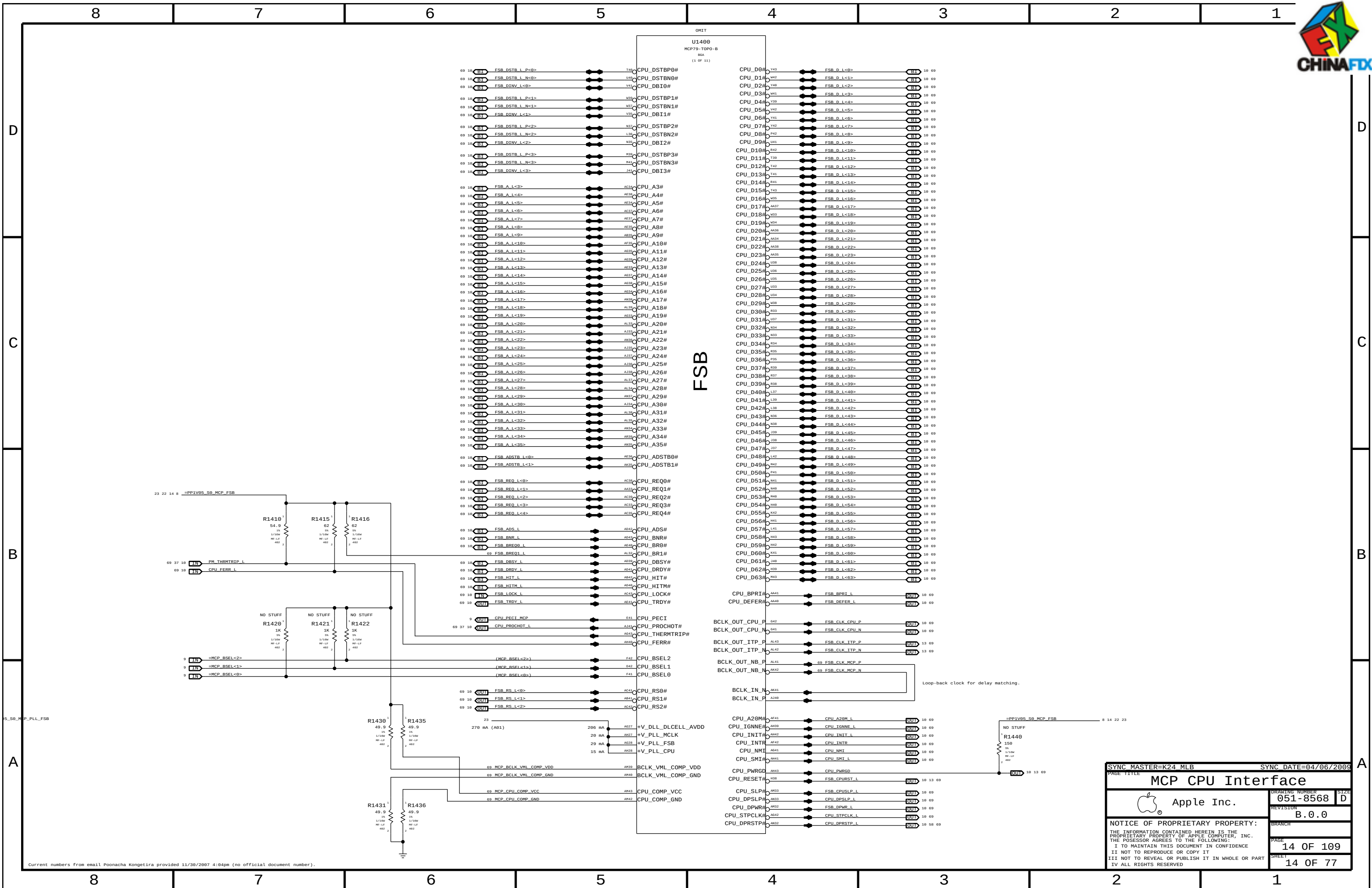
MCP79-specific pinout

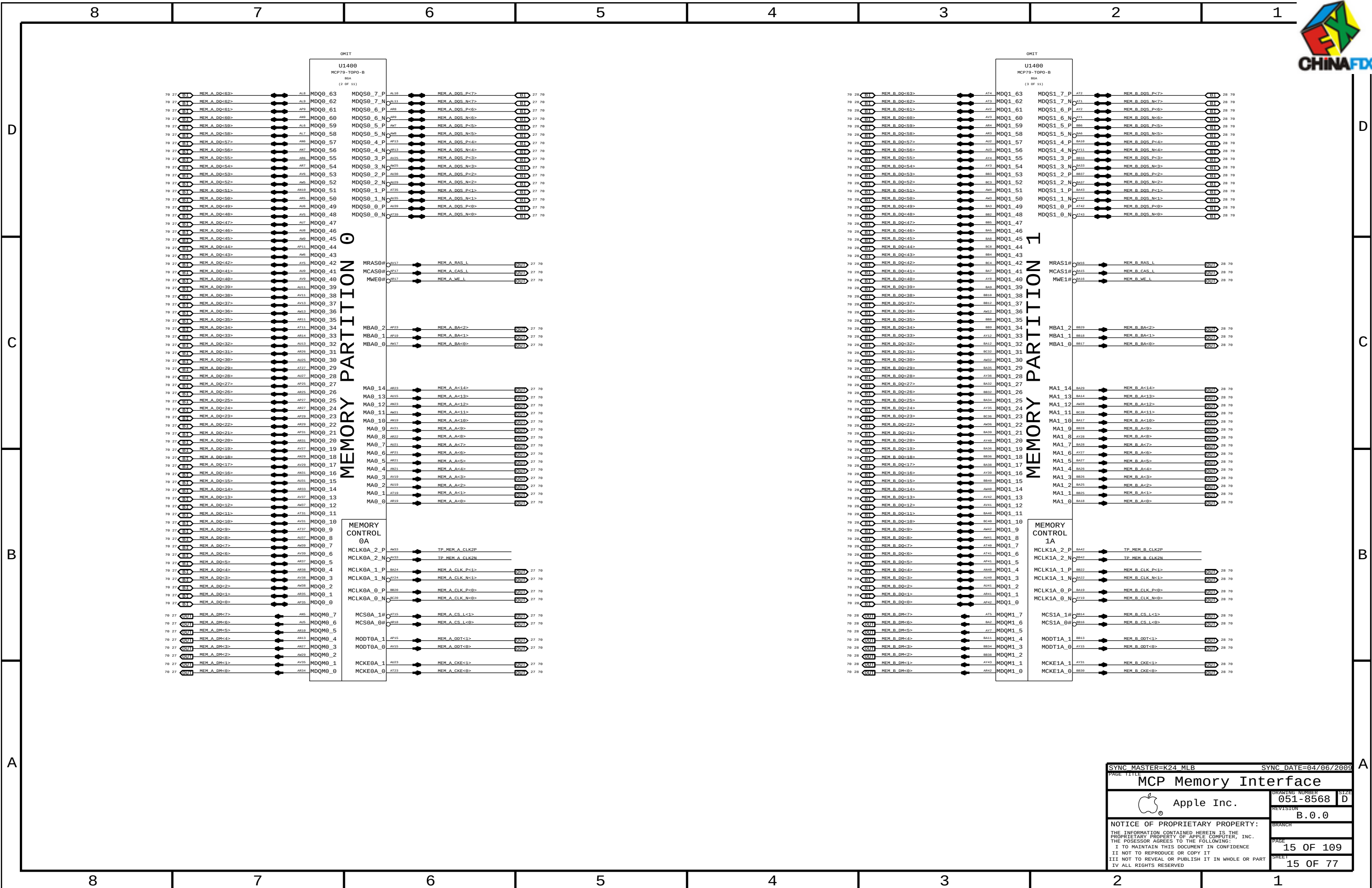


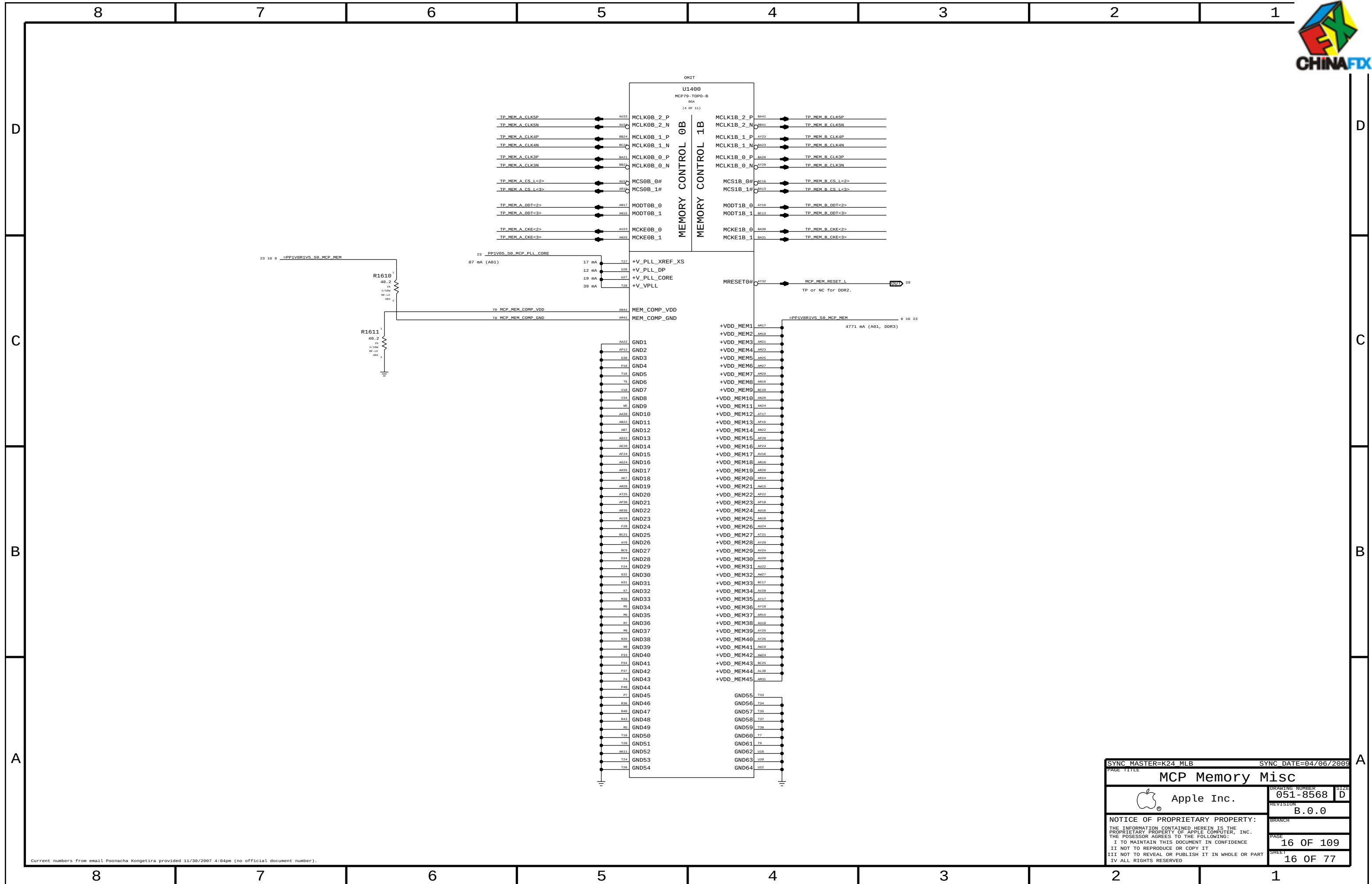
Direction of XDP module

Please avoid any obstructions
ON 000-NUMBERED SIDE OF J1300

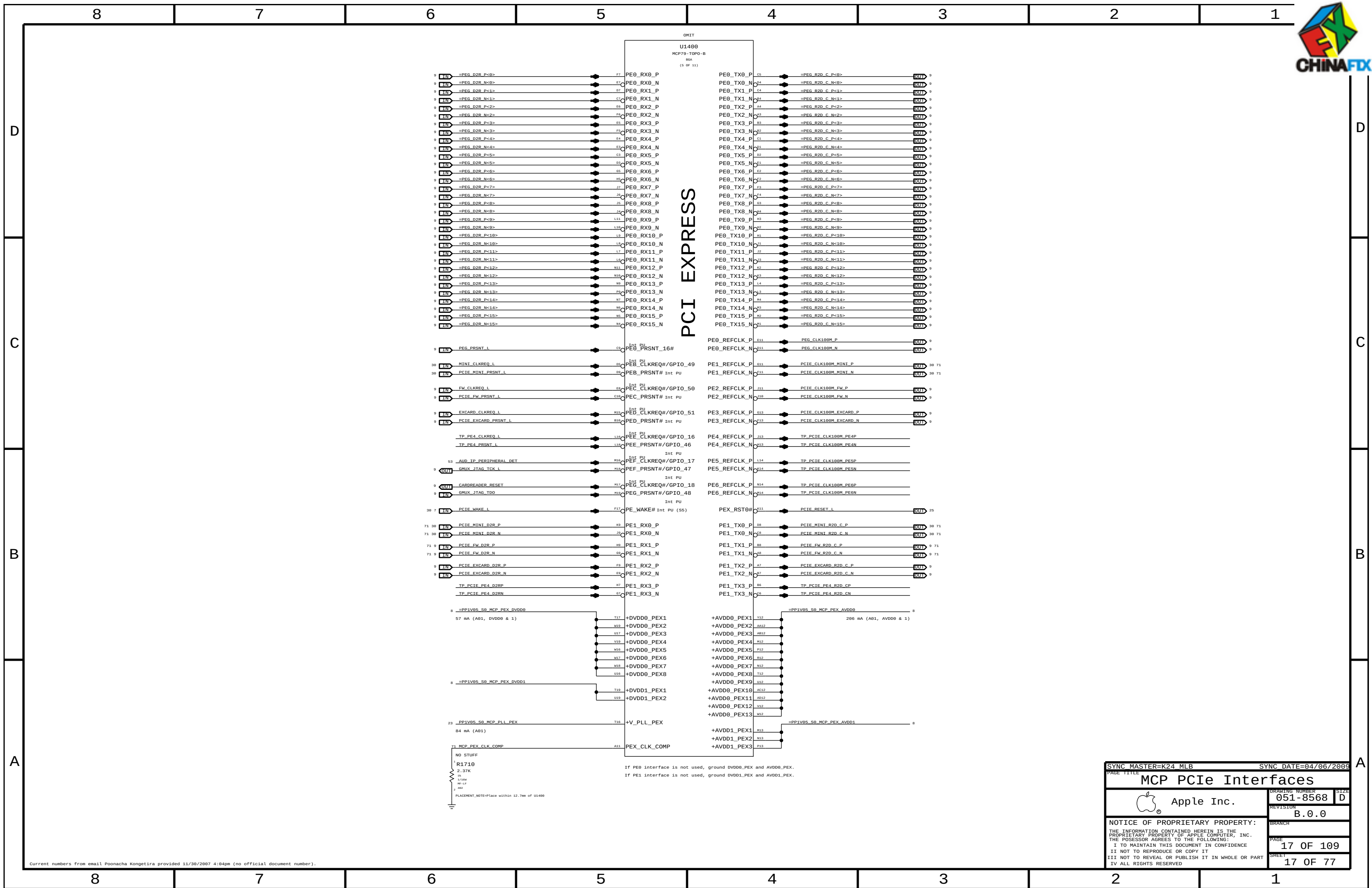
SYNC MASTER=K24 MLB		SYNC DATE=02/25/2009	
PAGE TITLE		eXtended Debug Port(MiniXDP)	
Apple Inc.		DRAWING NUMBER	051-8568
		REVISION	B.0.0
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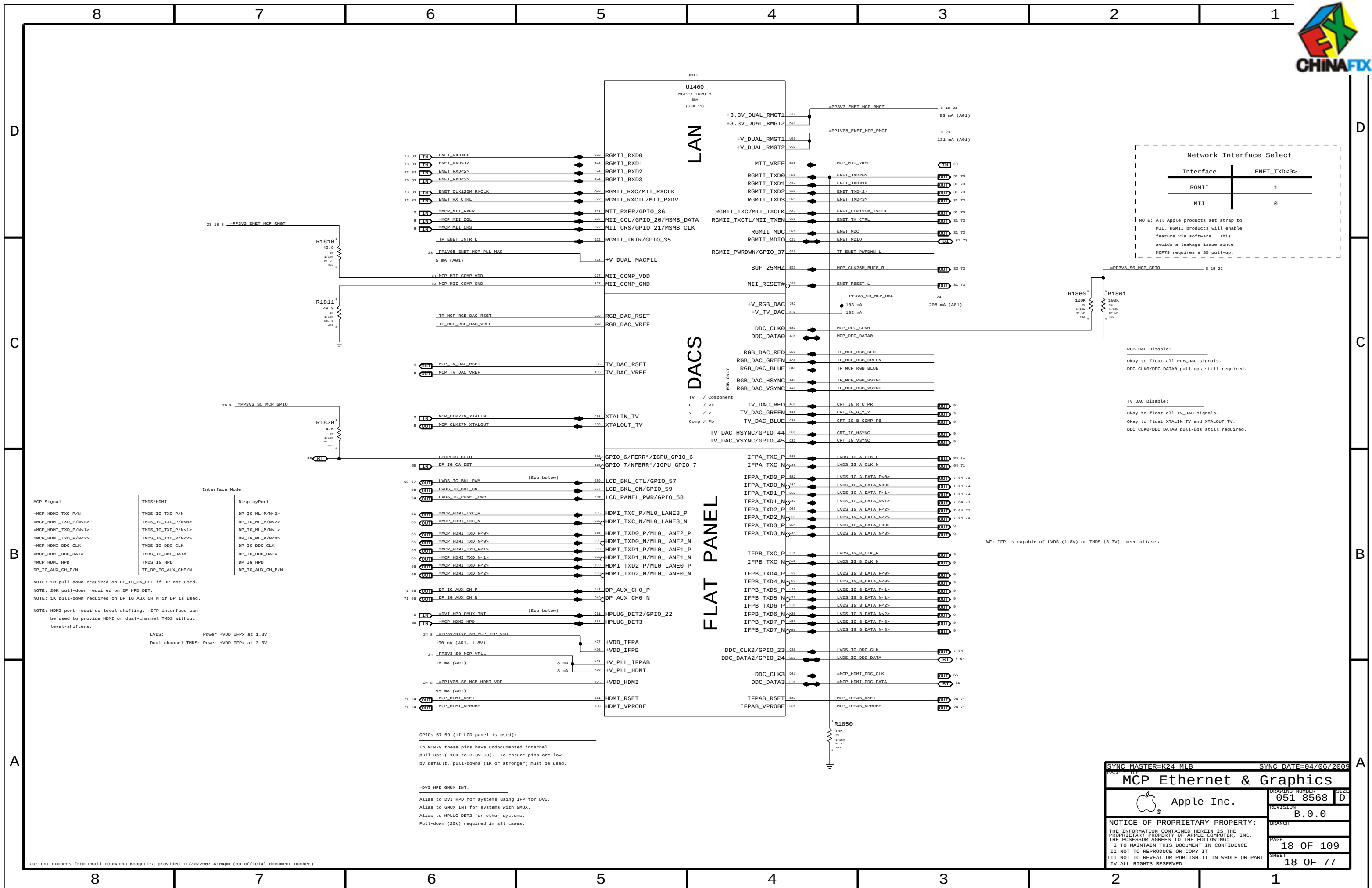




SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE			
MCP Memory		Misc	
 Apple Inc.		DRAWING NUMBER	051-8568
		SIZE	D
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SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE			
MCP PCIe Interfaces			
 Apple Inc.		DRAWING NUMBER	051-8568
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Network Interface Select	
Interface	ENET_TXD<0>
RGMII	1
MII	0

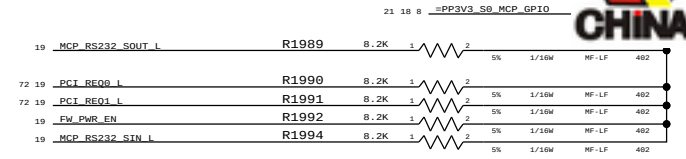
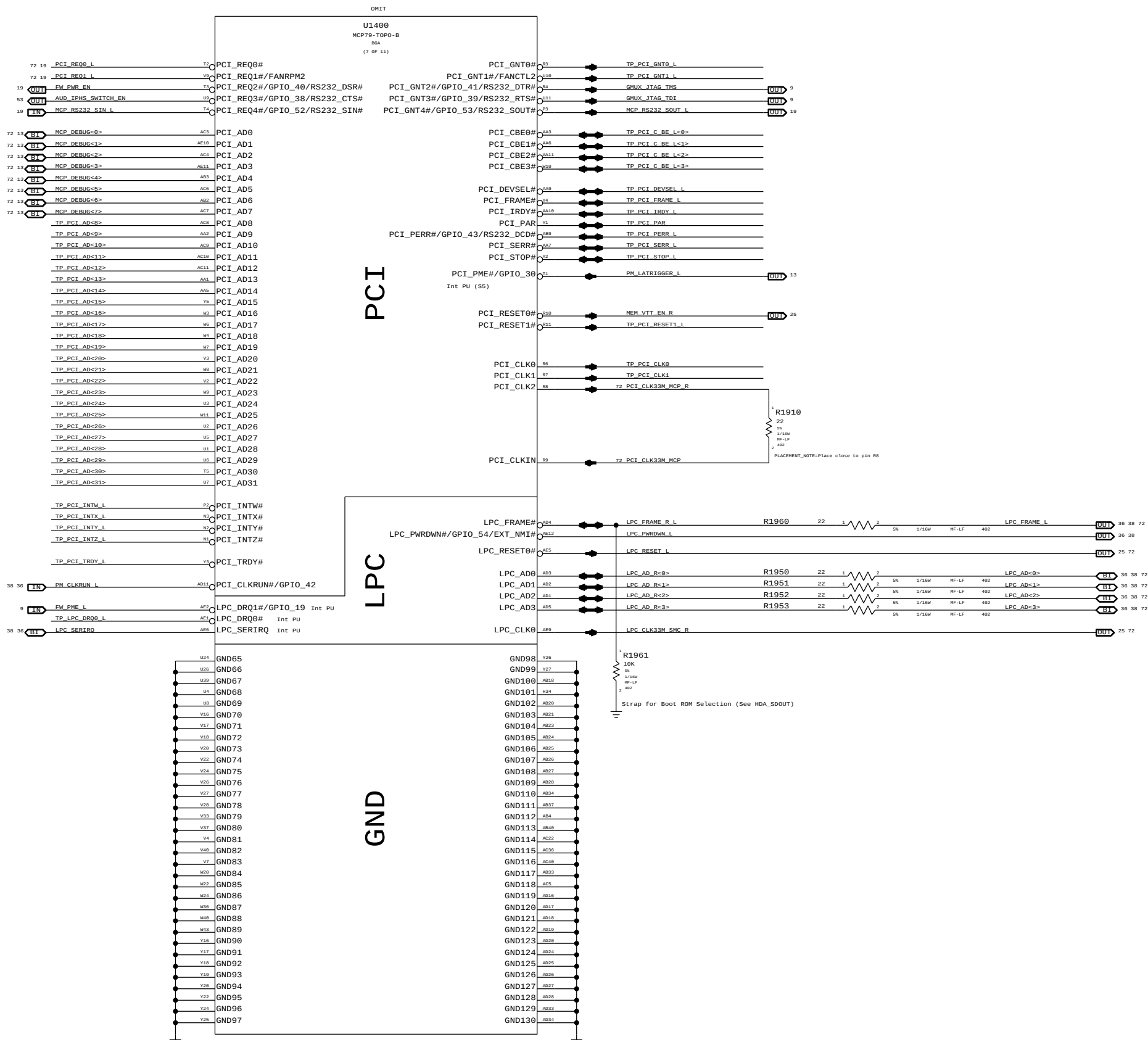
NOTE: All Apple products set strap to MII, RGMII products will enable feature via software. This avoids a leakage issue since MCP79 requires a S5 pull-up.

RGB DAC Disable:
 Okay to float all RGB_DAC signals.
 DDC_CLK0/DDC_DATA0 pull-ups still required.

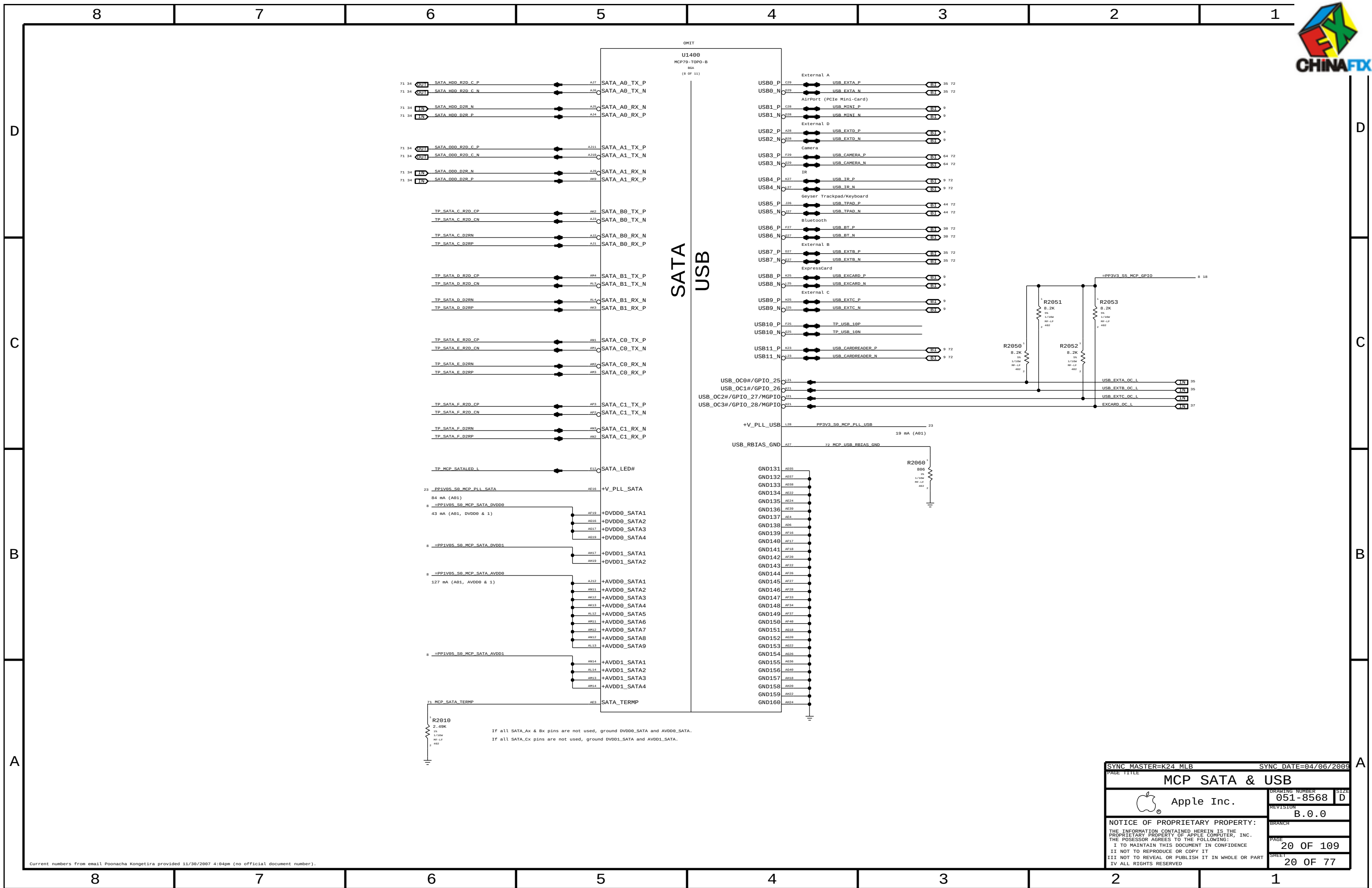
TV DAC Disable:
 Okay to float all TV_DAC signals.
 Okay to float XTALIN_TV and XTALOUT_TV.
 DDC_CLK0/DDC_DATA0 pull-ups still required.

WF: IFP is capable of LVDS (1.8V) or TMDS (3.3V), need aliases

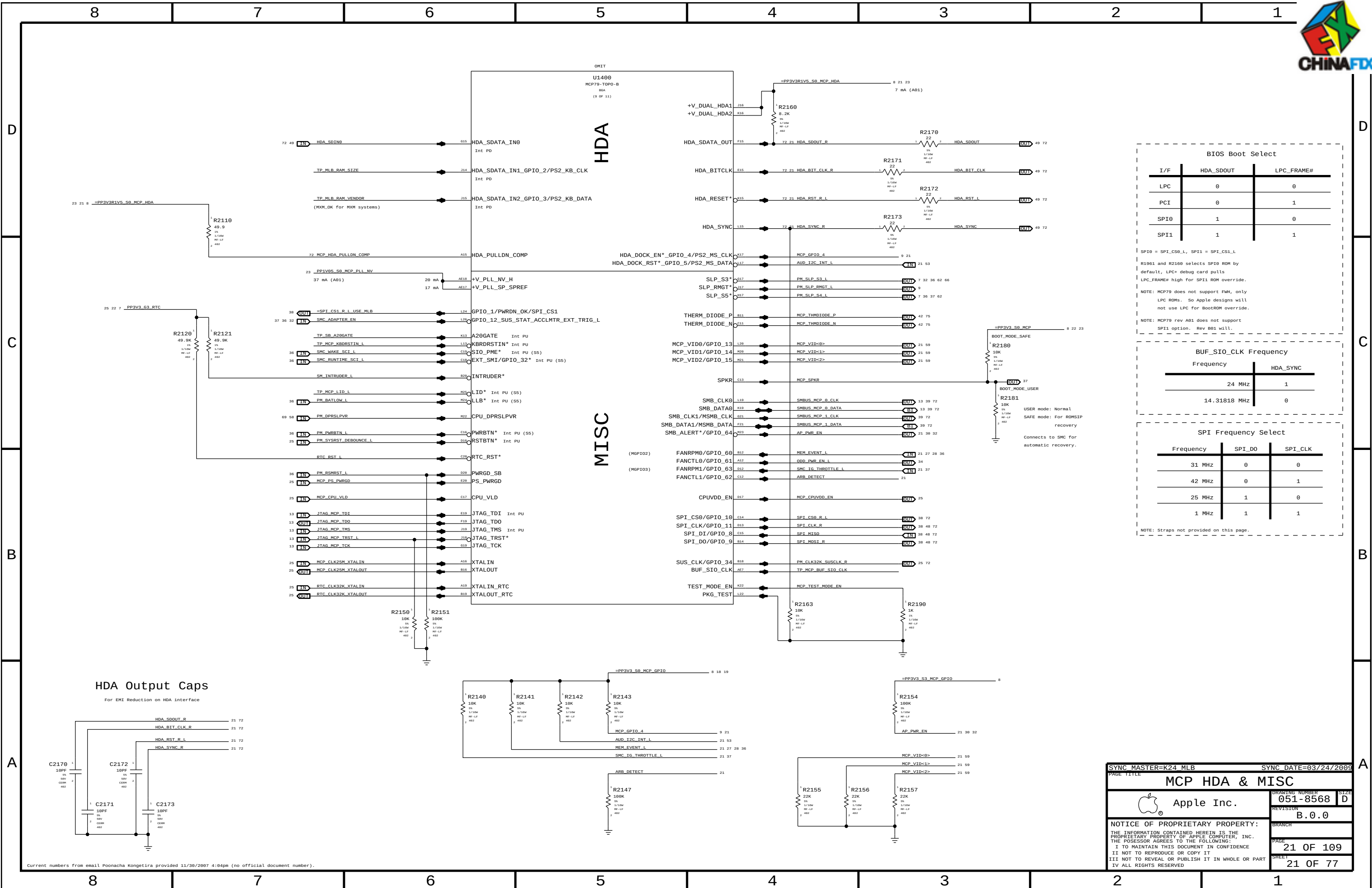
SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE		MCP Ethernet & Graphics	
Apple Inc.		DRAWING NUMBER	051-8568
		REVISION	B.0.0
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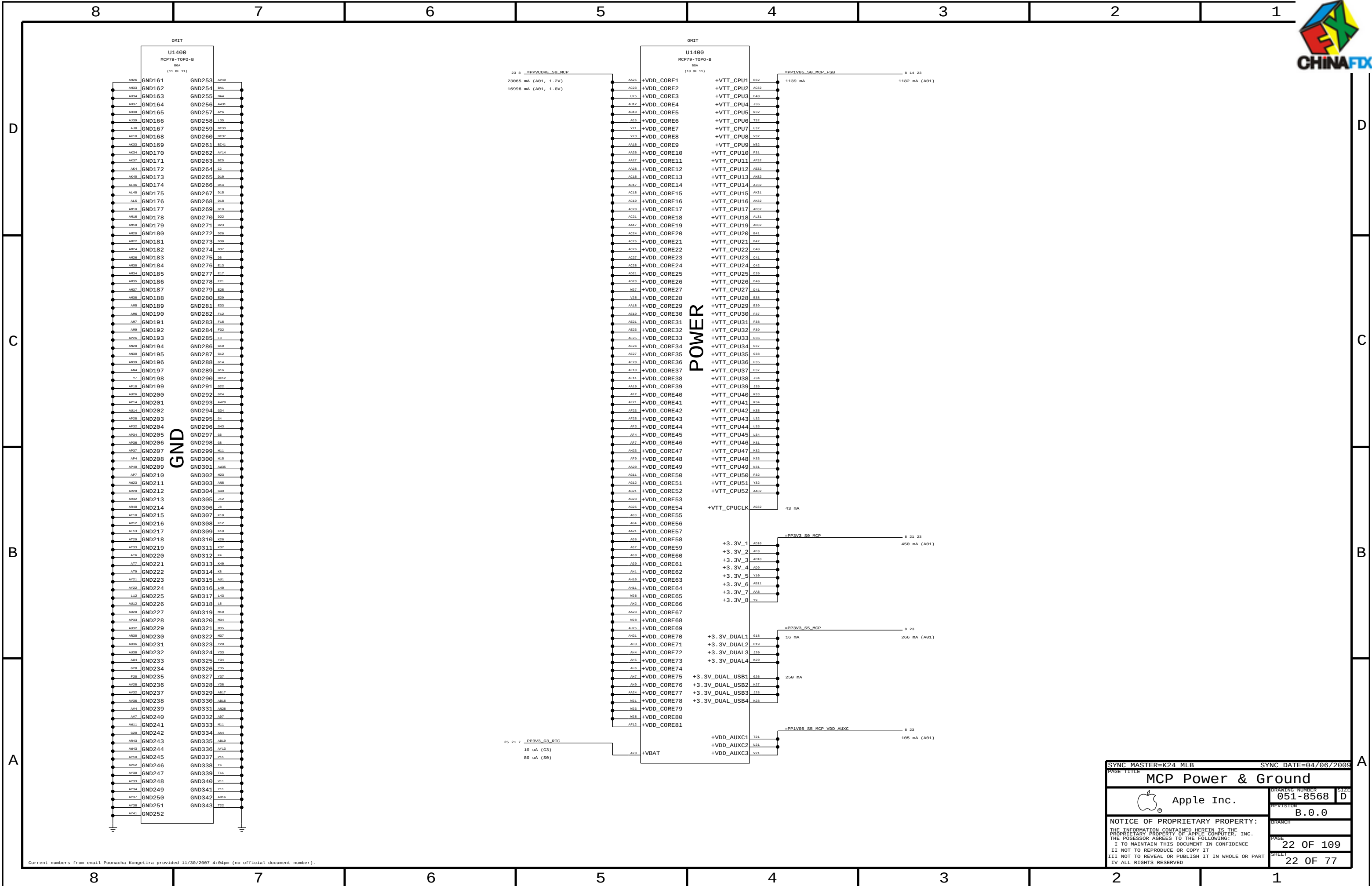


SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE		MCP PCI & LPC	
Apple Inc.		DRAWING NUMBER	051-8568
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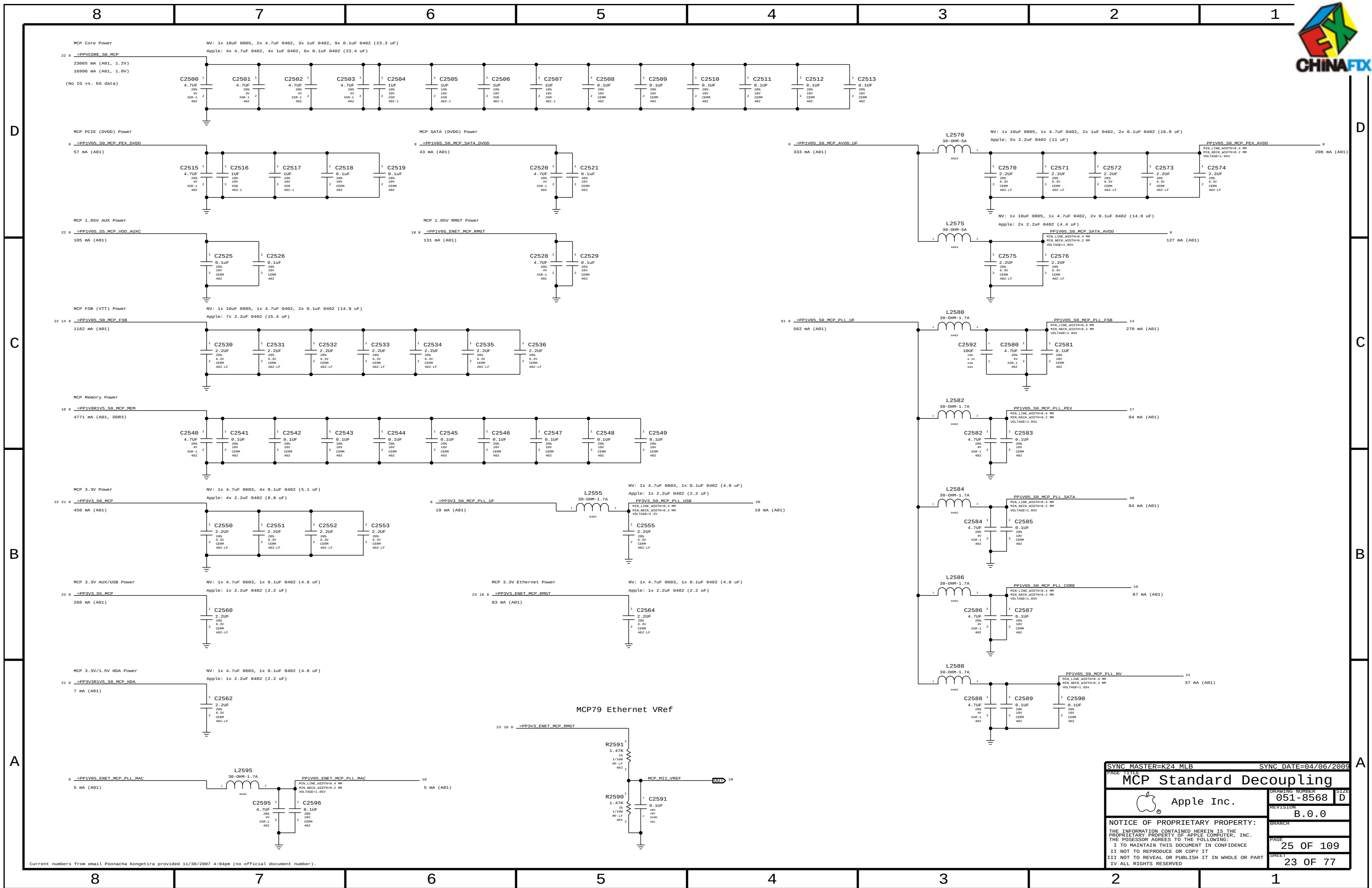


SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE		MCP SATA & USB	
		DRAWING NUMBER	051-8568
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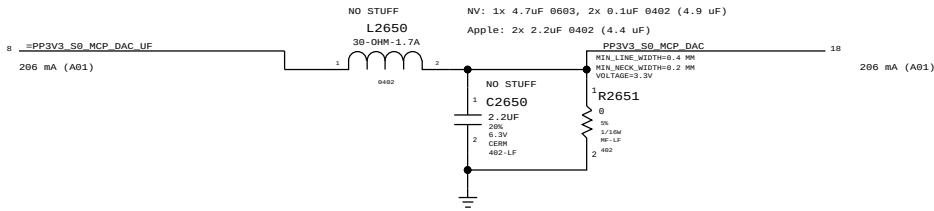
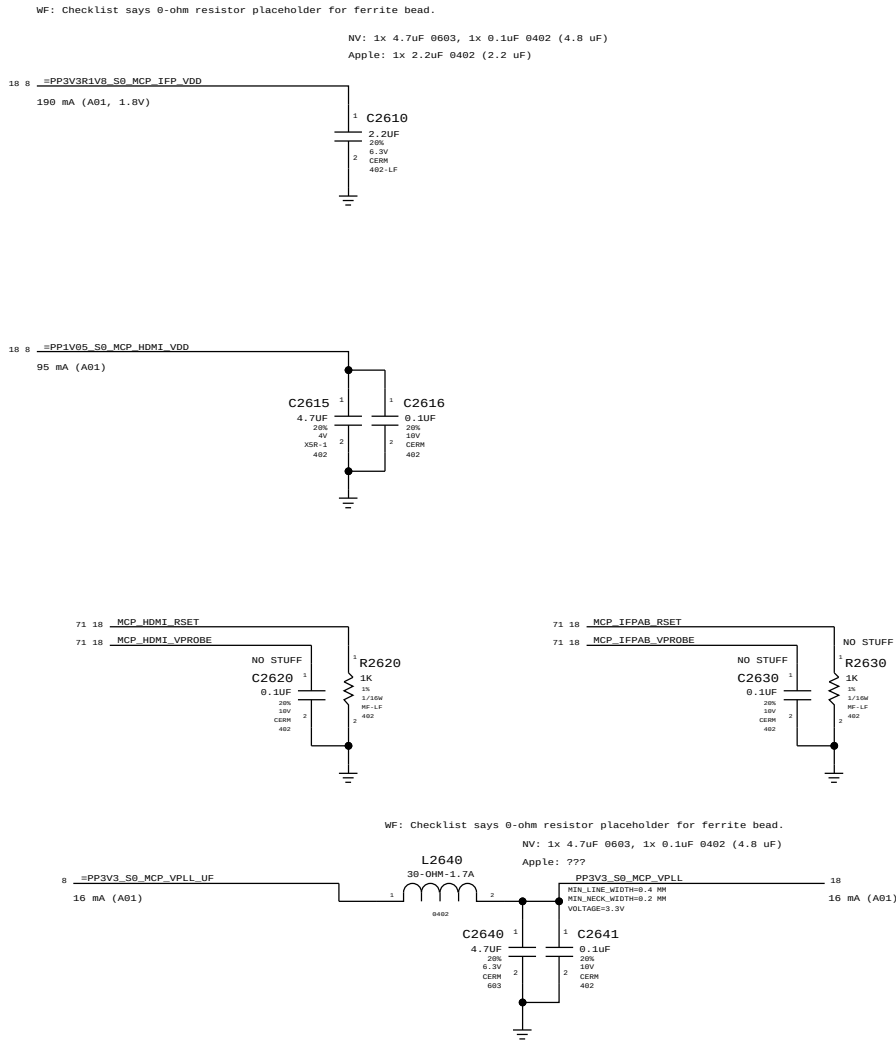




SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE		MCP Power & Ground	
		DRAWING NUMBER	051-8568
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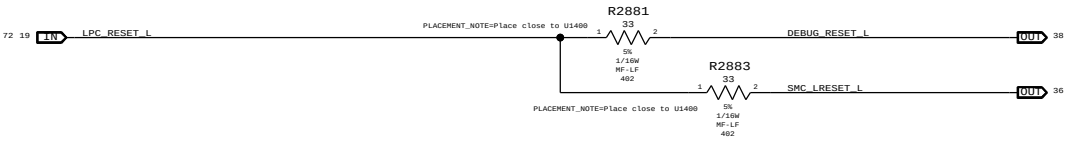
SYNC FROM T18
REMOVE MCP 27MHZ CRYSTAL CRICUIT SINCE NOT SUPPORTING TV-OUT
REMOVE DAC TERMINATIONS R2665,C2665 AND R2670 TO R2672
NOSTUFF PP3V3_S0_MCP_DAC RAIL COMPONENTS (L2650 AND C2650)
CHANGE C2651 TO R2651 TO GND PP3V3_S0_MCP_DAC
REMOVE HDCP ROMS

SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE		MCP Graphics Support	
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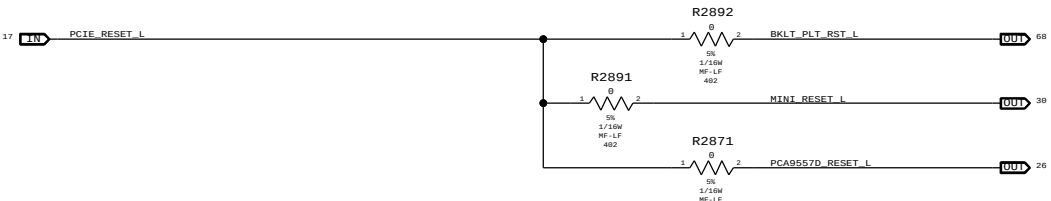


Platform Reset Connections

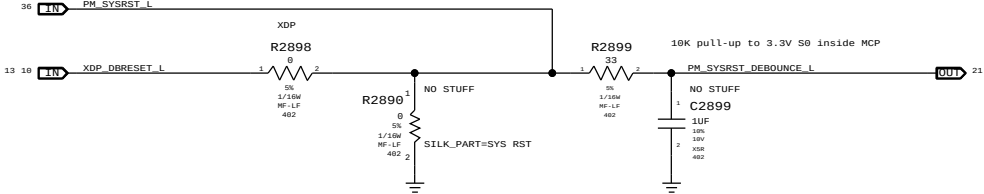
LPC Reset (Unbuffered)



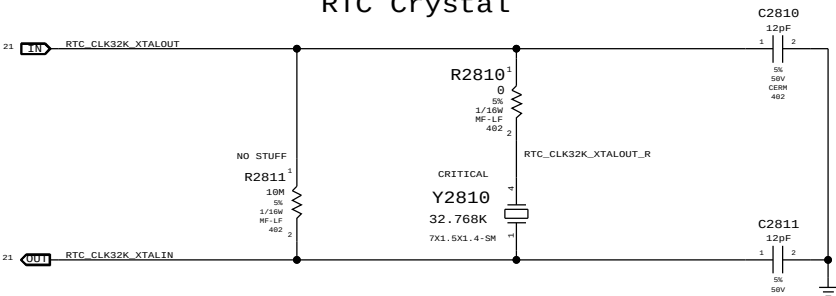
PCIE Reset (Unbuffered)



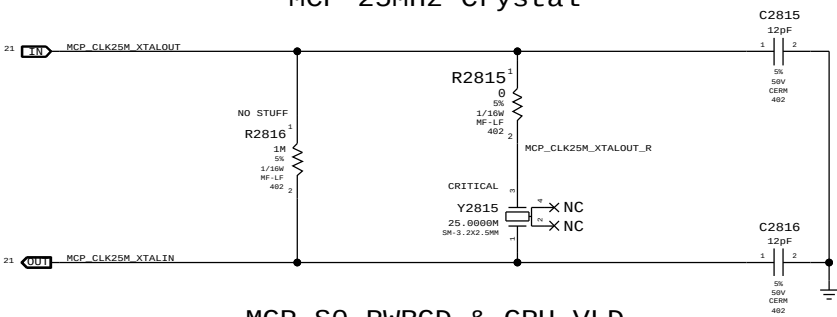
Reset Button



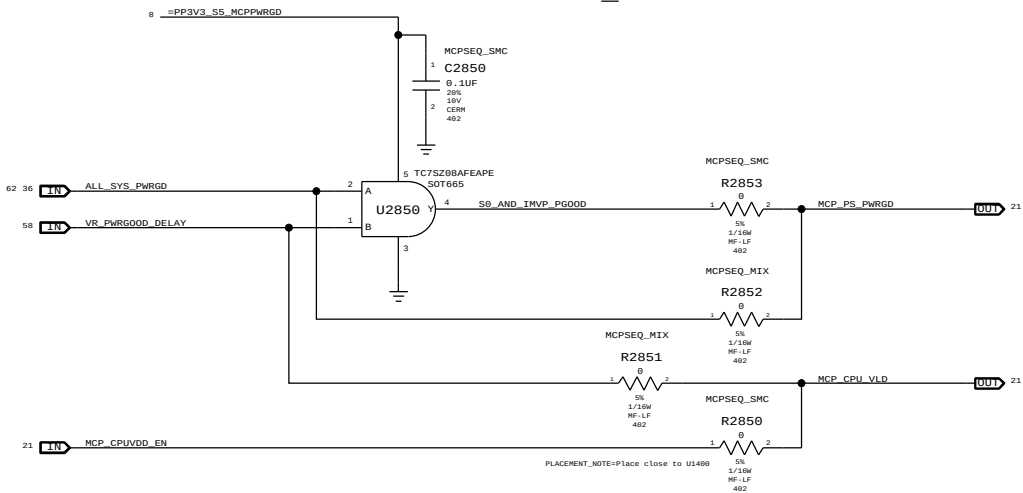
RTC Crystal



MCP 25MHz Crystal



MCP S0 PWRGD & CPU_VLD



SYNC FROM T18
CHANGE RESET BUTTON TO RESET PADS
REMOVE UNUSED PCIE RESET SIGNALS
REMOVE R2824 AND NET PCI_CLK33M_SLOT_A
CHANGE RTC COIN CELL TO LDO & SUPERCAP
ALIAS MEM_VTT_EN TO =DDRVTT_EN
CHANGE Y2810 AND U2850 TO SMALLER PARTS

SYNC MASTER=K24 MLB		SYNC DATE=02/15/2009	
PAGE TITLE		SB Misc	
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Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PP3V3_S5_VREFMRGN
- =PPVTT_S3_DDR_BUF

Signal aliases required by this page:

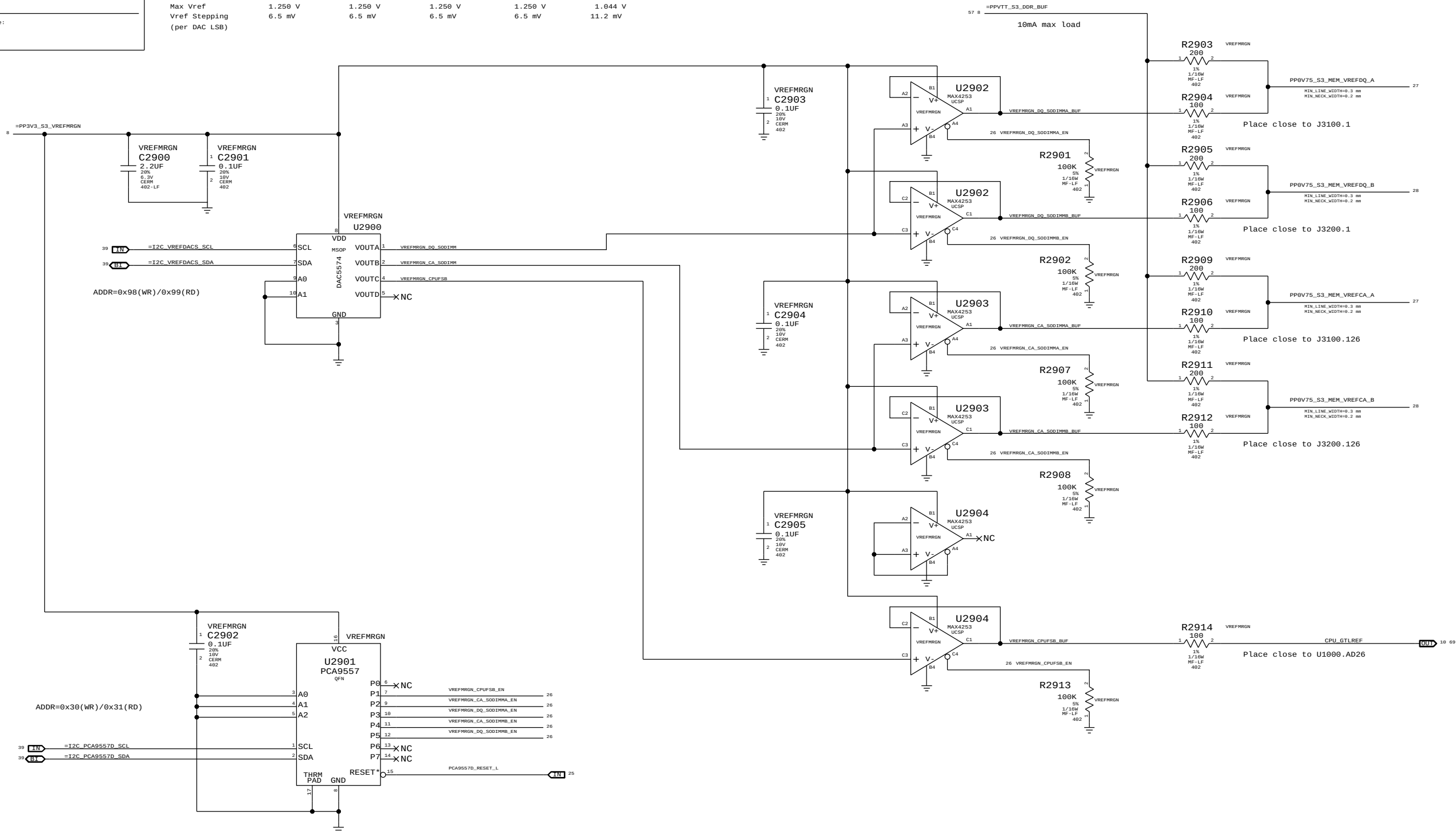
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- VREFMRGN
- NO_VREFMRGN

	MEM A VREF DQ		MEM A VREF CA		MEM B VREF DQ		MEM B VREF CA		CPU FSB VREF
DAC channel	A	B	A	B	A	B	C		
Min DAC code	0x00	0x00	0x00	0x00	0x00	0x00	0x00		
Max DAC code	0x87	0x87	0x87	0x87	0x87	0x87	0x55		
Max sink I	-3.75 mA	-3.75 mA	-3.75 mA	-3.75 mA	-3.75 mA	-3.75 mA	-0.91 mA		
Max source I	5 mA	5 mA	5 mA	5 mA	5 mA	5 mA	0.52 mA		
Nominal Vref	0.75 V	0.75 V	0.75 V	0.75 V	0.75 V	0.75 V	0.70 V		
Min Vref	0.375 V	0.375 V	0.375 V	0.375 V	0.375 V	0.375 V	0.091 V		
Max Vref	1.250 V	1.250 V	1.250 V	1.250 V	1.250 V	1.250 V	1.044 V		
Vref Stepping (per DAC LSB)	6.5 mV	6.5 mV	6.5 mV	6.5 mV	6.5 mV	6.5 mV	11.2 mV		

S0-DIMM A and S0-DIMM B Vref settings should be margined separately (i.e. not simultaneously) due to current limitation of TPS51116 regulator.



Required zero ohm resistors when no VREF margining circuit stuffed

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	1	RES, MTL FILM, 0, 5%, 0402, SM, LF	R2903	CRITICAL	NO_VREFMRGN
116S0004	1	RES, MTL FILM, 0, 5%, 0402, SM, LF	R2905	CRITICAL	NO_VREFMRGN
116S0004	1	RES, MTL FILM, 0, 5%, 0402, SM, LF	R2909	CRITICAL	NO_VREFMRGN
116S0004	1	RES, MTL FILM, 0, 5%, 0402, SM, LF	R2911	CRITICAL	NO_VREFMRGN

SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE		FSB/DDR3 Vref Margining	
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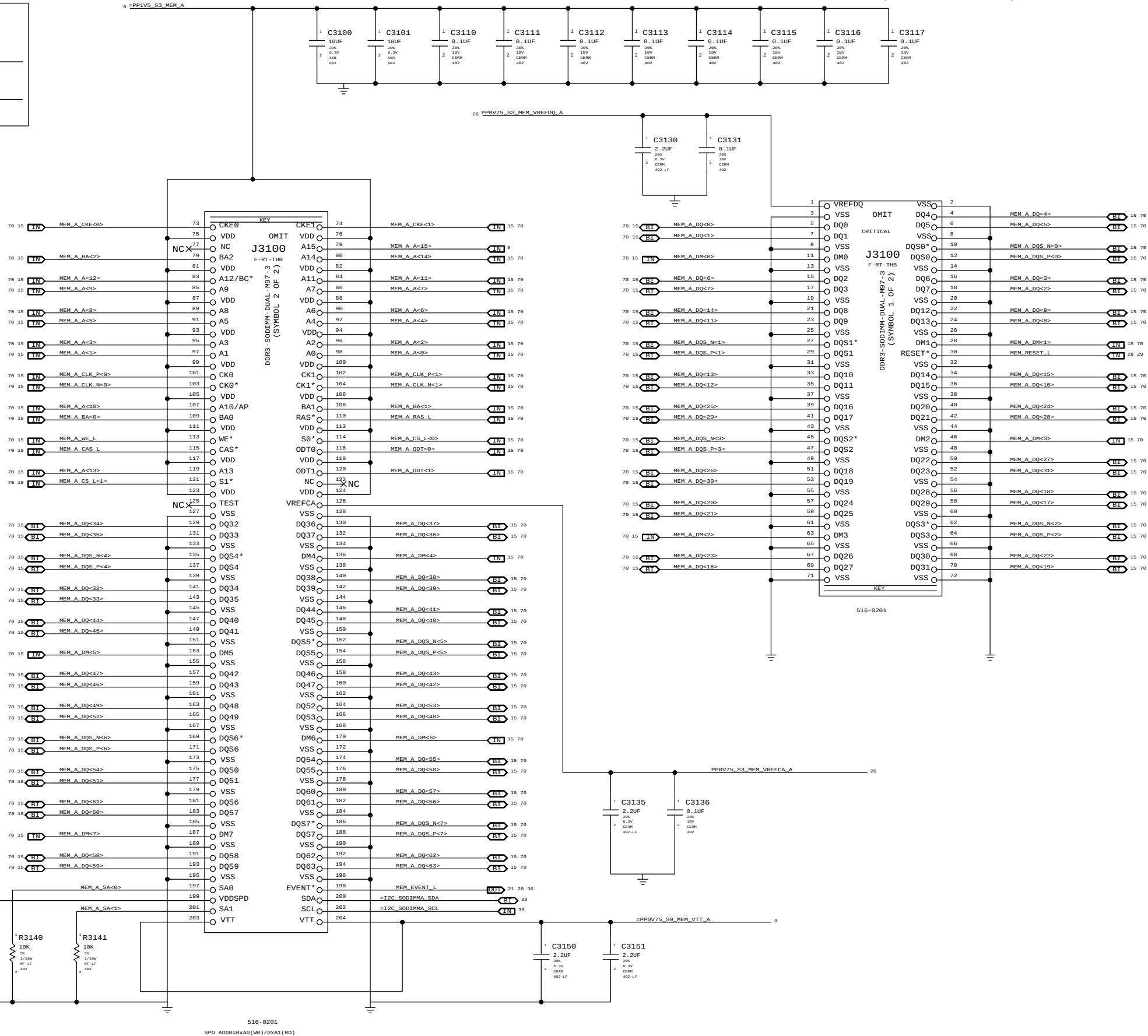
Page Notes

Power aliases required by this page:
- =PP1V5_S3_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S3_MEM_VTT_A
- =PPSPD_S3_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

BOM options provided by this page:
(NONE)

DDR3 DECOUPLING AND GROUND RETURN CAPS (CONNECTOR SIDE)



"Factory" (top) slot

SYNC MASTER=K24 MLB		SYNC DATE=02/05/2009	
PAGE TITLE			
DDR3 SO-DIMM Connector A			
 Apple Inc.		DRAWING NUMBER	051-8568
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Page Notes

Power aliases required by this page:

- =PP1V5_S3_MEM_B
- =PP1V5_S3_MEM_B
- =PP0V75_S3_MEM_VTT_B
- =PPSPD_S3_MEM_B (2.5 - 3.3V)

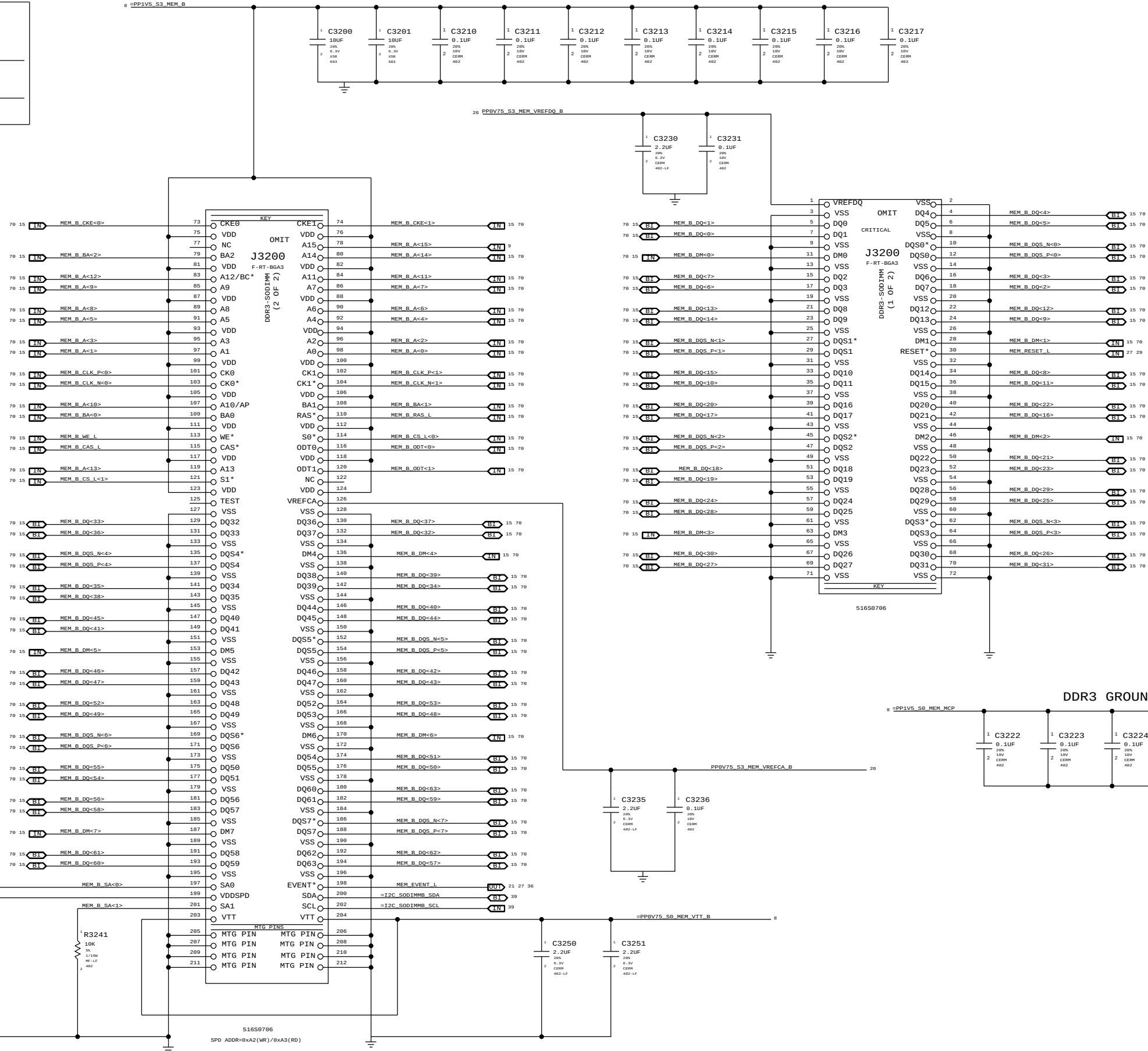
Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

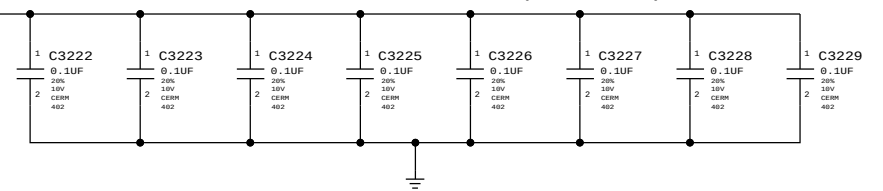
BOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GROUND RETURN CAPS (CONNECTOR SIDE)



DDR3 GROUND RETURN CAPS (MCP SIDE)



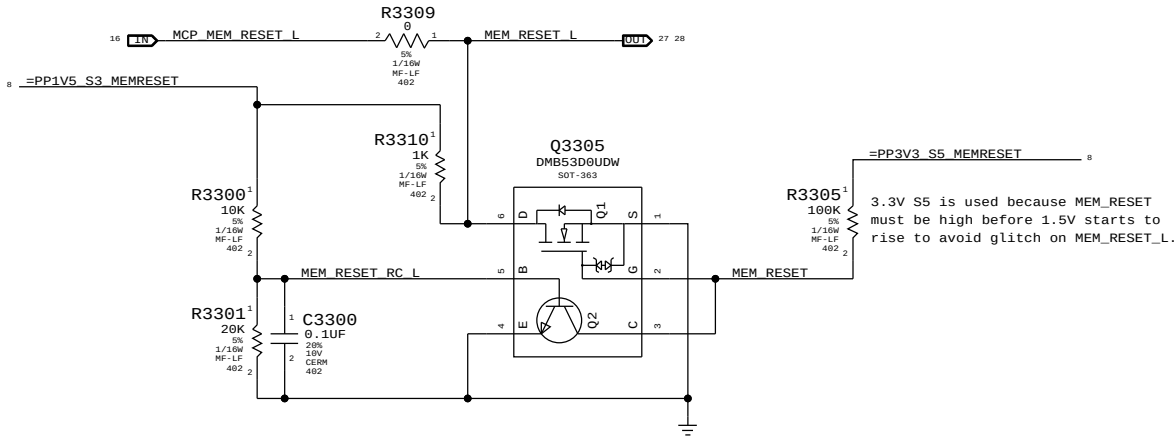
"Expansion" (bottom) slot

SYNC MASTER=K24 MLB		SYNC DATE=02/05/2009	
PAGE TITLE			
DDR3 S0-DIMM Connector B			
Apple Inc.		DRAWING NUMBER	051-8568
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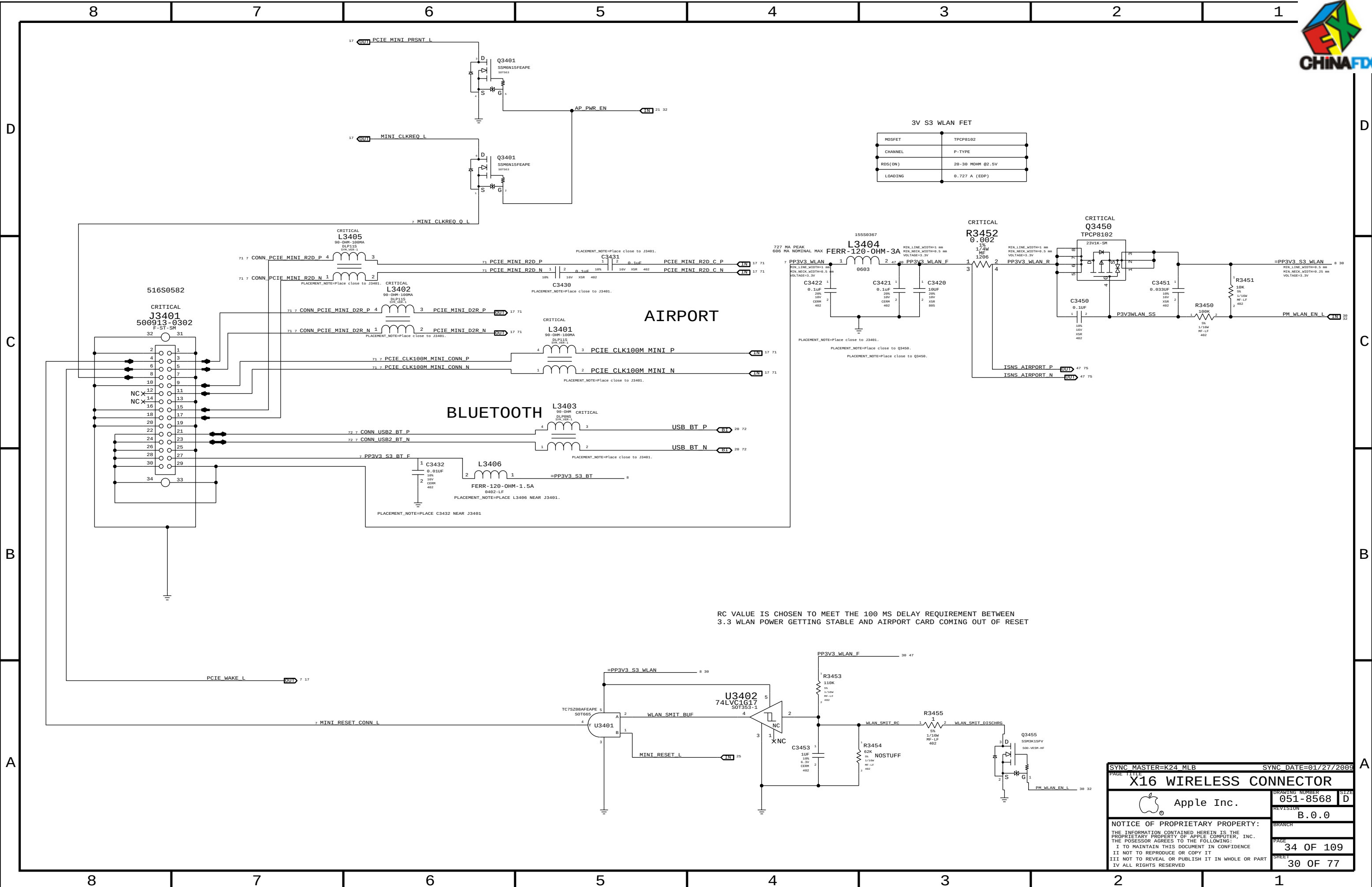
DDR3 RESET Support

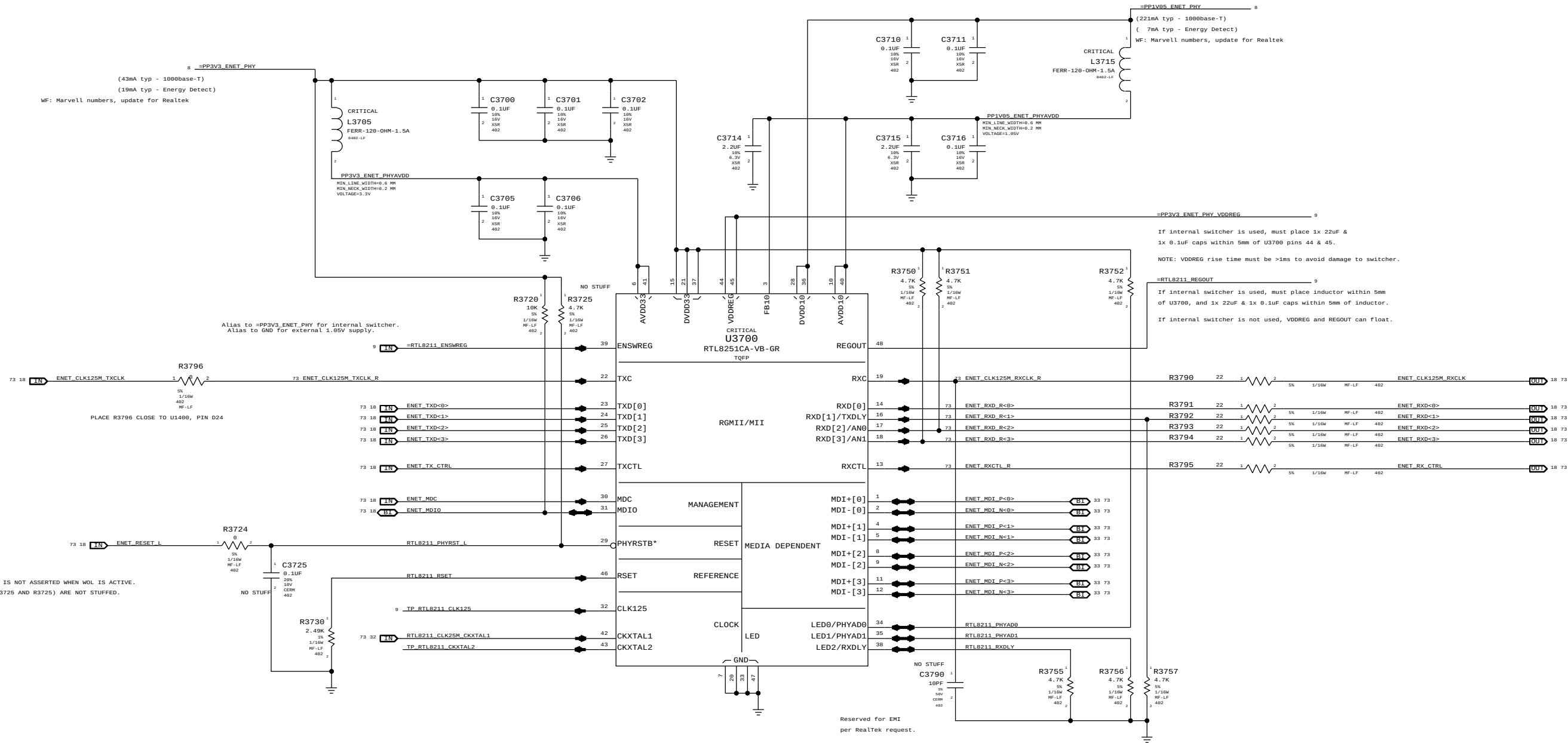
Required because MCP79 does not meet DDR3 spec power-up reset timing requirement.



3.3V S5 is used because MEM_RESET must be high before 1.5V starts to rise to avoid glitch on MEM_RESET_L.

SYNC_MASTER=K24_MLB		SYNC_DATE=04/06/2009	
PAGE TITLE			
DDR3 Support			
 Apple Inc.		DRAWING NUMBER	051-8568
		REVISION	B.0.0
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		PAGE	33 OF 109
		SHEET	29 OF 77

[illegible]



Configuration Settings:

PHYAD = 01 (PHY Address 00001)
 AN[1:0] = 11 (Full auto-negotiation)
 RXDLY = 0 (RXCLK transitions with data)
 TXDLY = 0 (No TXCLK Delay)

SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE		Ethernet PHY (RTL8211CL)	
DRAWING NUMBER		051-8568	SIZE D
REVISION		B.0.0	BRANCH
PAGE		37 OF 109	SHEET
SHEET		31 OF 77	

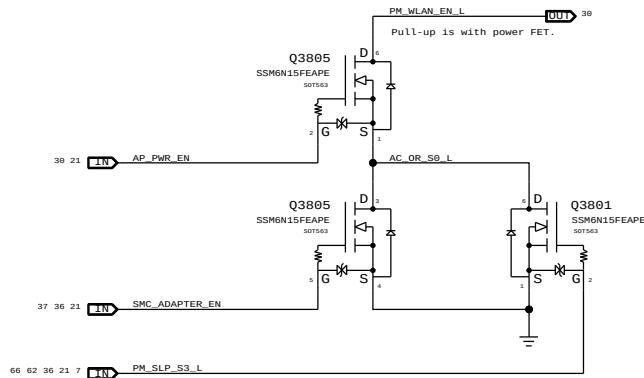
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WLAN Enable Generation

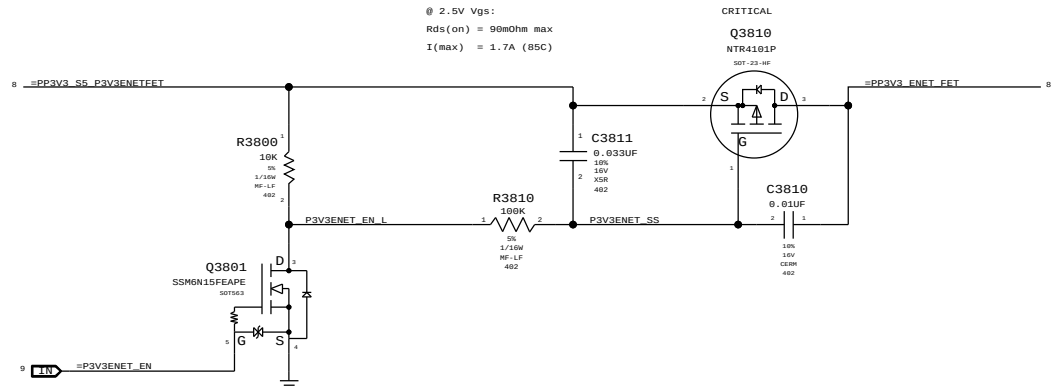
"WLAN" = ("S3" && "AP_PWR_EN" && ("AC" || "S6"))

NOTE: S3 term is guaranteed by S3 pull-up on open-drain AP_PWR_EN signal.



3.3V ENET FET

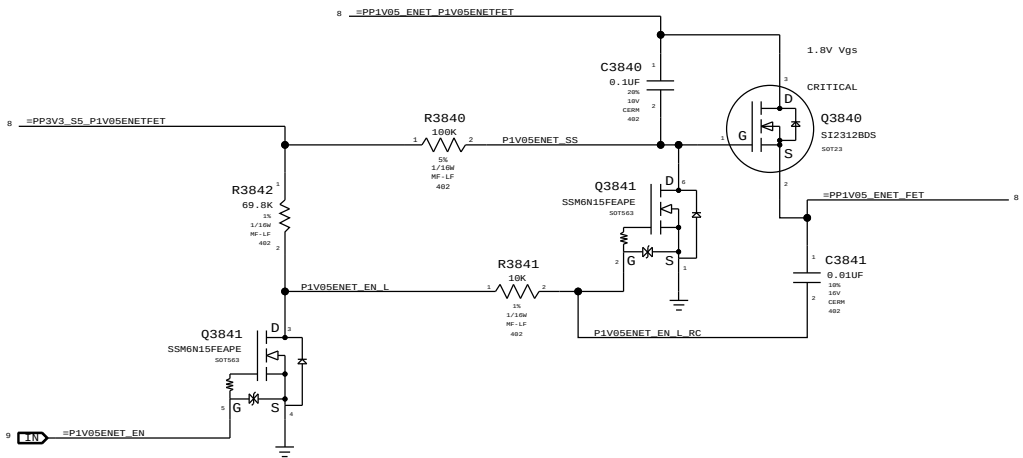
@ 2.5V Vgs:
Rds(on) = 90mOhm max
I(max) = 1.7A (85C)



MOBILE:

Recommend aliasing PM_SLP_RMG_T_L and
=P3V3ENET_EN. Nets separated on
ARB for alternate power options.

1.05V ENET FET

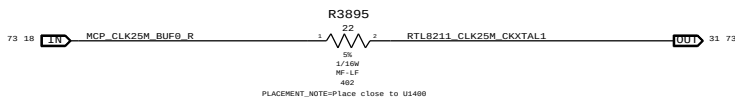


Non-ARB:

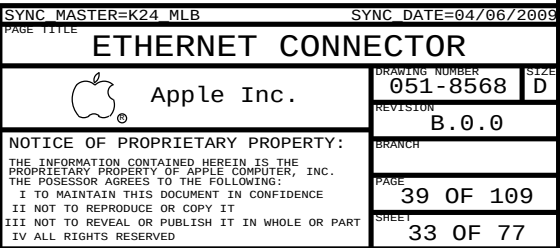
Recommend aliasing PM_SLP_RMG_T_L and
=P1V05ENET_EN. Nets separated on
ARB for alternate power options.

RTL8211 25MHz Clock

NOTE: MCP79 can provide 25MHz clock, but clock runs whenever RMGT rails are powered.
Designs must ensure PHY is powered whenever RMGT rails are, or use separate crystal.

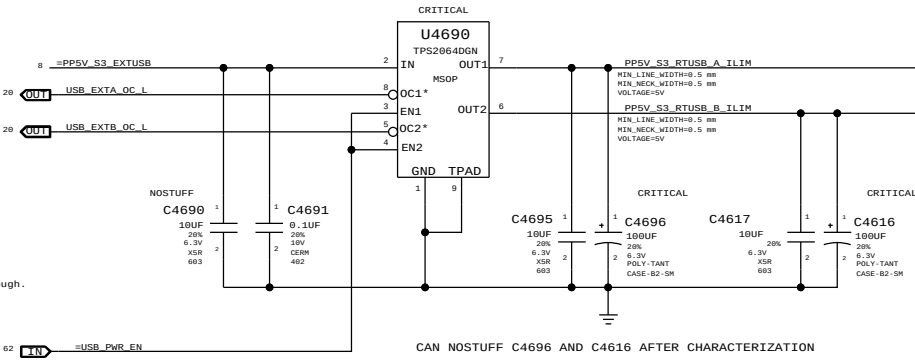


SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE		Ethernet & AirPort Support	
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		REVISION	B.0.0
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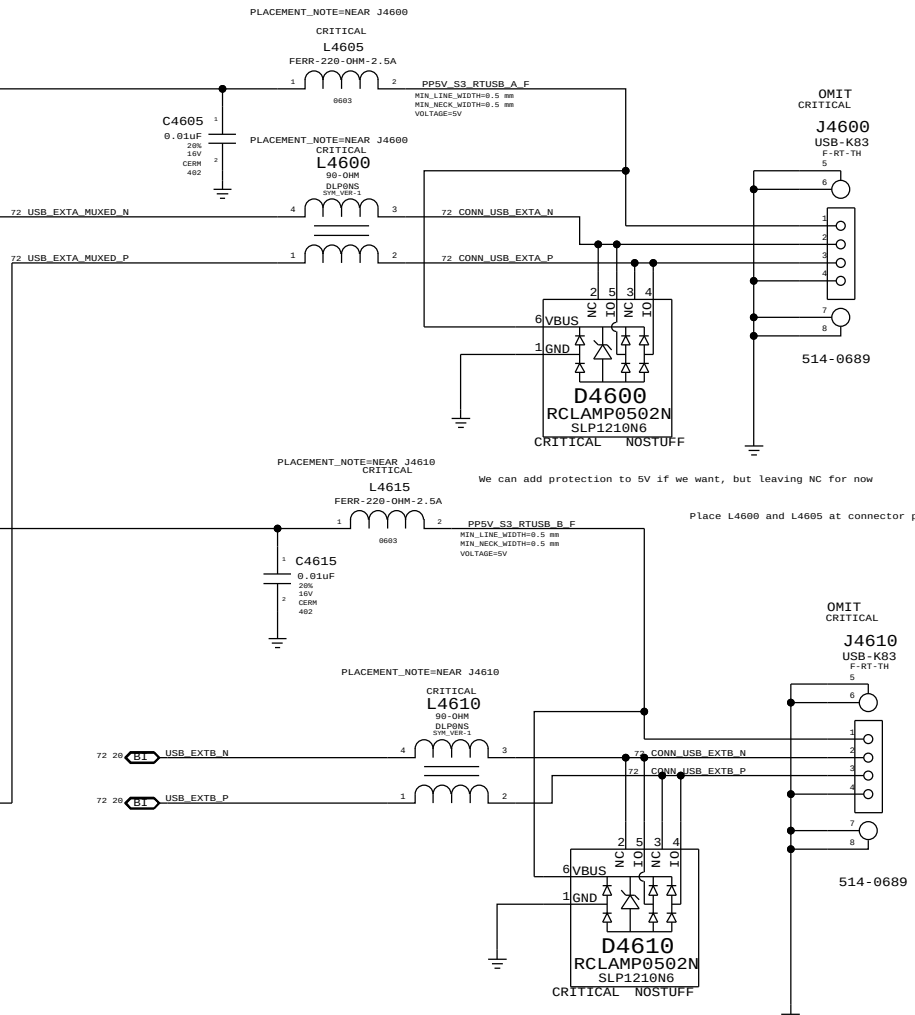


POR IS PLASTIC USB CONNECTOR PARTS BUT METAL PART'S SCHEMATIC AND CAD SYMBOLS HAVE BEEN USED AS ITS LAND PATTERN CAN ACCOMODATE BOTH TYPES

Port Power Switch

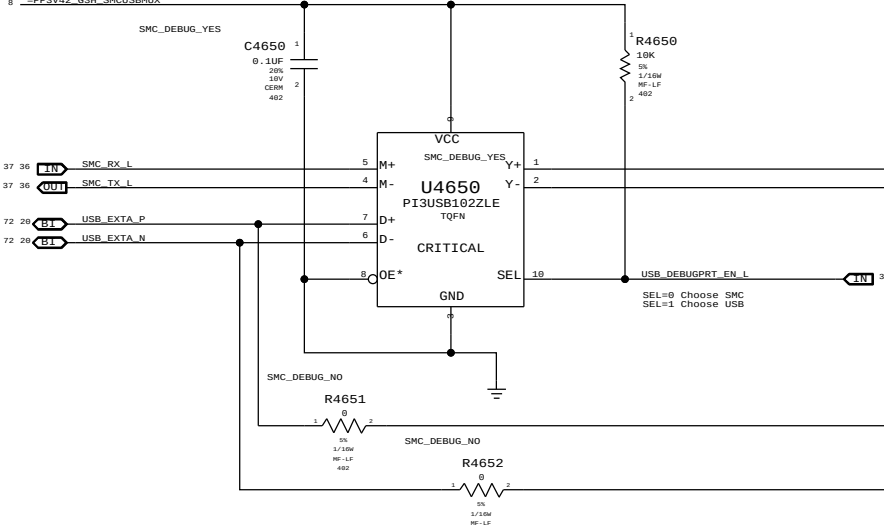


USB PORT A (FRONT PORT)



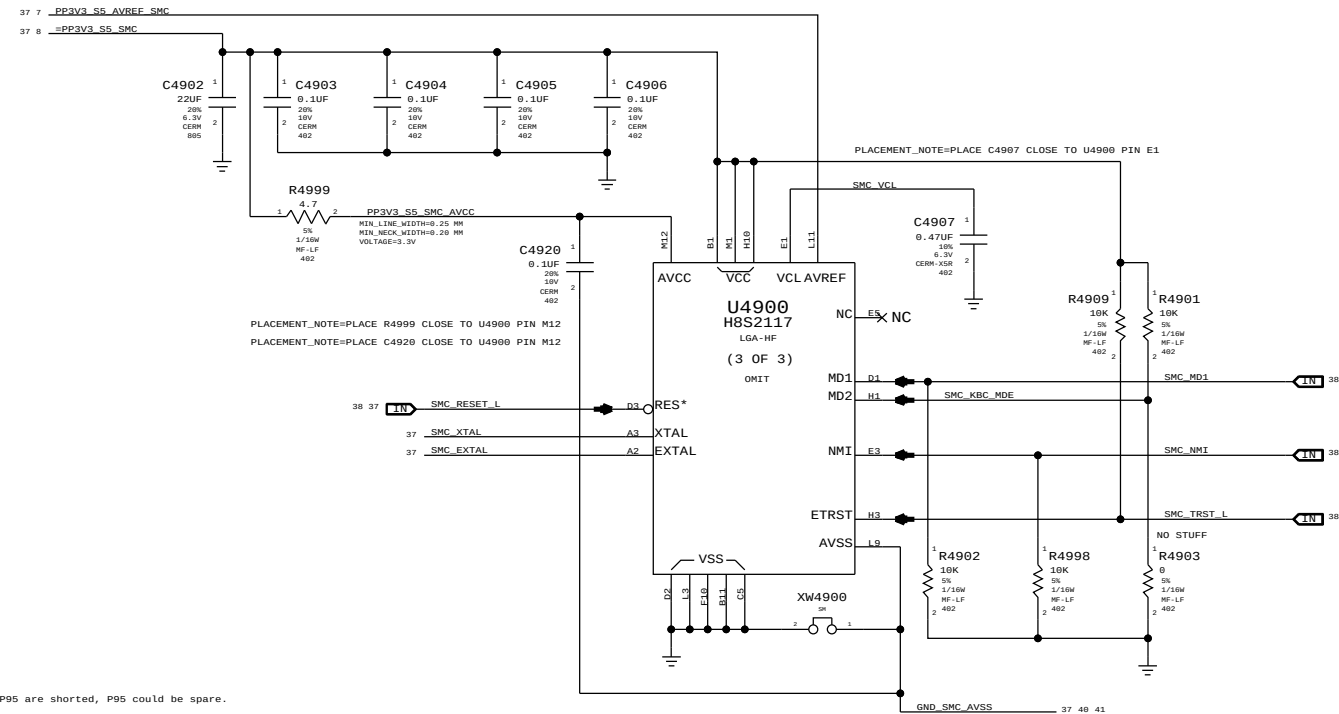
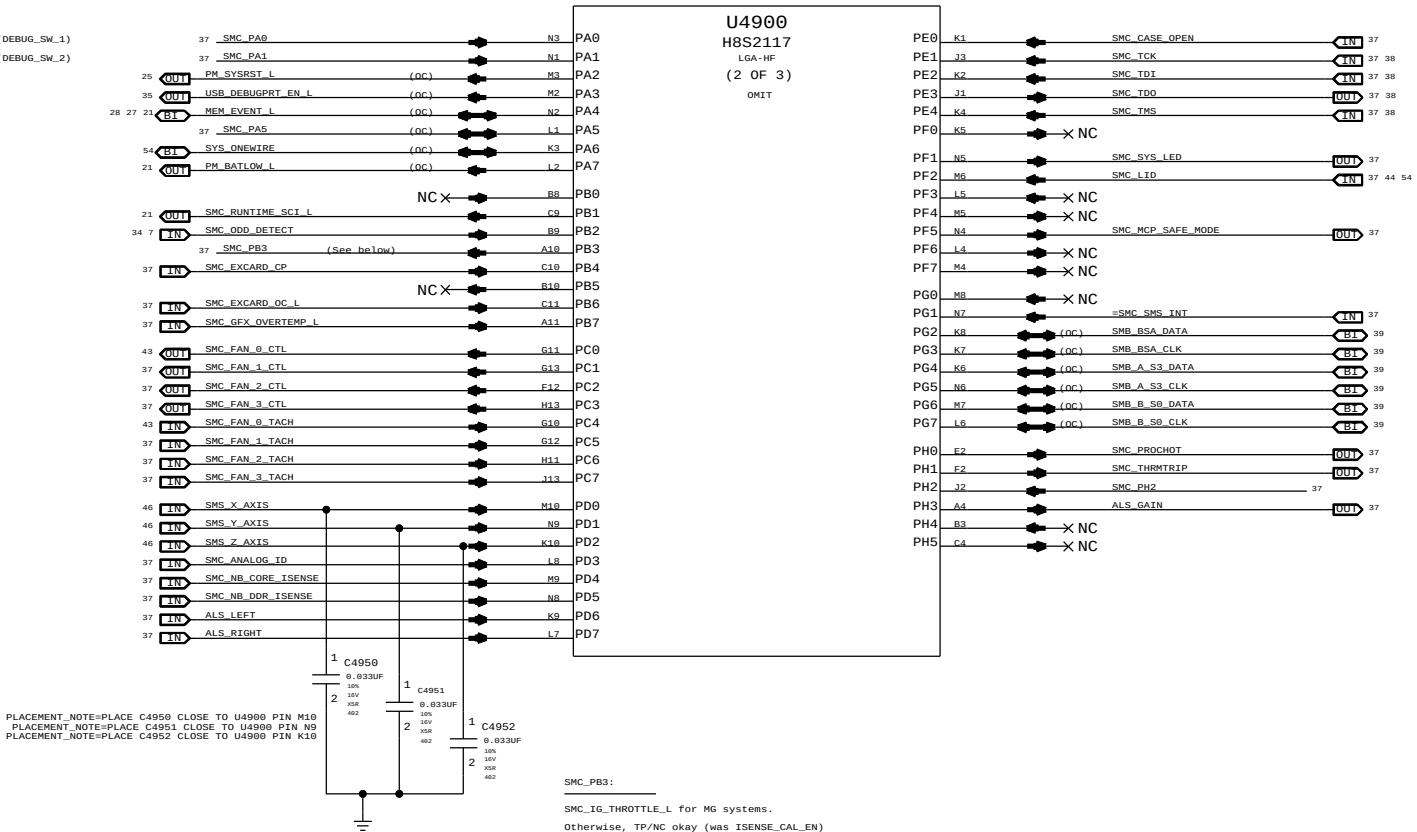
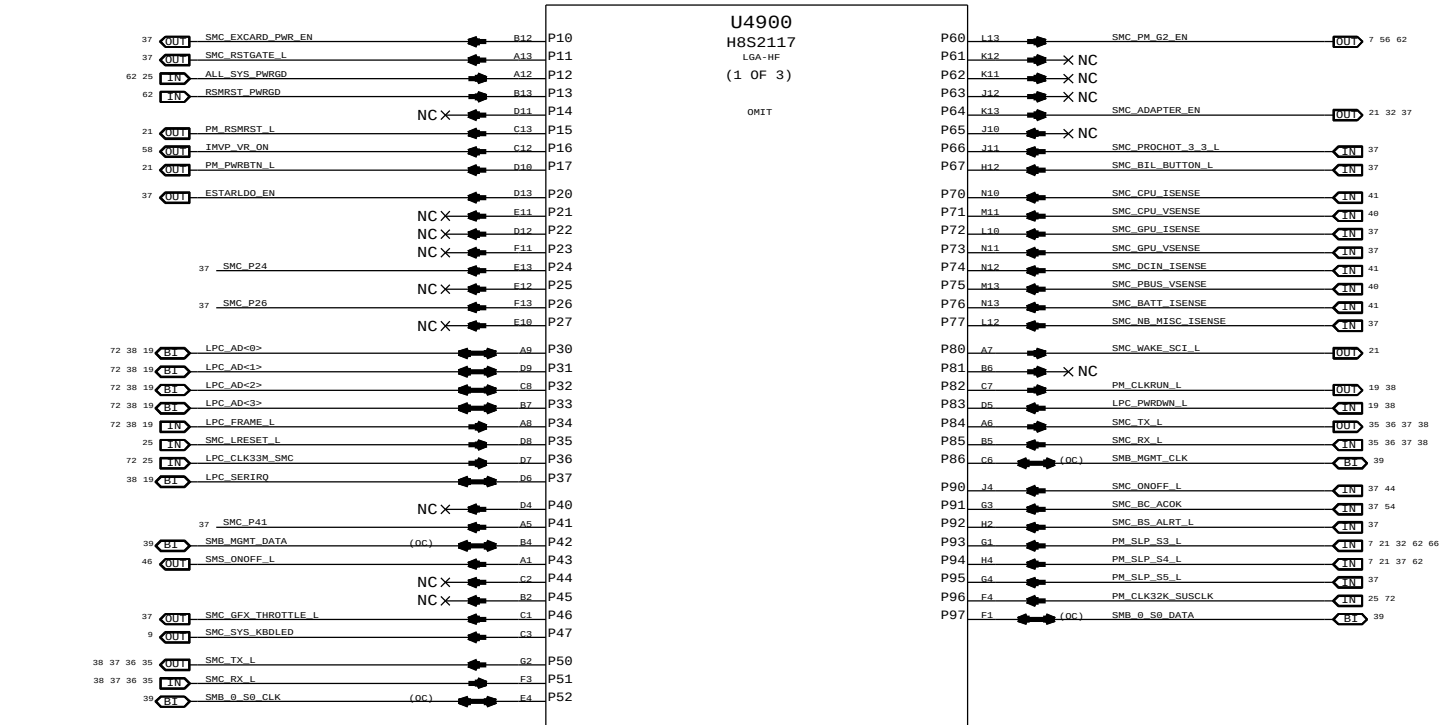
USB PORT B (BACK PORT)

USB/SMC Debug Mux



SYNC MASTER=K24 MLB		SYNC DATE=02/05/2009	
External USB Connectors		DRAWING NUMBER 051-8568	
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NOTE: Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

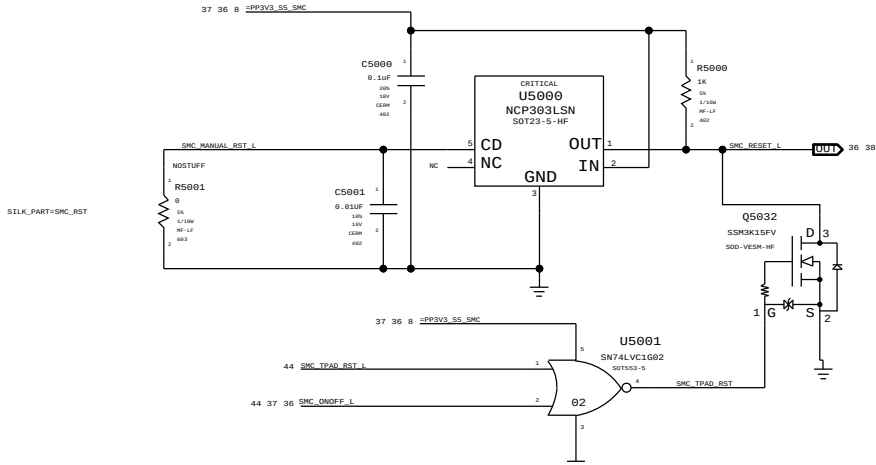


NOTE: P94 and P95 are shorted, P95 could be spare.

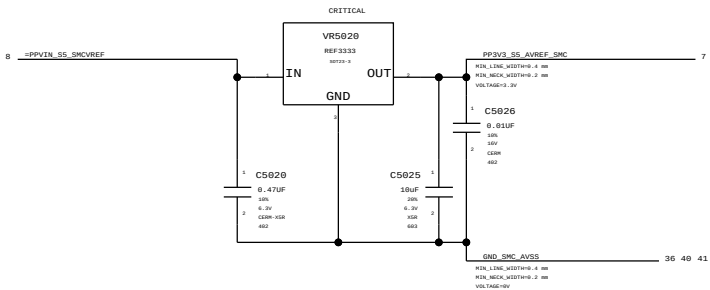
NOTE: SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

36	_SMC_BILL_BUTTON_L	==	NC_SMC_BILL_BUTTON_L	
36	_SMC_FAN_4_CTL	==	MAKE_BASE+TRUE	NC_SMC_FAN_1_CTL
36	_SMC_FAN_2_CTL	==	MAKE_BASE+TRUE	NC_SMC_FAN_2_CTL
36	_SMC_FAN_3_CTL	==	MAKE_BASE+TRUE	NC_SMC_FAN_3_CTL
36	_SMC_GFX_THROTTLE_L	==	MAKE_BASE+TRUE	SMC_GFX_THROTTLE_L
36	_ESTARLDO_EN	==	MAKE_BASE+TRUE	NC_ESTARLDO_EN
54 37 36	_SMC_RC_ACOK	==	==CHSR_ACOK	
36	_SMC_P24	==	TP_SMC_P24	
36	_SMC_P26	==	TP_SMC_P26	
36	_SMC_P41	==	TP_SMC_P41	
36	_SMC_NB_CORE_ISENSE	==	MAKE_BASE+TRUE	SMC_MCP_CORE_ISENSE
36	_SMC_NB_DDR_ISENSE	==	MAKE_BASE+TRUE	SMC_MCP_DDR_ISENSE
36	_ALS_LEFT	==	MAKE_BASE+TRUE	SMC_CPU_FSB_ISENSE
36	_SMC_GPU_VSENSE	==	MAKE_BASE+TRUE	SMC_MCP_VSENSE
36	_SMC_EXCARD_PWR_EN	==	MAKE_BASE+TRUE	TP_SMC_EXCARD_PWR_EN
36	_SMC_RSTGATE_L	==	MAKE_BASE+TRUE	TP_SMC_RSTGATE_L
36	_SMC_PB3	==	MAKE_BASE+TRUE	NC_SMC_PB3
36	_ALS_GAIN	==	MAKE_BASE+TRUE	NC_ALS_GAIN
36	_SMC_ANALOG_ID	==	MAKE_BASE+TRUE	NC_SMC_ANALOG_ID
36	_ALS_RIGHT	==	MAKE_BASE+TRUE	NC_ALS_RIGHT

SMC Reset "Button" / Brownout Detect

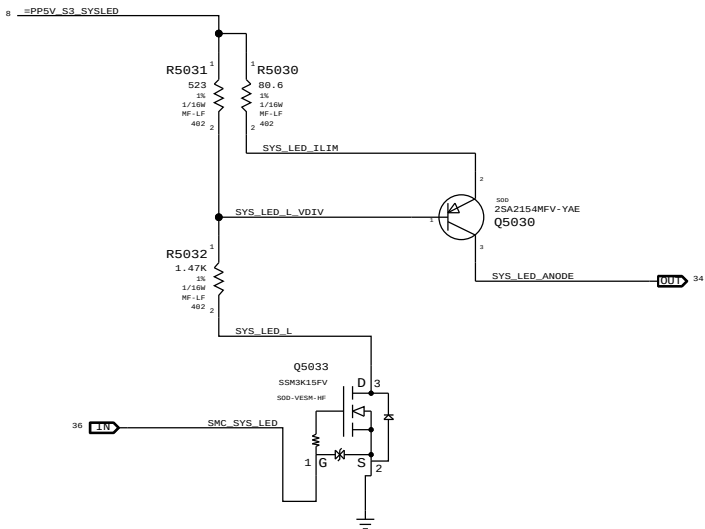


SMC AVREF Supply

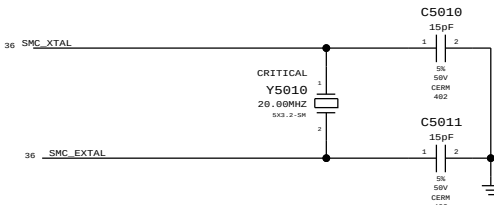


PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S1381	353S1912		ALL	ISL6002-33, INTERSIL

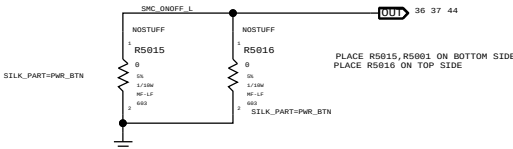
System (Sleep) LED Circuit



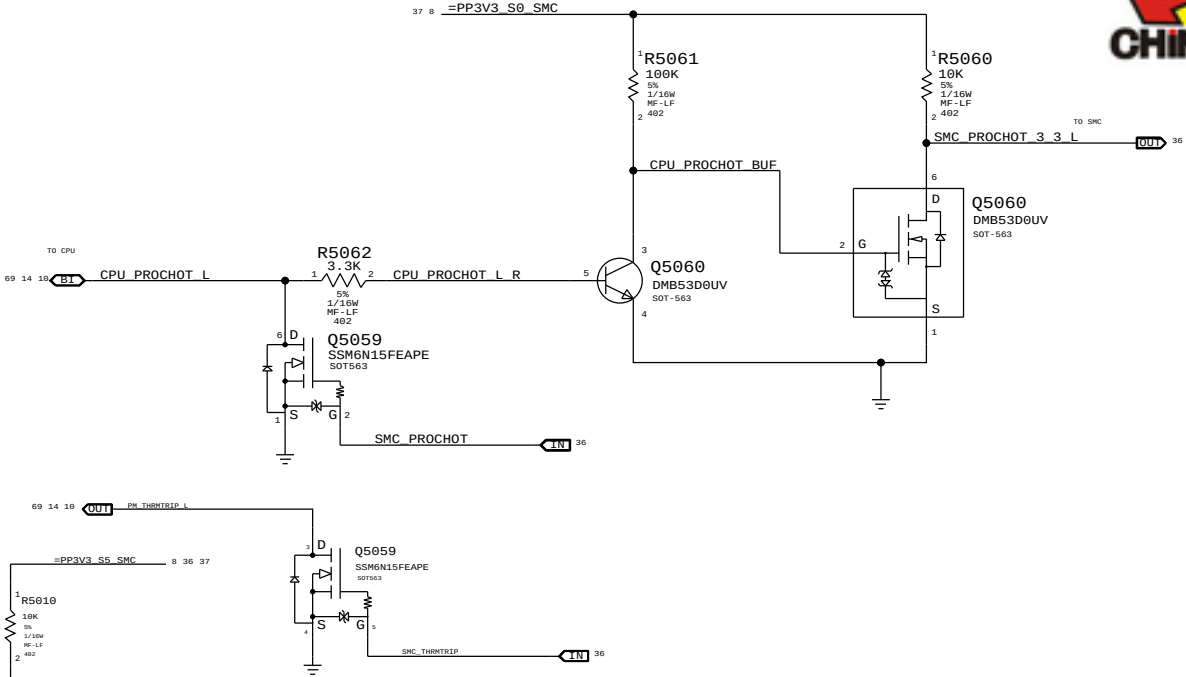
SMC Crystal Circuit



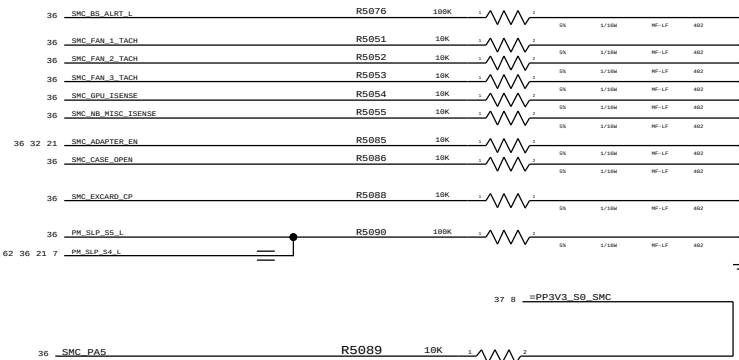
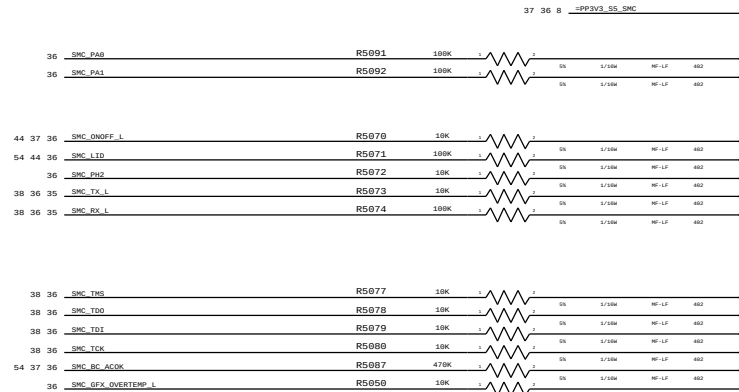
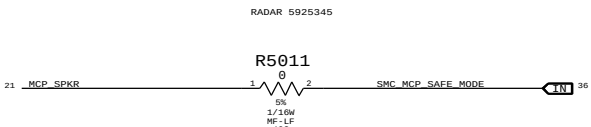
Debug Power "Button"



SMC FSB to 3.3V Level Shifting



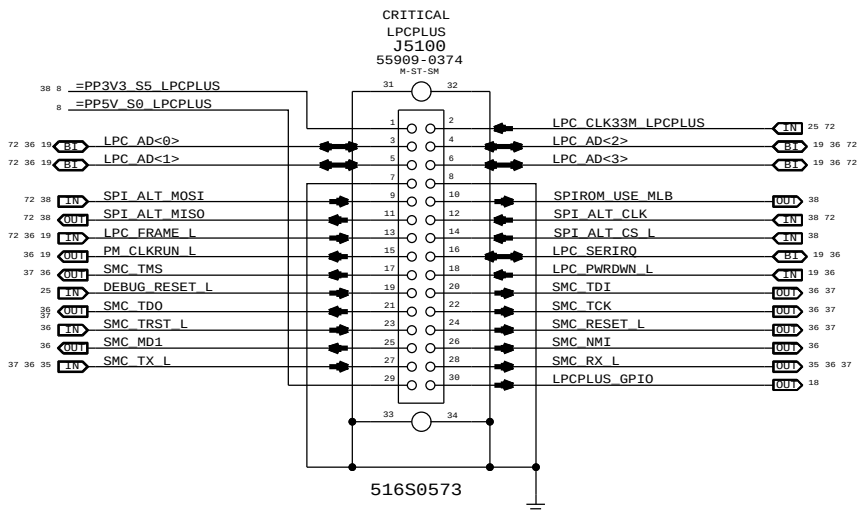
MCP_SAFE_MODE SIGNAL TO SUPPORT ROM FAILURE OVERRIDE



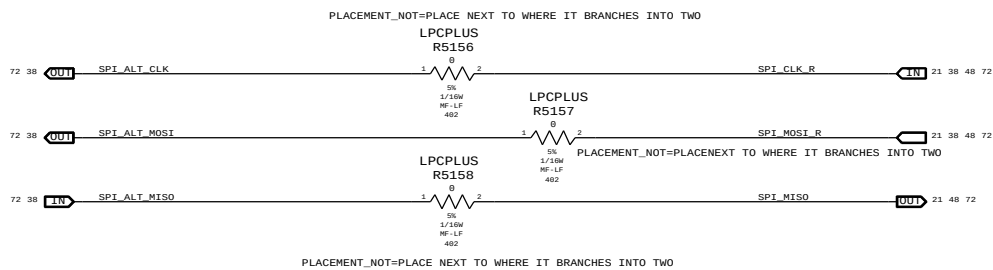
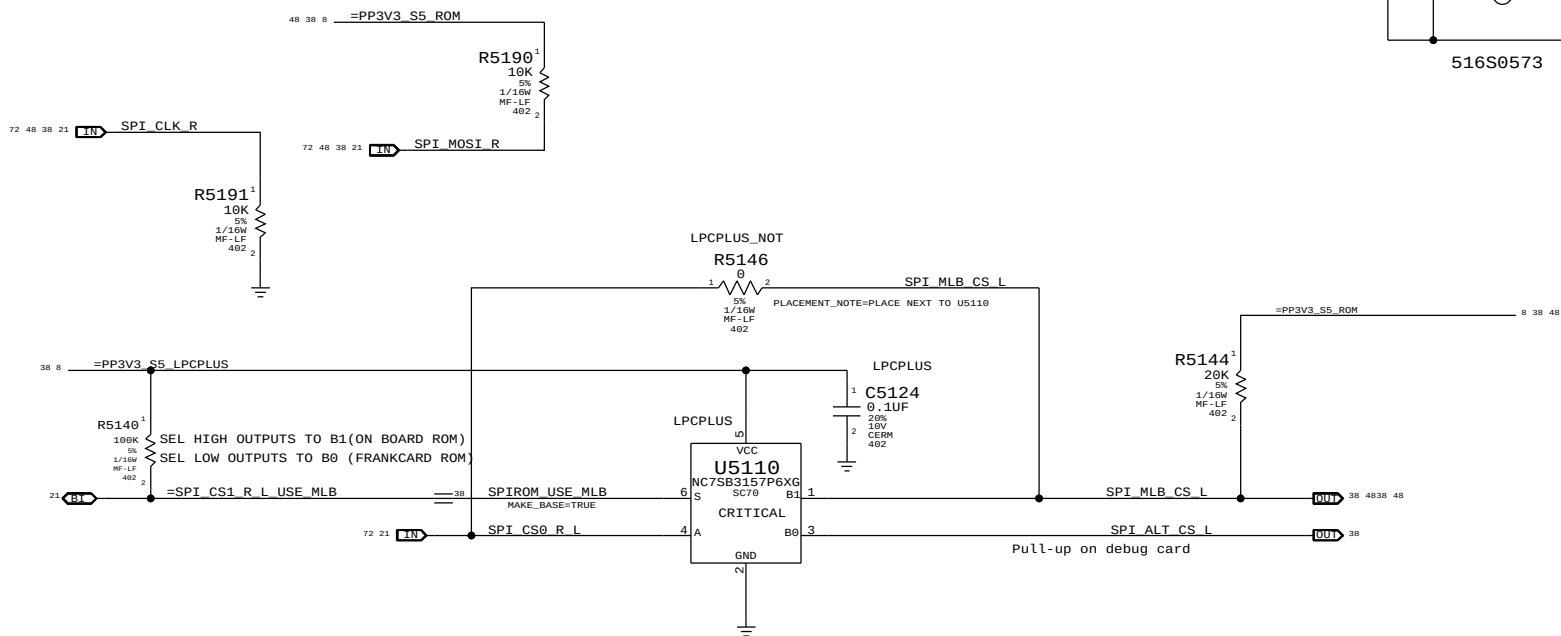
SYNC MASTER=K24 MLB	SYNC DATE=02/04/2009
PAGE TITLE	
SMC Support	
Apple Inc.	DRAWING NUMBER 051-8568
REVISION B.0.0	SIZE D
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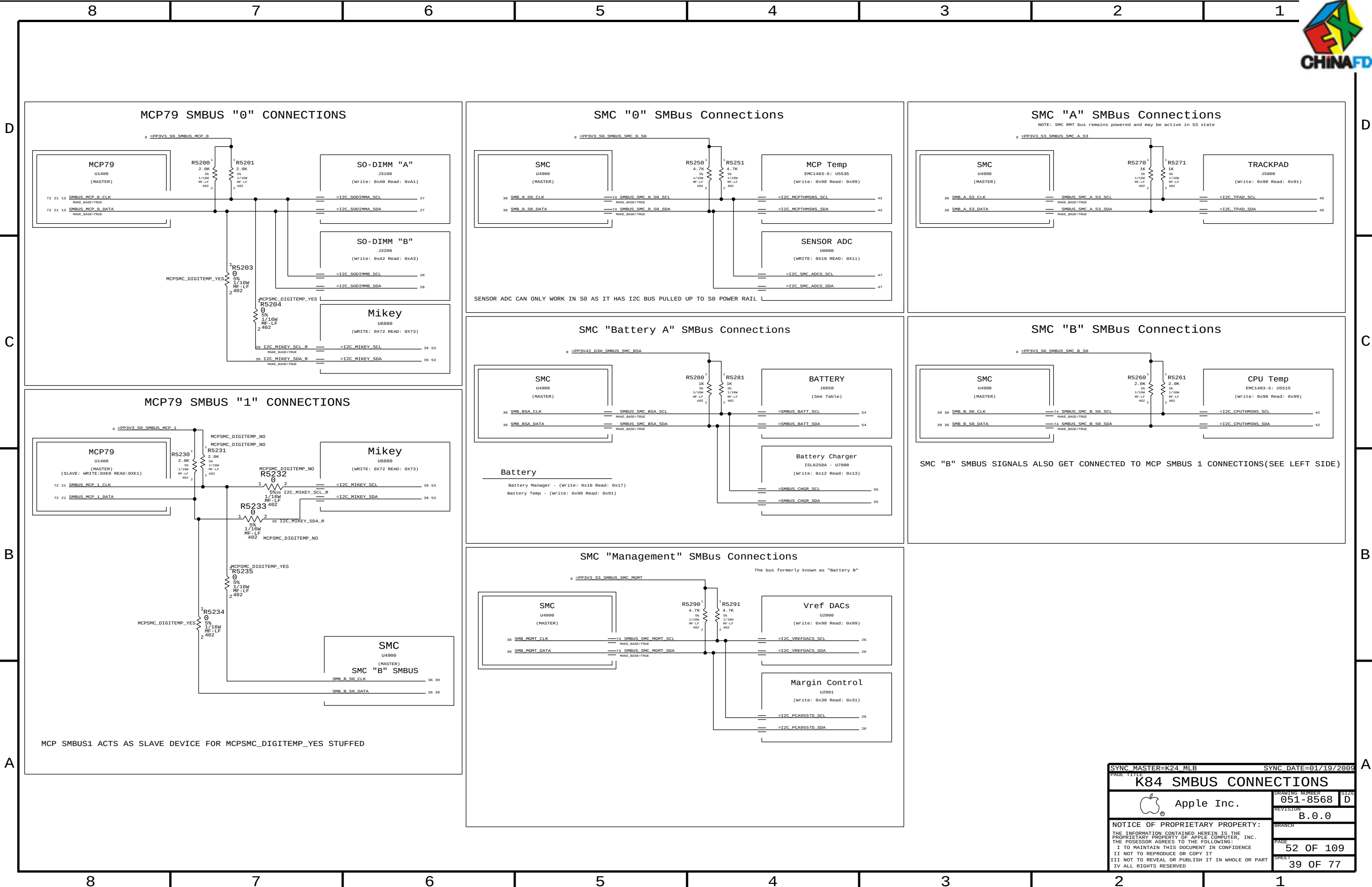
LPC+SPI Connector

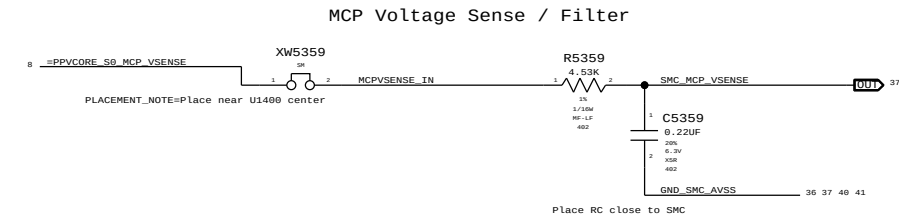
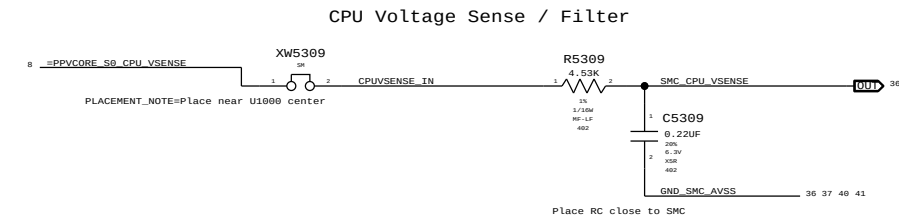


Alternate SPI ROM Support

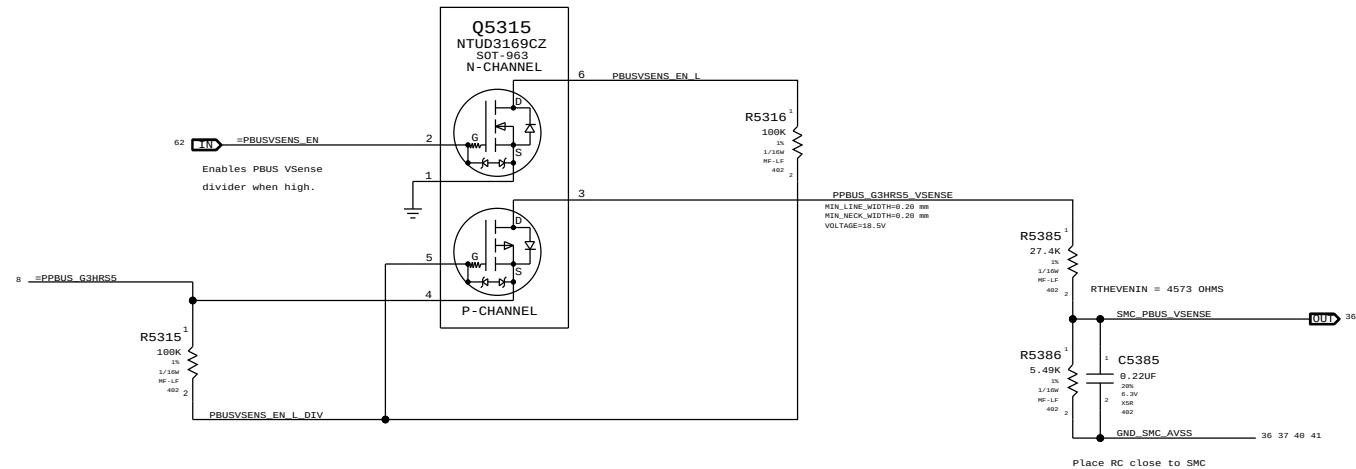


SYNC MASTER=K24_MLB		SYNC DATE=02/15/2009	
PAGE TITLE			
LPC+SPI Debug Connector			
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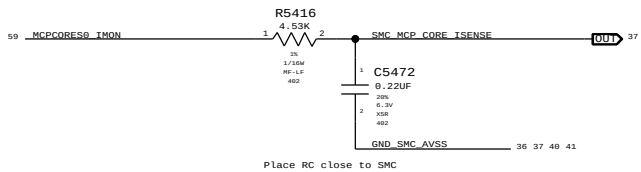
PBUS VOLTAGE SENSE ENABLE & FILTER



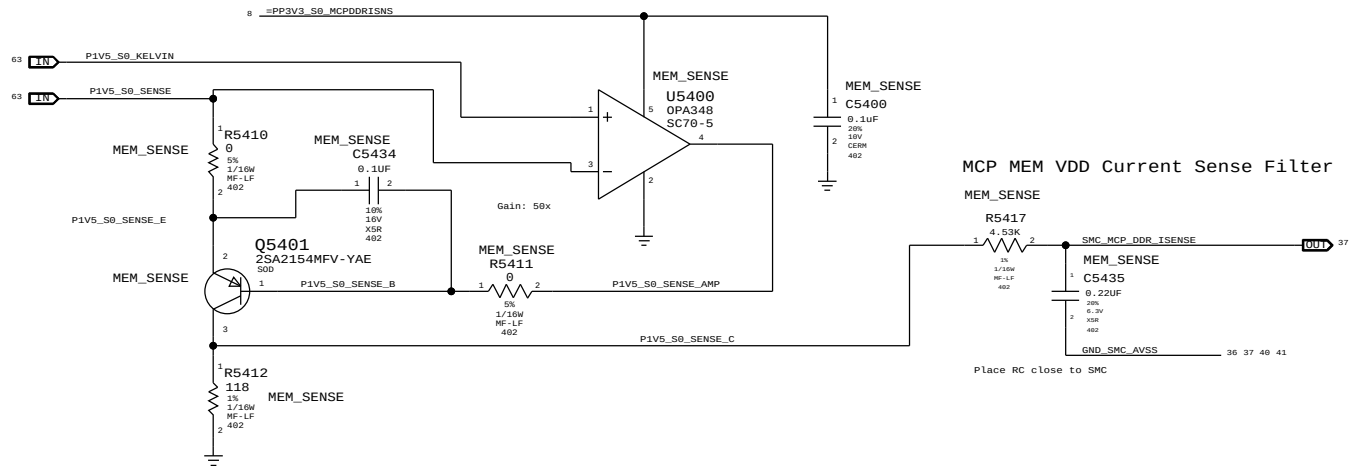
SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE		VOLTAGE SENSING	
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		PAGE	53 OF 109
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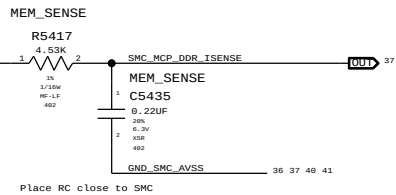
MCP VCore Current Sense Filter



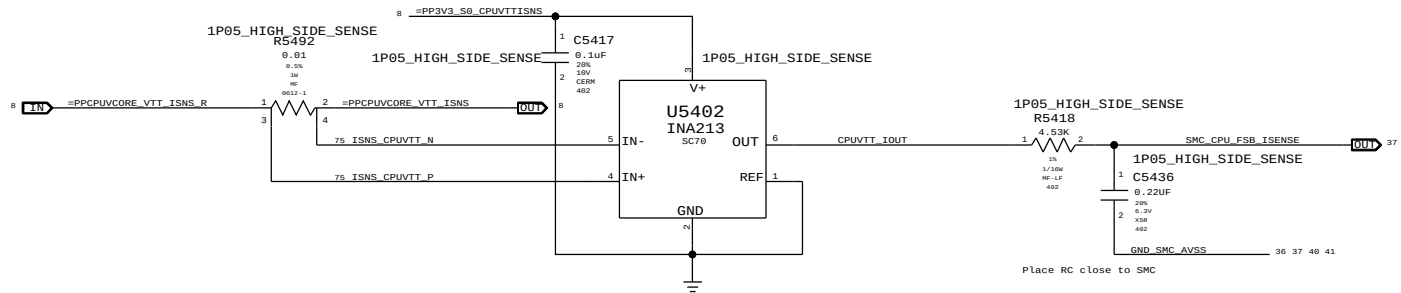
MCP MEM VDD Current Sense



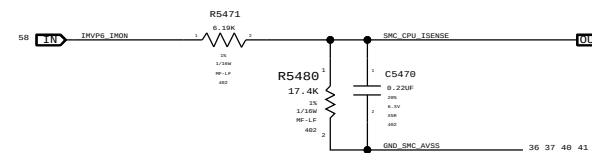
MCP MEM VDD Current Sense Filter



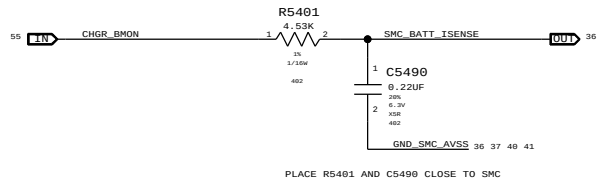
CPU 1.05V AND CPU VCore HIGH SIDE CURRENT SENSE



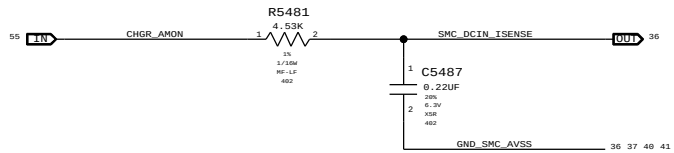
CPU VCore Load Side Current Sense / Filter



DC-IN (BMON) CURRENT SENSE



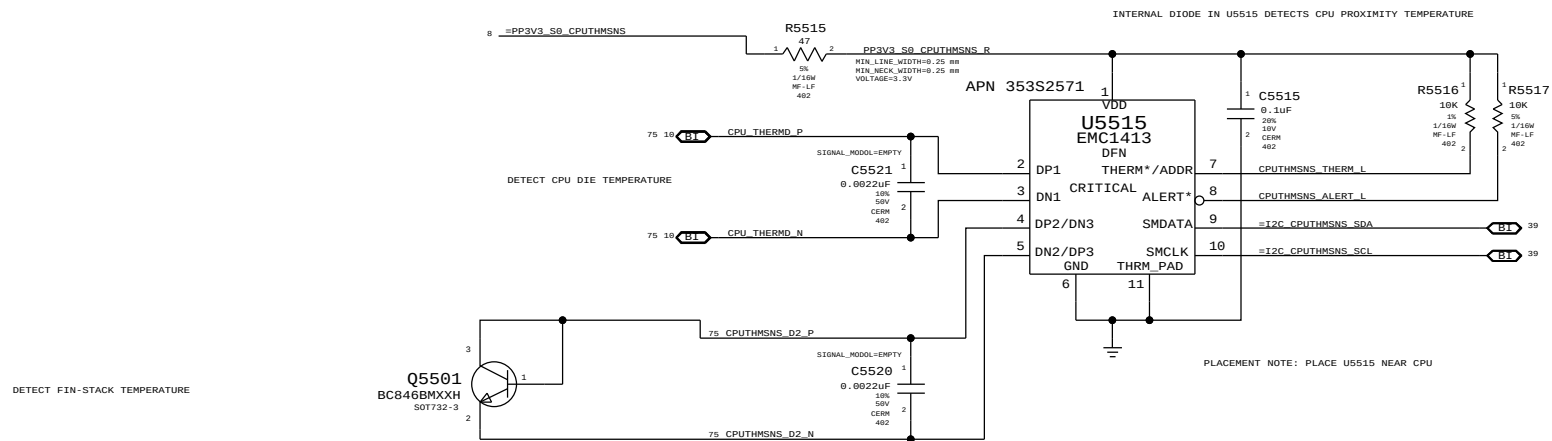
DC-IN (AMON) CURRENT SENSE



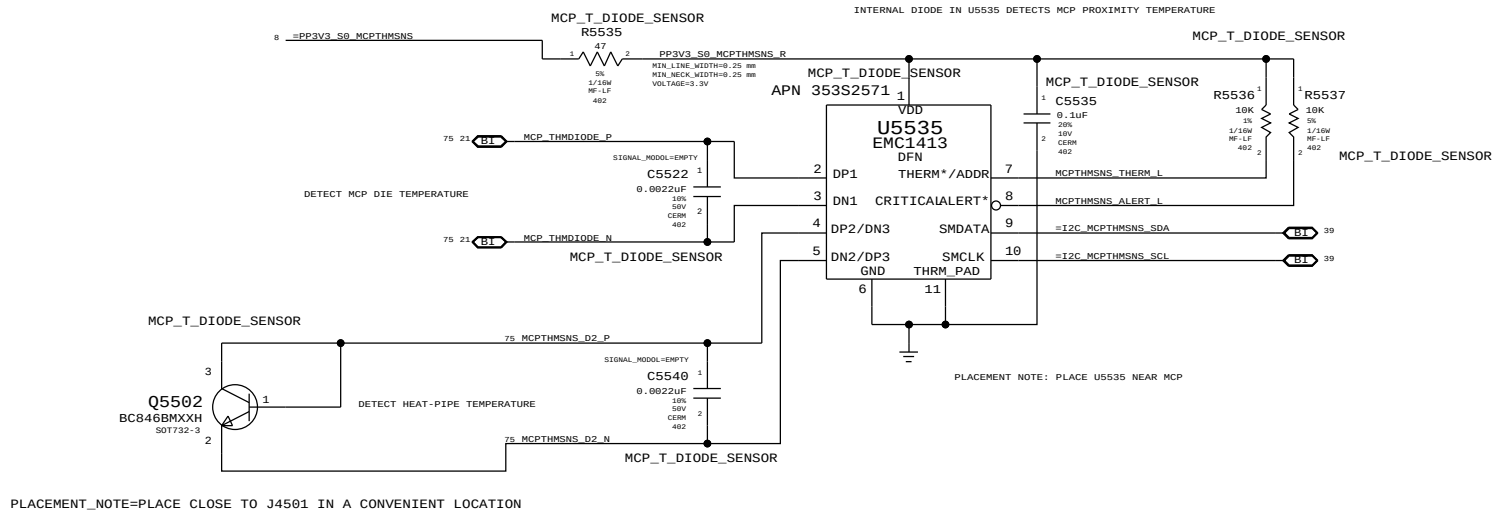
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PAGE TITLE		Current Sensing	
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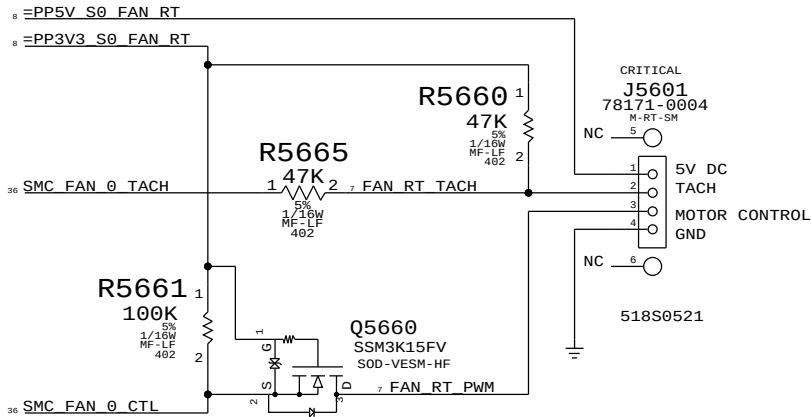
CPU T-Diode Thermal Sensor



MCP T-Diode Thermal Sensor



SYNC MASTER=K24 MLB		SYNC DATE=02/04/2009	
PAGE TITLE		Thermal Sensors	
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SYNC_MASTER=K24_MLB		SYNC_DATE=04/06/2009	
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		REVISION	B.0.0
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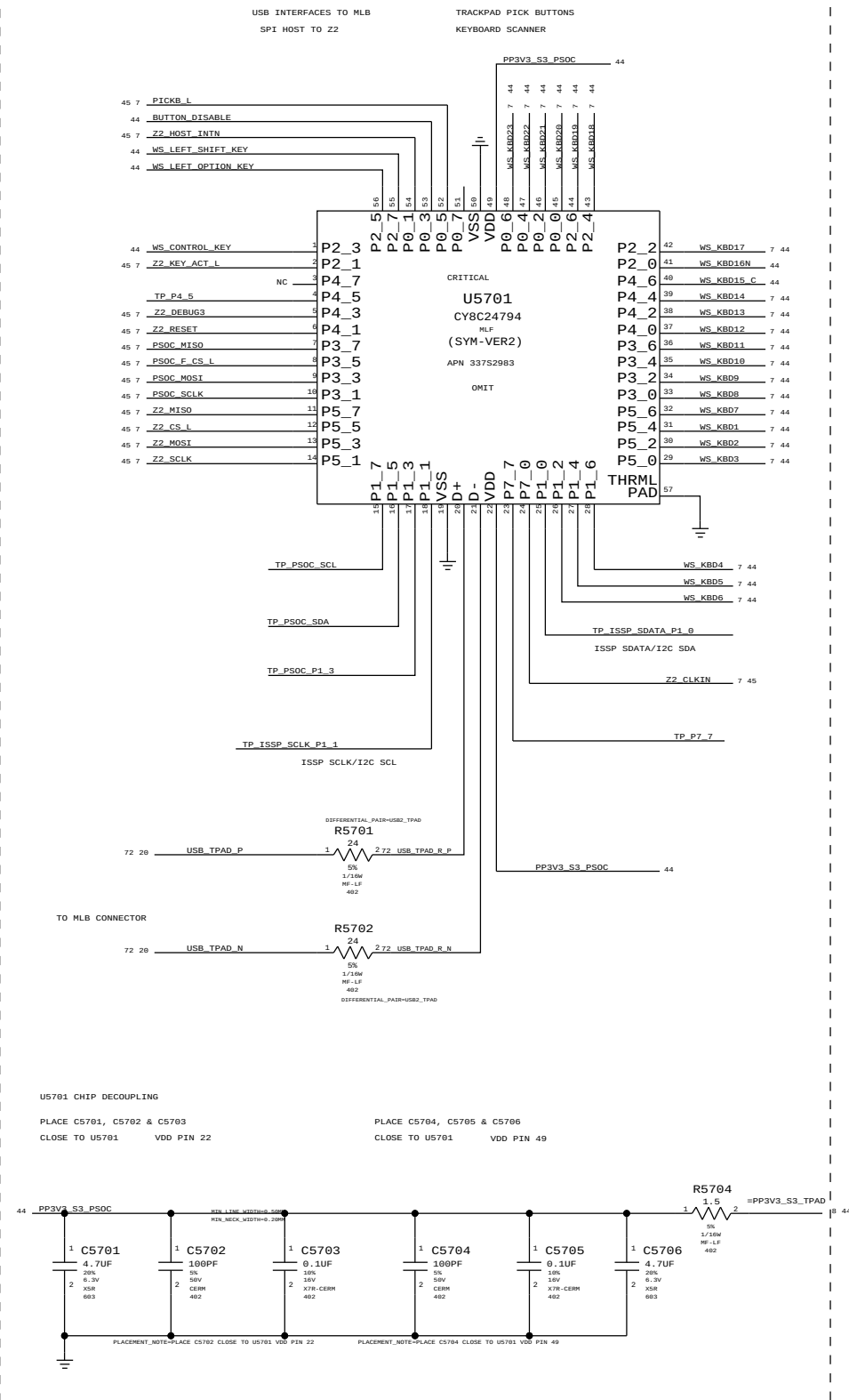
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3

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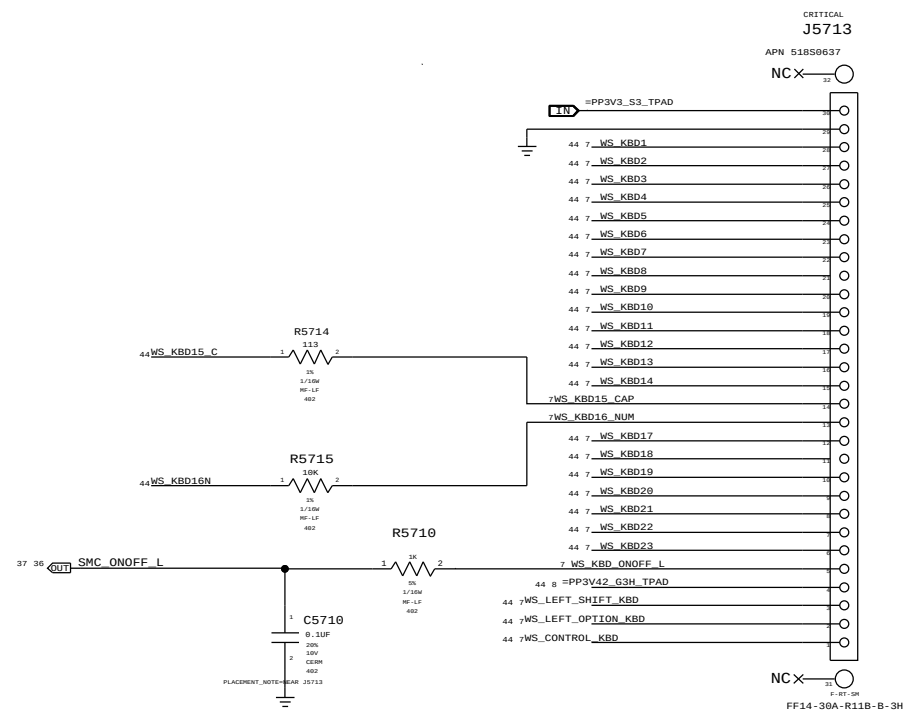
1

PSOC USB CONTROLLER

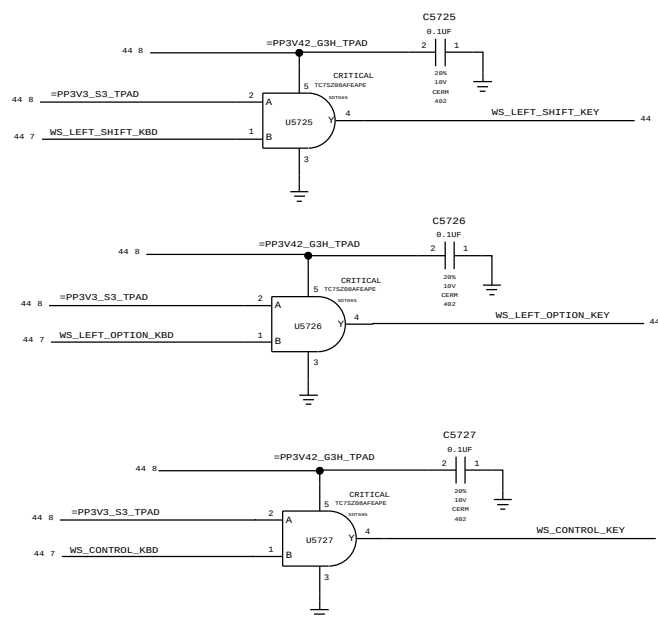


IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TP182	V+	180A	2.55 OHM	0.8255 V	0.255E-6 W
3V3 LDO	VDD	880A	10 OHM	0.284 V	16.32E-6 W
PSOC	VOUT	680A MAX	0.2 OHM	0.812 V	0.72E-6 W
	VDD	880A (TYP)	1.5 OHM	0.812 V	90E-6 W
		140A (MAX)		0.821 V	294E-6 W
1.8V BOOSTER	VIN	40A (MAX)	4.7 OHM	0.9188 V	75.2E-6 W

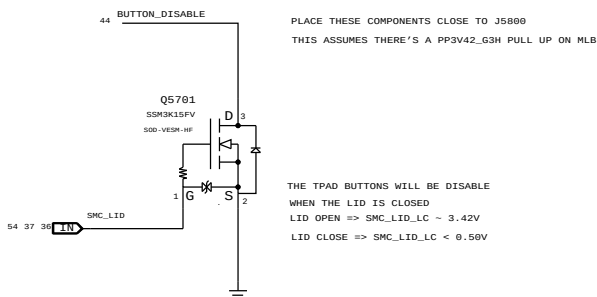
KEYBOARD CONNECTOR



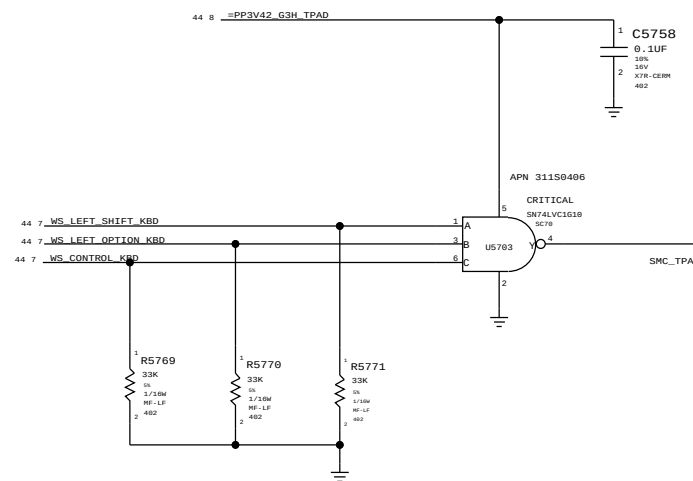
ISOLATION CIRCUIT



TPAD BUTTONS DISABLE



SMC_MANUAL_RESET LOGIC



Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
31158406	31158447		ALL	SWP PART AS ALTERNATE

SYNC MASTER=K24 MLB		SYNC DATE=03/04/2009	
PAGE TITLE			
WELLSPRING 1			
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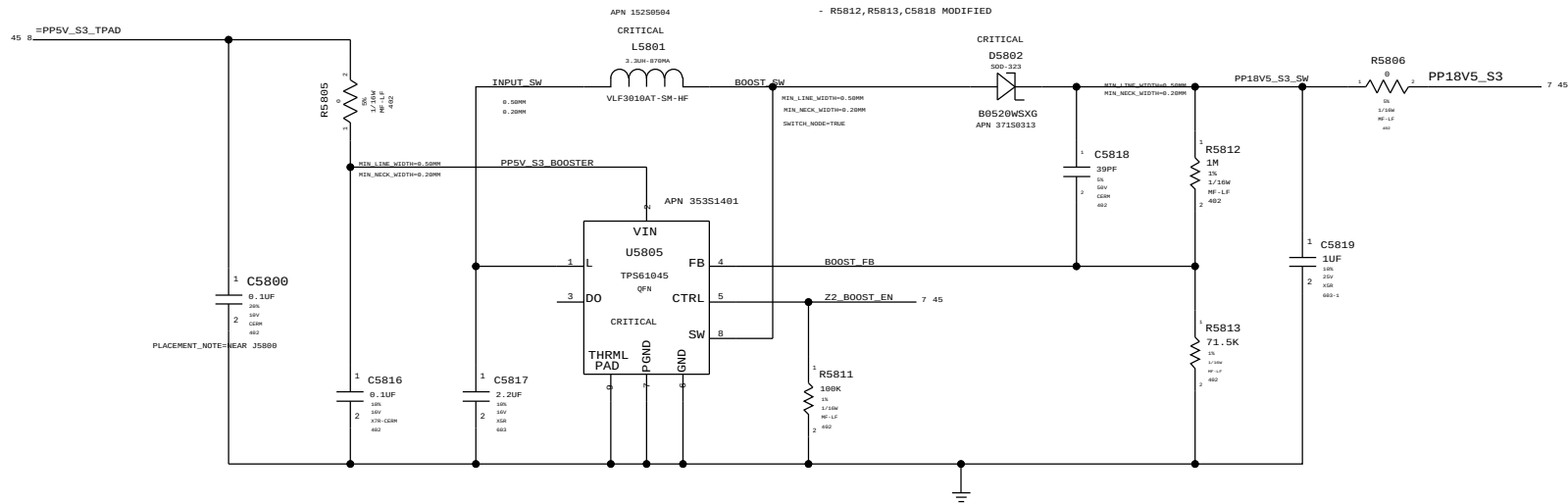
2

1

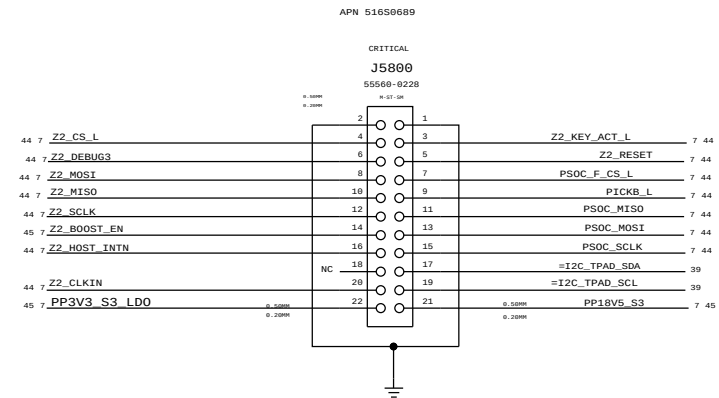


BOOSTER +18.5VDC FOR SENSORS

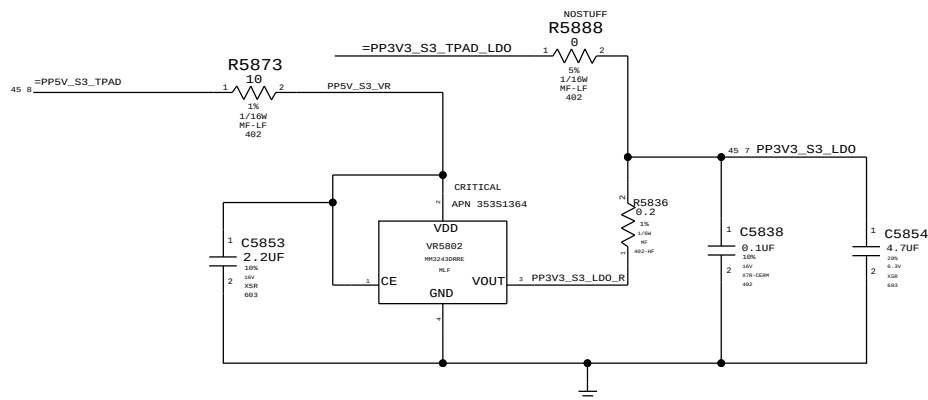
- BOOSTER DESIGN CONSIDERATION:
- POWER CONSUMPTION
 - DROOP LINE REGULATION
 - RIPPLE TO MEET ERS
 - 100-300 KHZ CLEAN SPECTRUM
 - STARTUP TIME LESS THAN 2MS
 - R5812, R5813, C5818 MODIFIED



IPD FLEX CONNECTOR



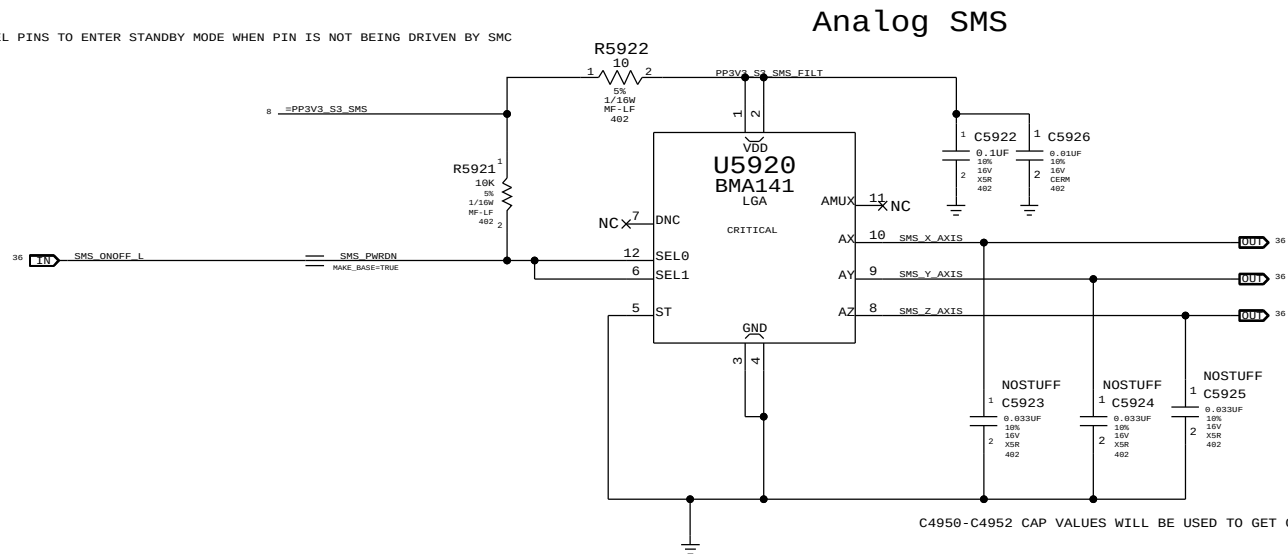
3V3 LDO FOR IPD



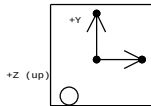
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Apple Inc.		DRAWING NUMBER	051-8568
		REVISION	B.0.0
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R5921 PULLS UP SEL PINS TO ENTER STANDBY MODE WHEN PIN IS NOT BEING DRIVEN BY SMC



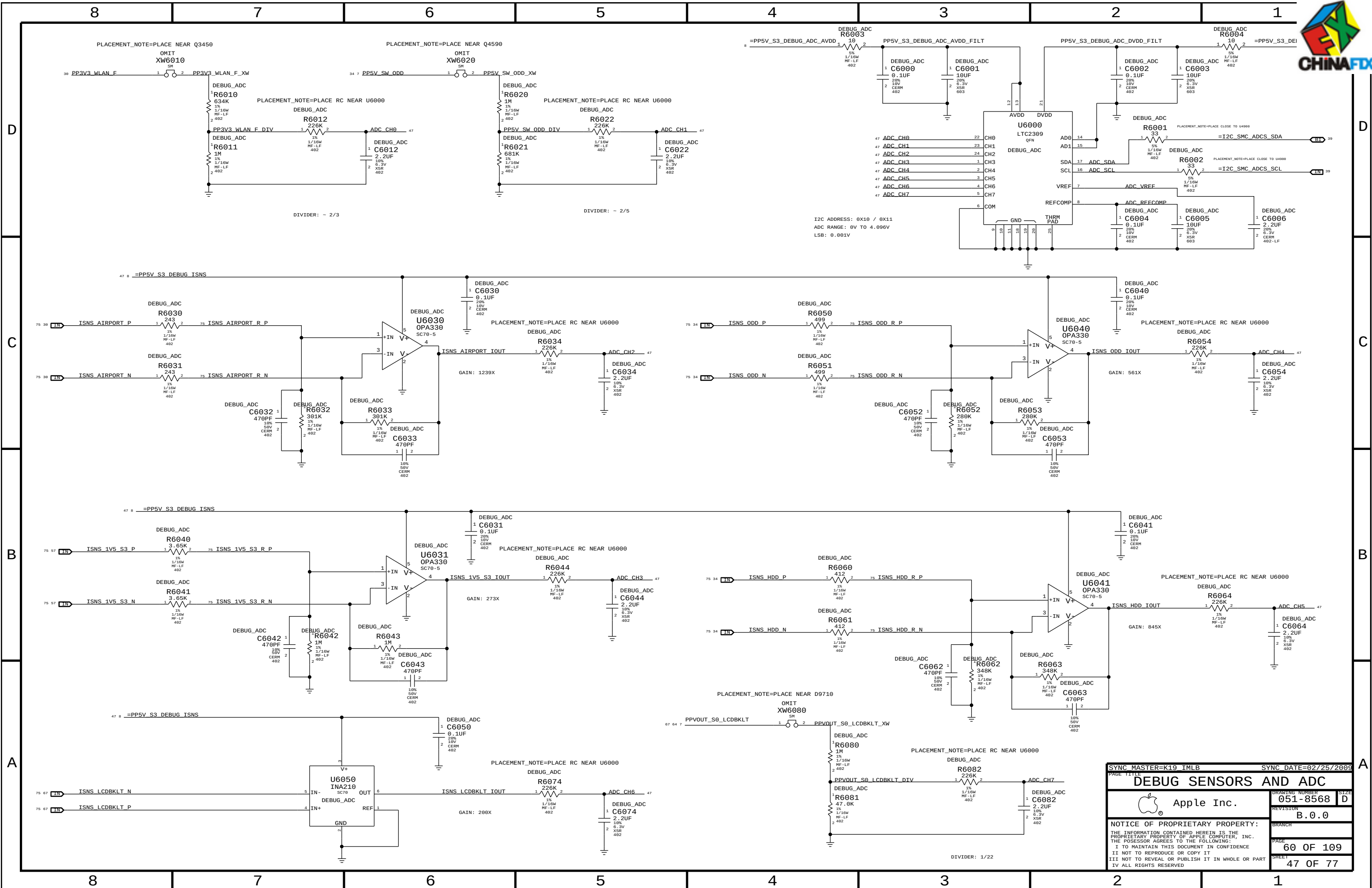
Desired orientation when placed on board top-side:



Front of system

Circle indicates pin 1 location when placed in correct orientation

PAGE TITLE		SMS	
	DRAWING NUMBER		051-8568
	REVISION		B.0.0
	BRANCH		
	PAGE		59 OF 109
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D

D

C

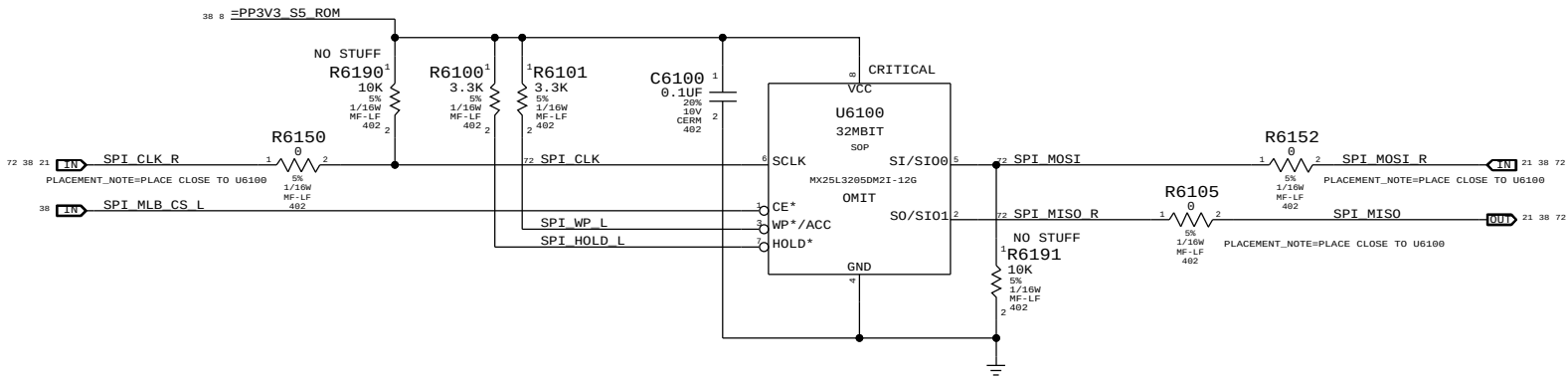
C

B

B

A

A



MCP79 SPI Frequency Select		
Frequency	SPI_MOSI	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

25MHz is selected with R5190 and R5191
Any of the 4 frequencies can be selected
with R6190, R6191, R5190 and R5191

SYNC MASTER=K24_MLB

SYNC DATE=02/15/2009

PAGE TITLE

SPI ROM

 Apple Inc.

DRAWING NUMBER
051-8568

SIZE
D

REVISION
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PAGE
61 OF 109

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48 OF 77

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8

7

6

5

4

3

2

1

8

7

6

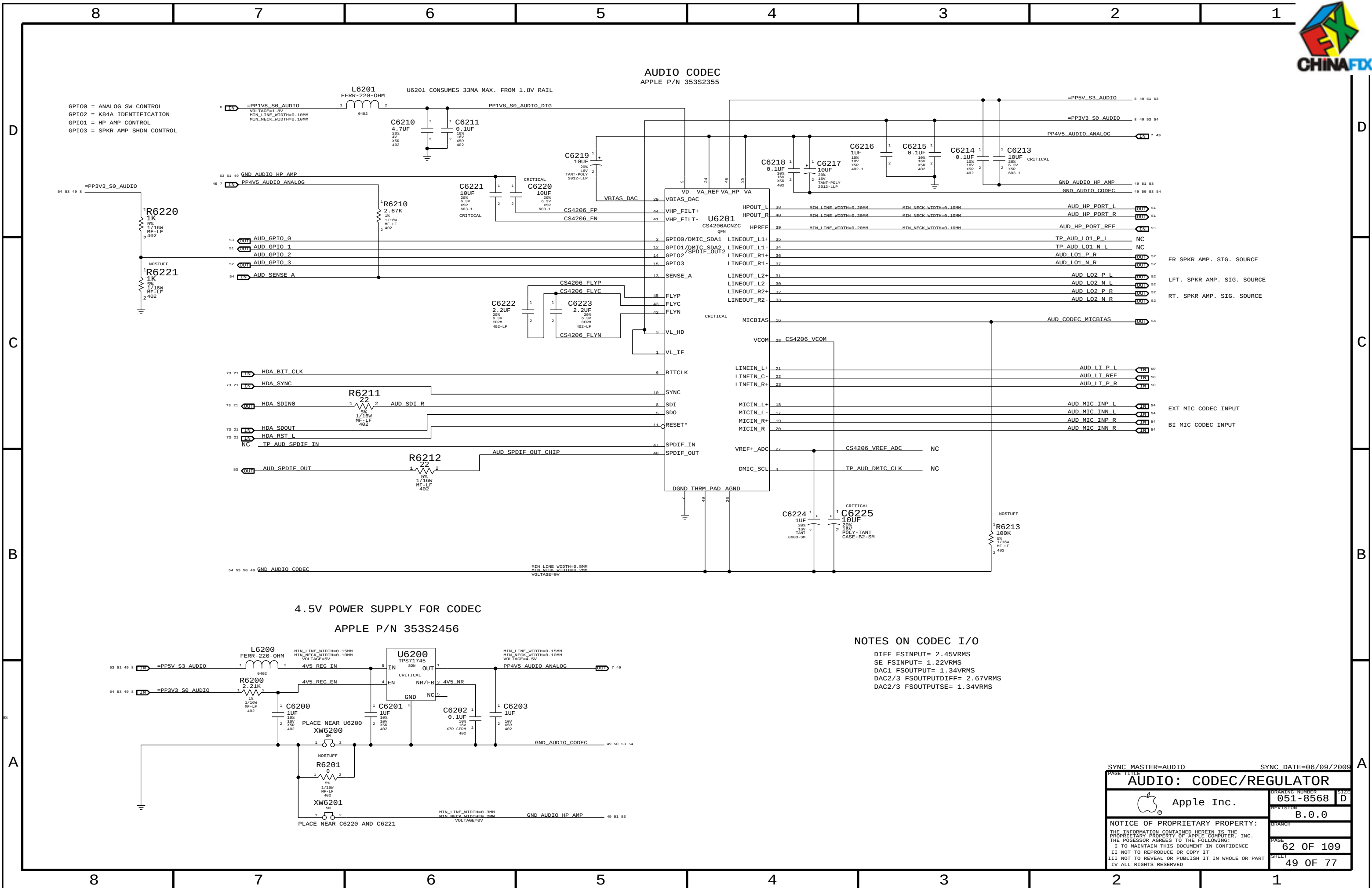
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2

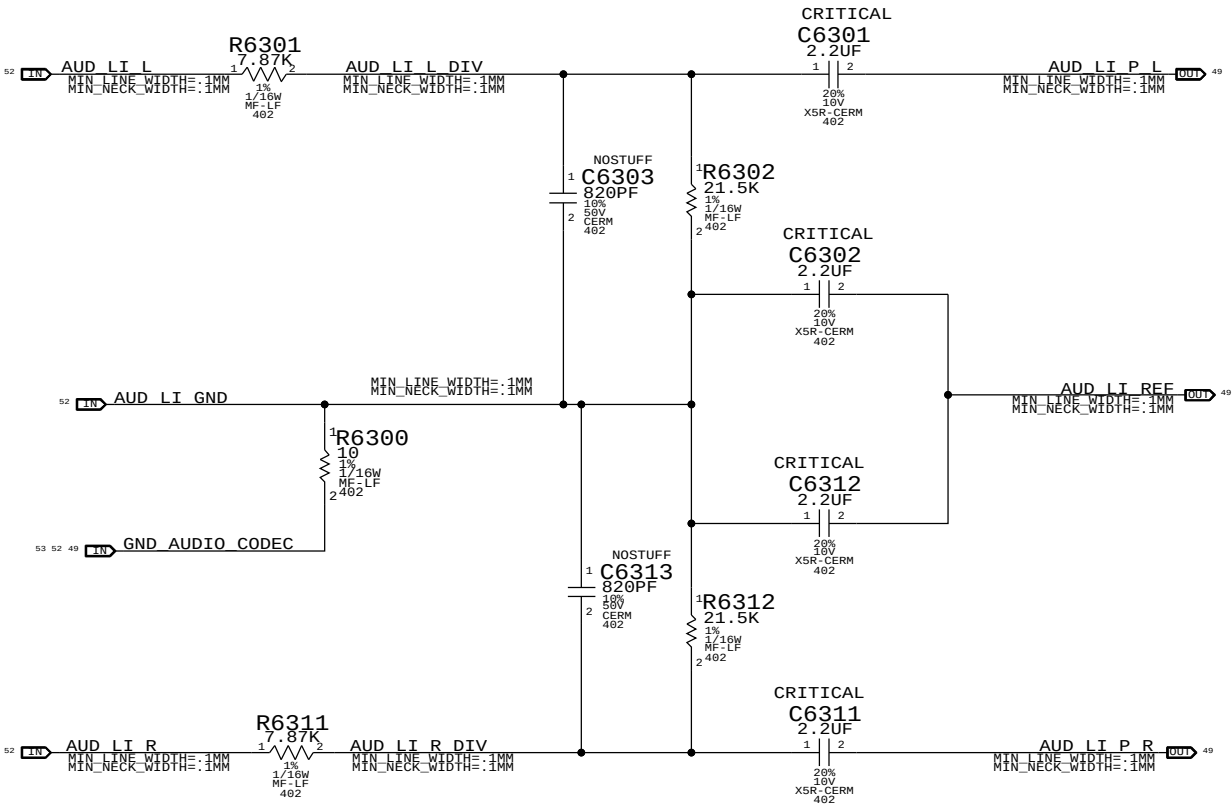
1





LINE INPUT VOLTAGE DIVIDER

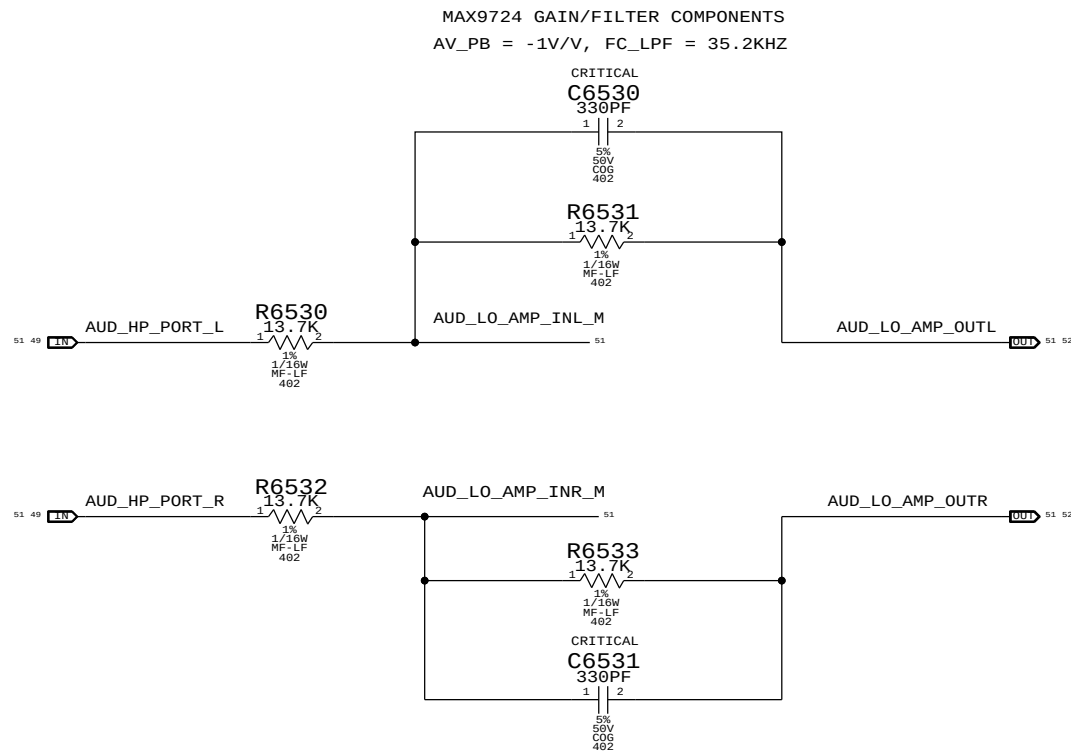
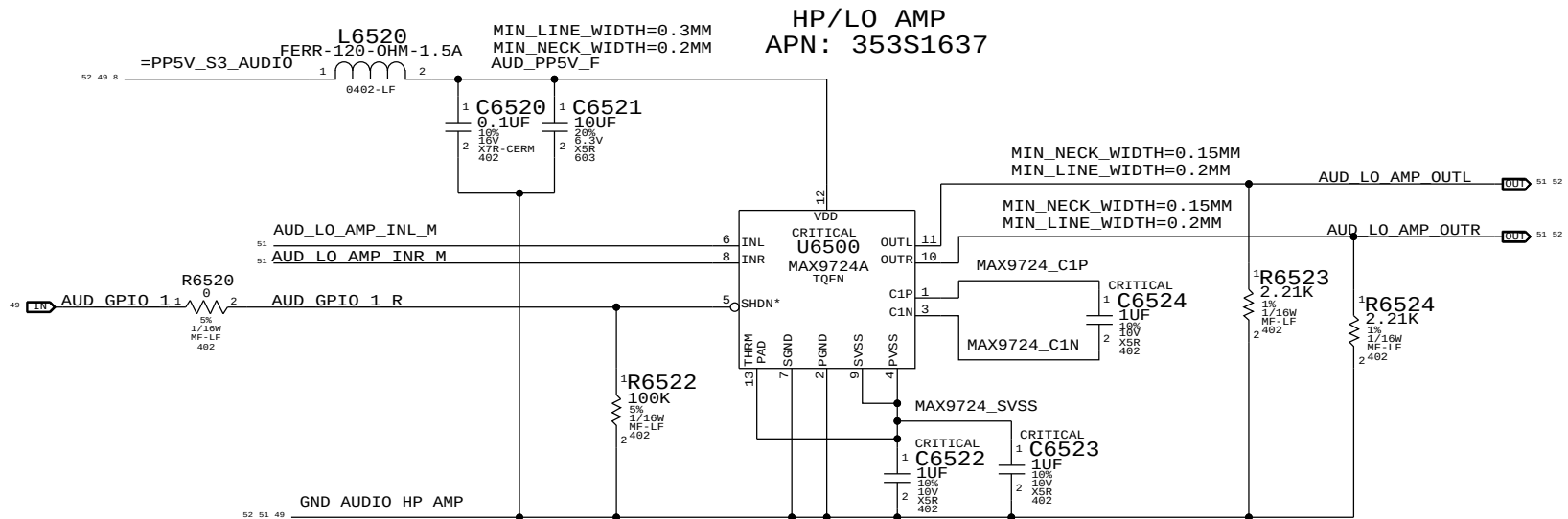
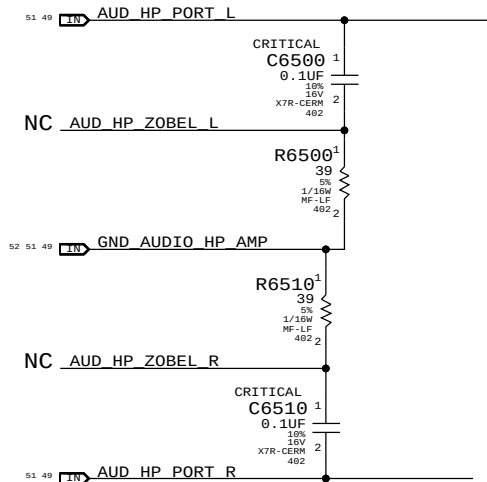
CODEC RIN = 20K OHMS
NET RIN = 10.36K OHMS (INCLUDING PULL-DOWNS AT ANALOG SWITCH COM PINS)
FC_HP = 3.6 HZ
FC_LP = 43KHZ
VIN = 2VRMS, CODEC VIN = 1.14 VRMS



PAGE TITLE		
AUDIO: LINE INPUT FILTER		
 Apple Inc.	DRAWING NUMBER	051-8568
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ZOBEL NETWORK & 1ST ORDER DAC FILTER PLACEHOLDER



SYNC MASTER=AUDIO		SYNC DATE=06/09/2009	
PAGE TITLE		AUDIO: HEADPHONE FILTER	
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		PAGE	65 OF 109
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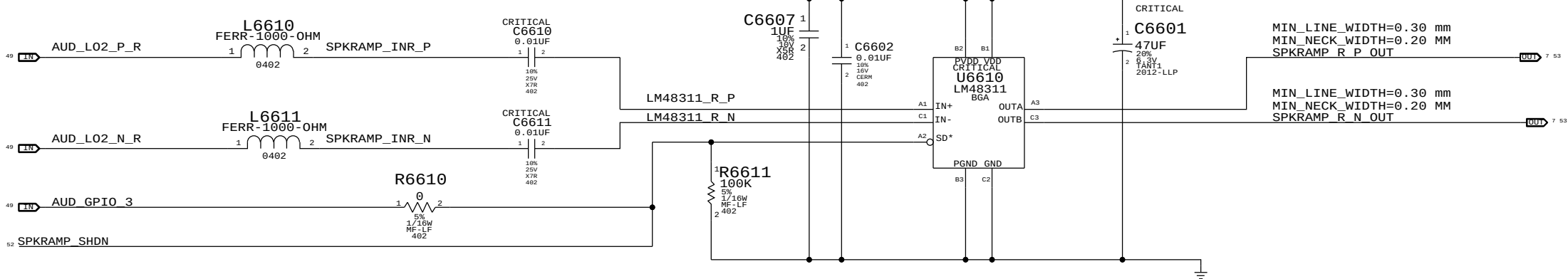


SATELLITE 796Hz < HPF FC < 936Hz
SUB 80 Hz < HPF FC < 94 Hz
GAIN 6DB (2V/V)
SPRK AMP. INPUT REFERRED CLIP POINT = ~-6dBFS

ALIAS OF PP5V_S3_REG, MIN_LINE_WIDTH=0.60MM, MIN_NECK_WIDTH=0.20MM

52 8 =PP5V_S3_AUDIO_AMP

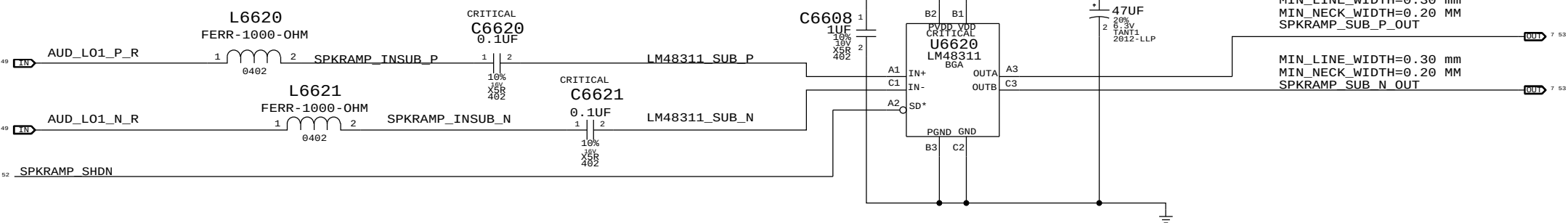
APN: 353S2621



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52 8 =PP5V_S3_AUDIO_AMP

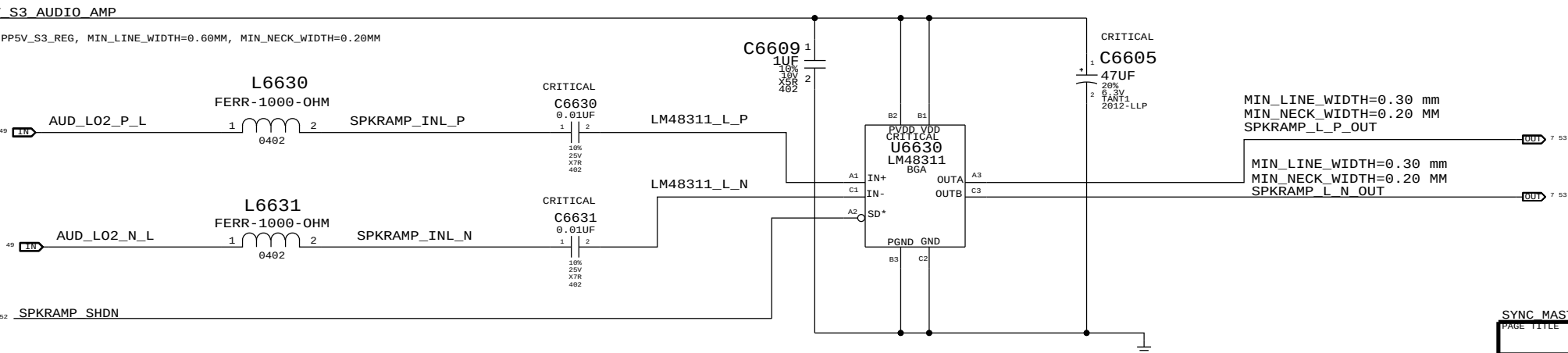
APN: 353S2621



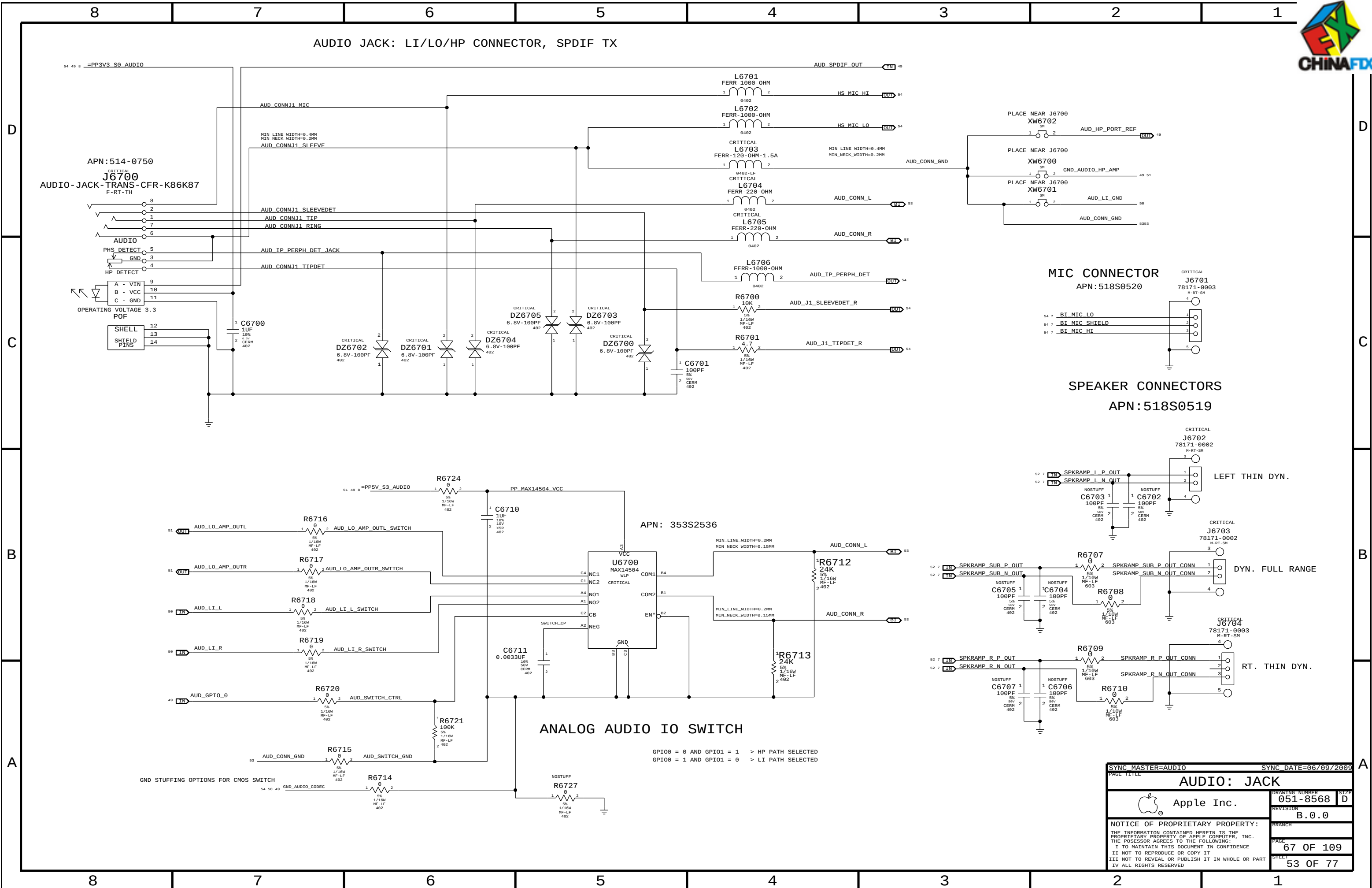
52 8 =PP5V_S3_AUDIO_AMP

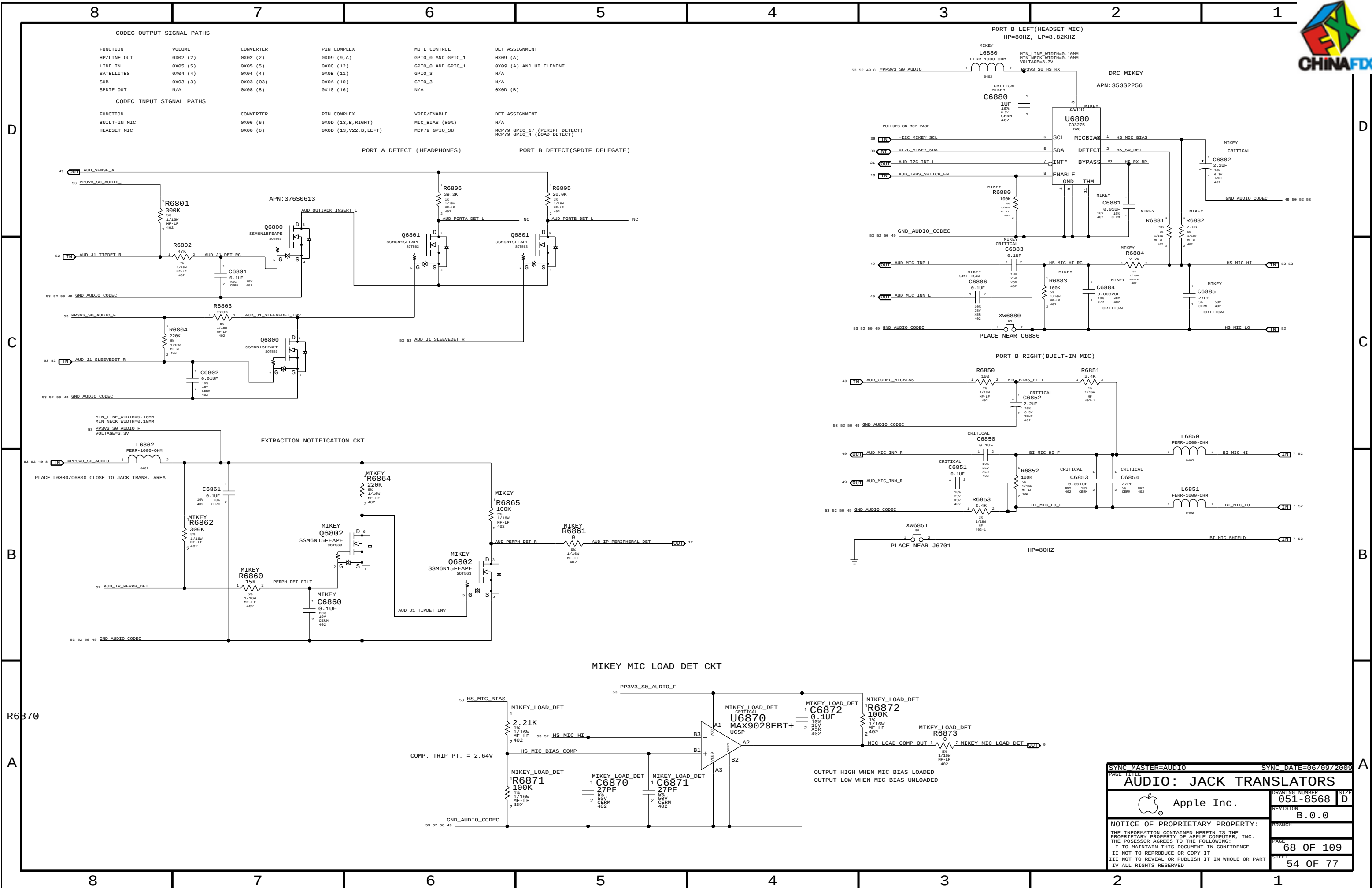
ALIAS OF PP5V_S3_REG, MIN_LINE_WIDTH=0.60MM, MIN_NECK_WIDTH=0.20MM

APN: 353S2621



SYNC MASTER=AUDIO		SYNC DATE=02/16/2016	
PAGE TITLE			
AUDIO0: SPEAKER AMP			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-8568	D
		REVISION	
		B.0.0	
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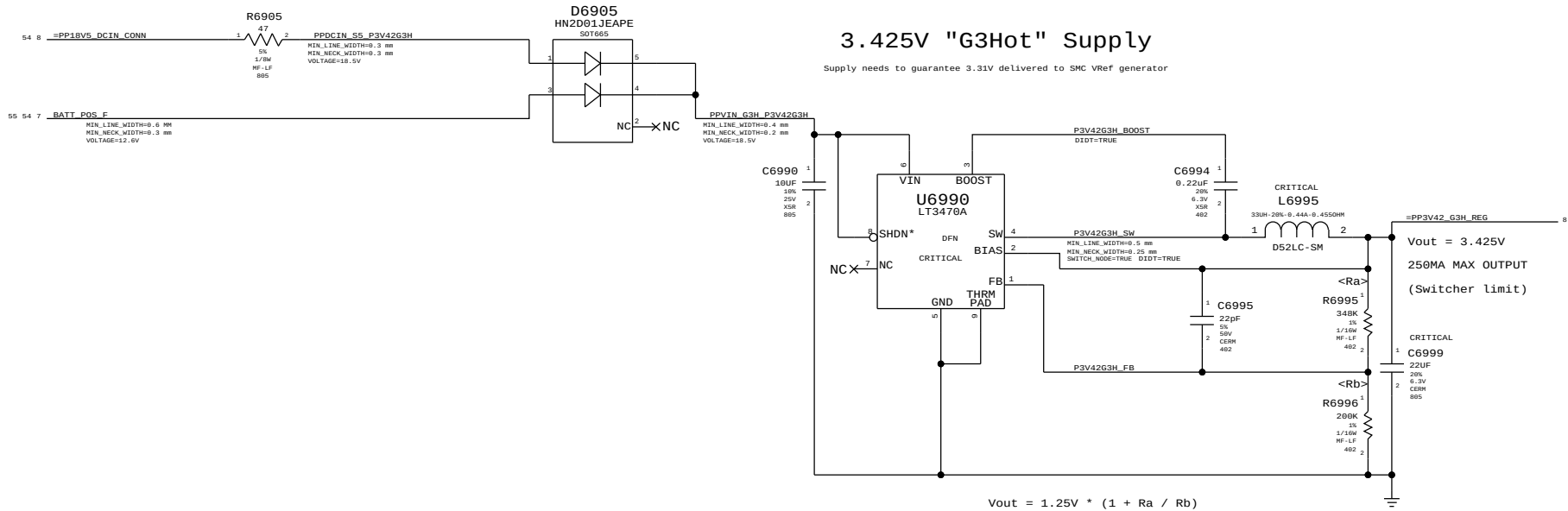
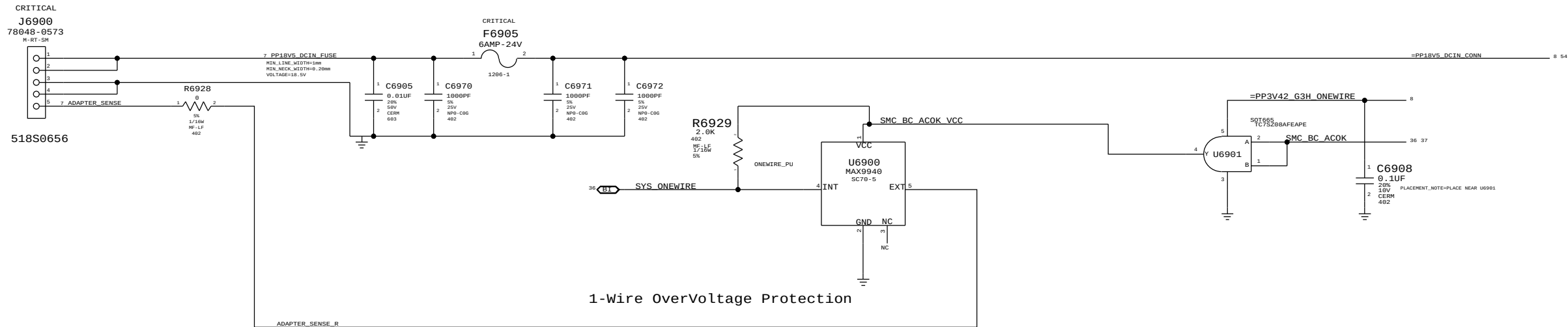




SYNC MASTER=AUDIO		SYNC DATE=06/09/2009	
PAGE TITLE		DRAWING NUMBER	
AUDIO: JACK TRANSLATORS		051-8568	
Apple Inc.		REVISION	
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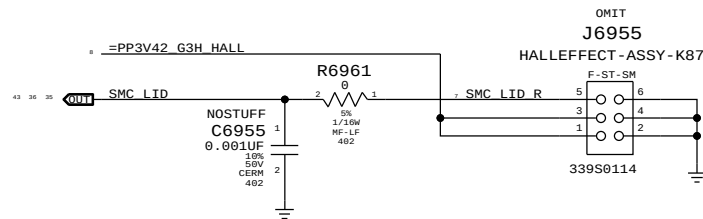
MagSafe DC Power Jack



HALL EFFECT ASSEMBLY

Assembly APN: 339S0114

- BOM: 639-0680
- PCBF: 820-2801
- MCO: 056-3515
- Conn APN:518S0788



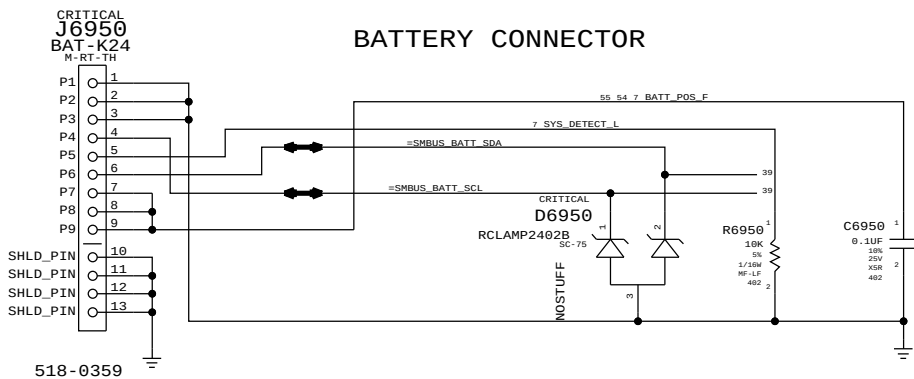
PROTO 0: STUFFING K84 CONNECTOR ONTO MODIFIED K84 PADS

PROTO 1: STUFFING K87 HALL EFFECT ASSEMBLY ONTO K87 PADS

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
607-6831	1	SUB ASSY - HALL EFFECT, K88 K87	J6955	CRITICAL	

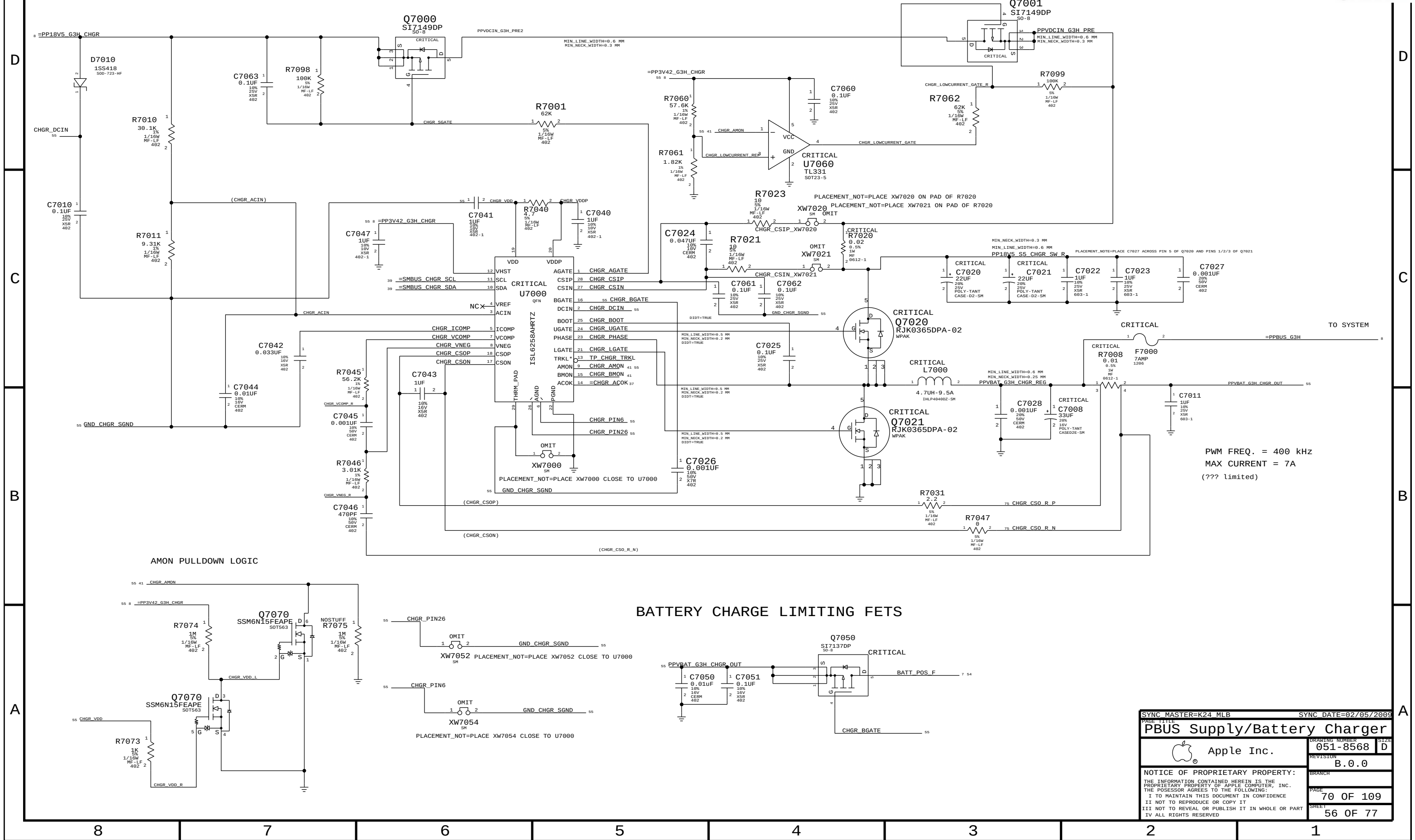
PN: 607-6831 for WCPM. PN: 339S0114 for schematic/board layout

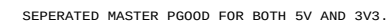
BATTERY CONNECTOR



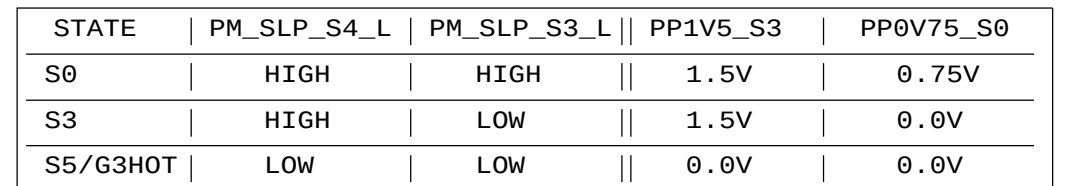
SYNC MASTER=K24 MLB		SYNC DATE=02/05/2009	
PAGE TITLE		DC-In & Battery Connectors	
Apple Inc.		DRAWING NUMBER	051-8568
		REVISION	B.0.0
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PBUS SUPPLY / BATTERY CHARGER

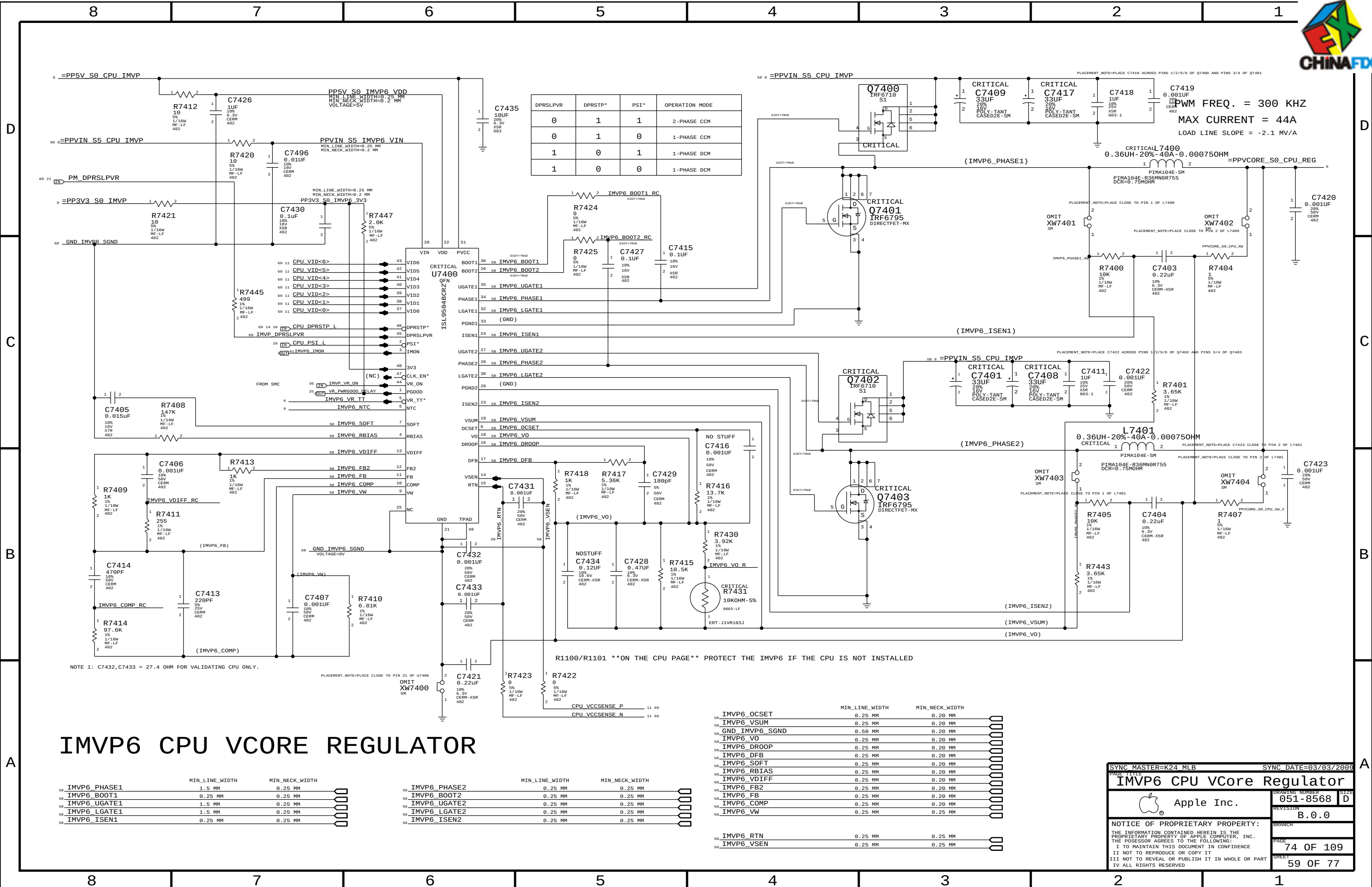


$$V_{OUT} = (2 * RC / RD) + 2$$


PAGE TITLE		5V/3.3V SUPPLY	
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	SIZE	D	
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PAGE TITLE			
1.5V/0.75V DDR3 SUPPLY			
	Apple Inc.		DRAWING NUMBER 051-8568
			SIZE D
		REVISION B.0.0	
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		PAGE 73 OF 109	
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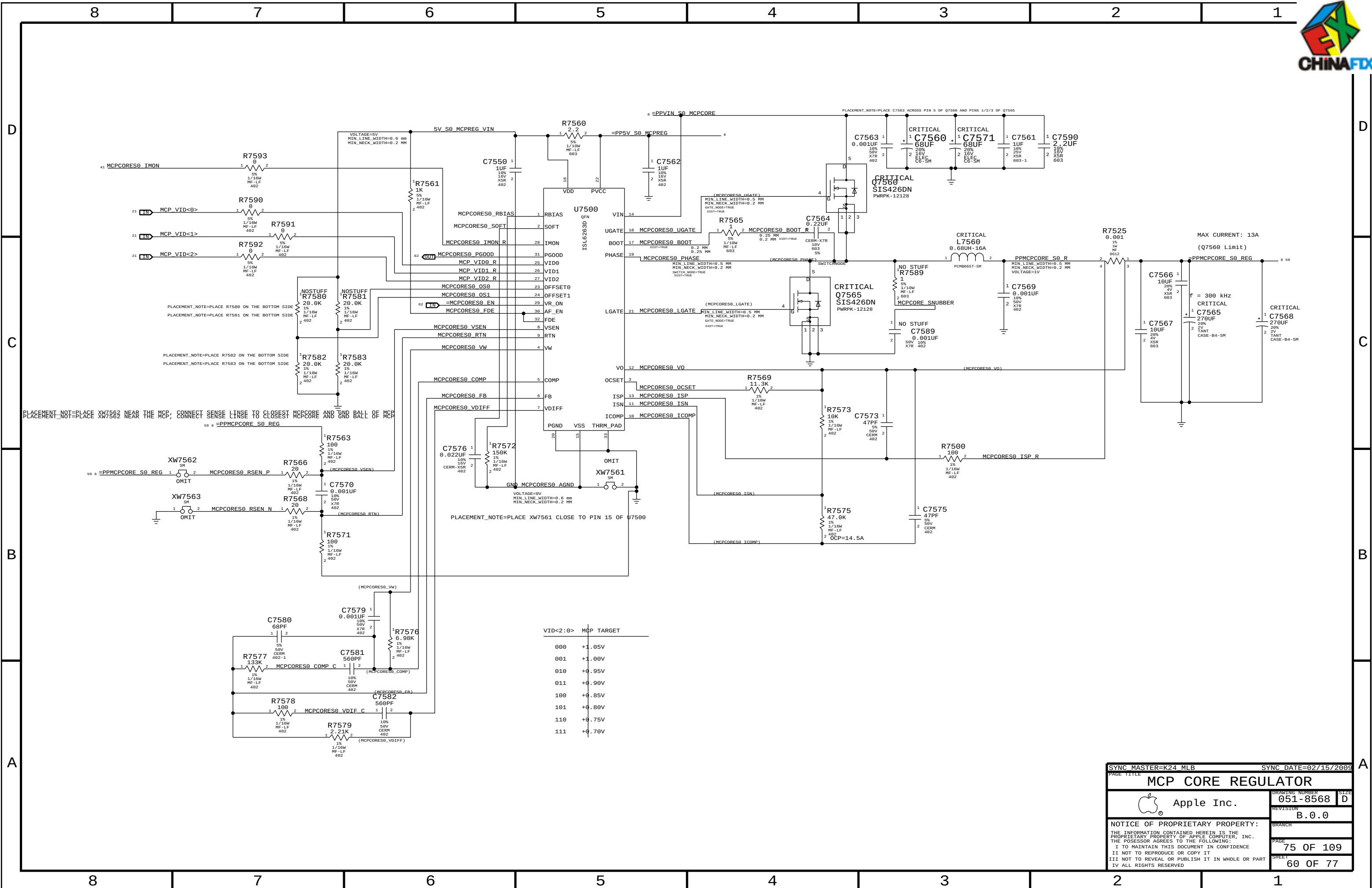


	MIN_LINE_WIDTH	MIN_NECK_WIDTH	
IMVP6_PHASE1	1.5 MM	0.25 MM	
IMVP6_BOOT1	0.25 MM	0.25 MM	
IMVP6_UGATE1	1.5 MM	0.25 MM	
IMVP6_LGATE1	1.5 MM	0.25 MM	
IMVP6_ISEN1	0.25 MM	0.25 MM	

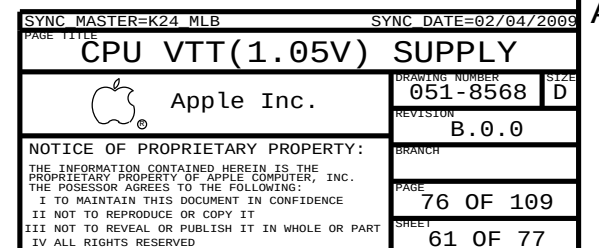
	MIN_LINE_WIDTH	MIN_NECK_WIDTH
58 IMVP6_PHASE2	0.25 MM	0.25 MM
58 IMVP6_BOOT2	0.25 MM	0.25 MM
58 IMVP6_UGATE2	0.25 MM	0.25 MM
58 IMVP6_LGATE2	0.25 MM	0.25 MM
58 IMVP6_ISEN2	0.25 MM	0.25 MM

	MIN_LINE_WIDTH	MIN_NECK_WIDTH	
59 IMPV6_OCSET	0.25 MM	0.20 MM	
59 IMPV6_VSUM	0.25 MM	0.20 MM	
59 GND_IMPV6_SGND	0.50 MM	0.20 MM	
59 IMPV6_VO	0.25 MM	0.20 MM	
59 IMPV6_DR00P	0.25 MM	0.20 MM	
59 IMPV6_DFB	0.25 MM	0.20 MM	
59 IMPV6_SOFT	0.25 MM	0.20 MM	
59 IMPV6_RBIAS	0.25 MM	0.20 MM	
59 IMPV6_VDIFF	0.25 MM	0.20 MM	
59 IMPV6_FB2	0.25 MM	0.20 MM	
59 IMPV6_FB	0.25 MM	0.20 MM	
59 IMPV6_COMP	0.25 MM	0.20 MM	
59 IMPV6_VW	0.25 MM	0.25 MM	
59 IMPV6_RTN	0.25 MM	0.25 MM	
59 IMPV6_VSEN	0.25 MM	0.25 MM	

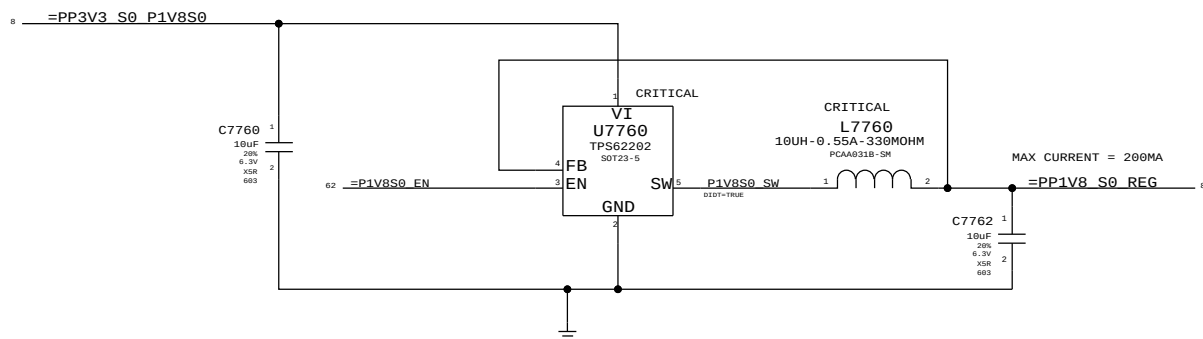
SYNC MASTER=K24 MLB		SYNC DATE=03/03/2009	
PAGE TITLE			
IMVP6 CPU VCore Regulator			
 Apple Inc.		DRAWING NUMBER 051-8568	
		SIZE D	
		REVISION B.0.0	
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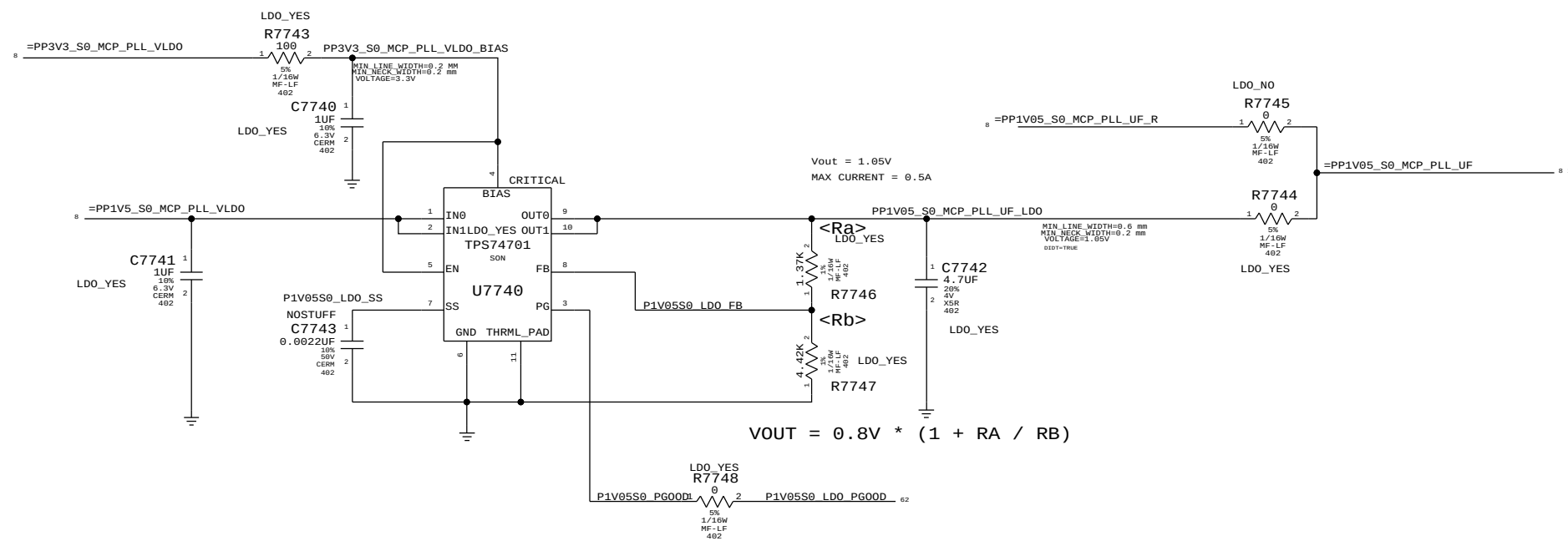
VID<2:0>	MCP TARGET
000	+1.05V
001	+1.00V
010	+0.95V
011	+0.90V
100	+0.85V
101	+0.80V
110	+0.75V
111	+0.70V



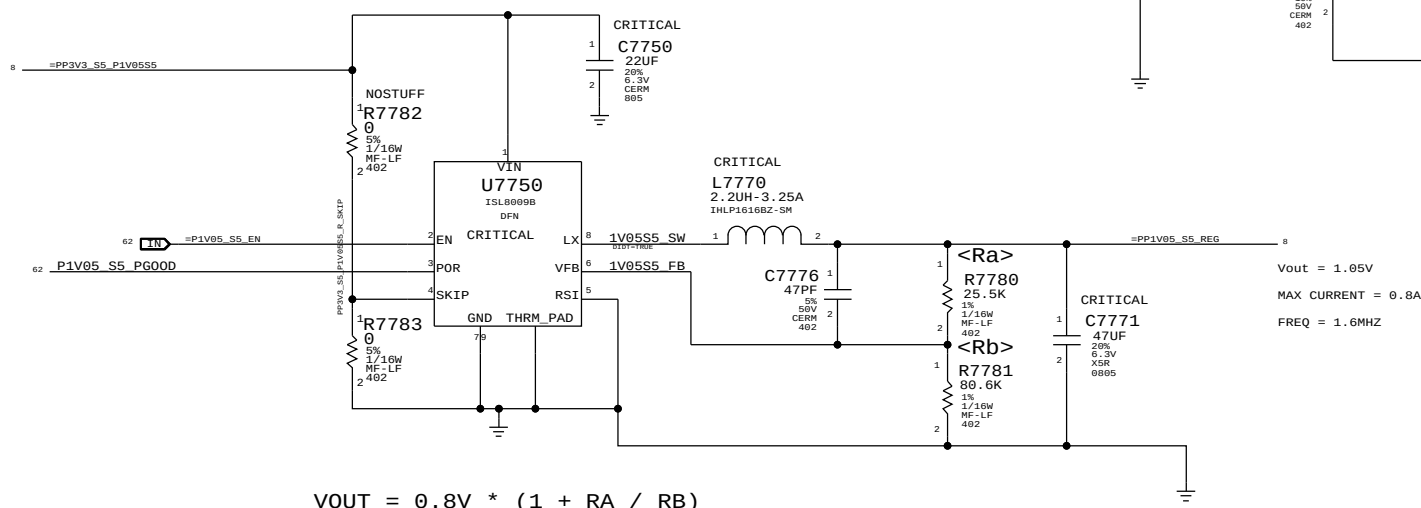
1.8V S0 SWITCHER



1.05V S0 PLL LDO



MCP 1.05V S5 (AUXC) SUPPLY

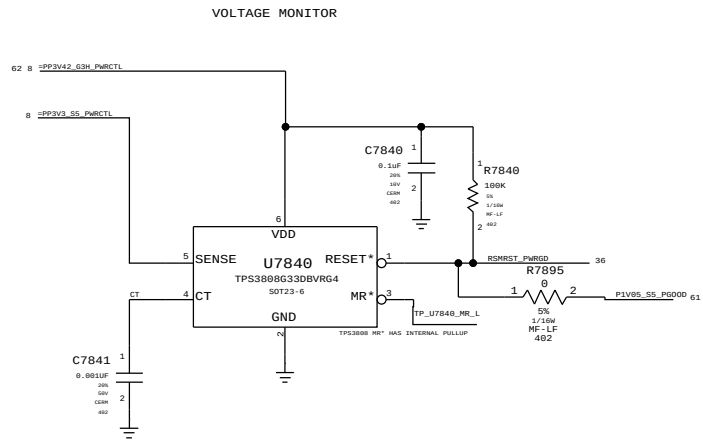
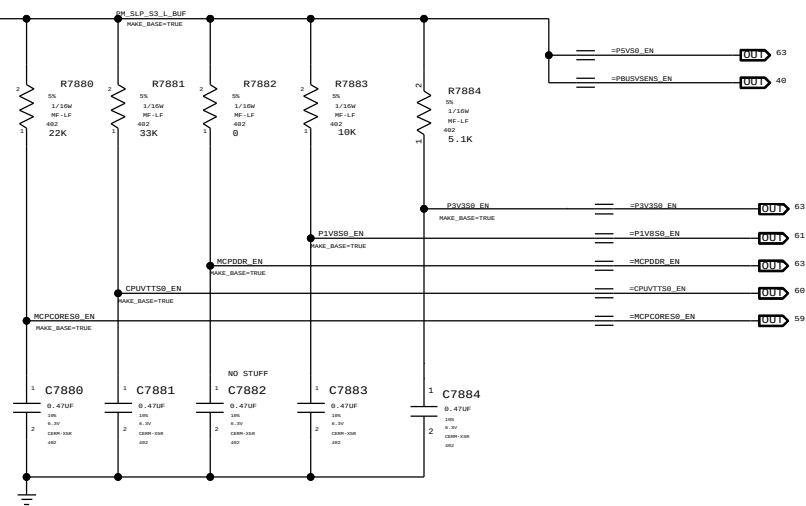
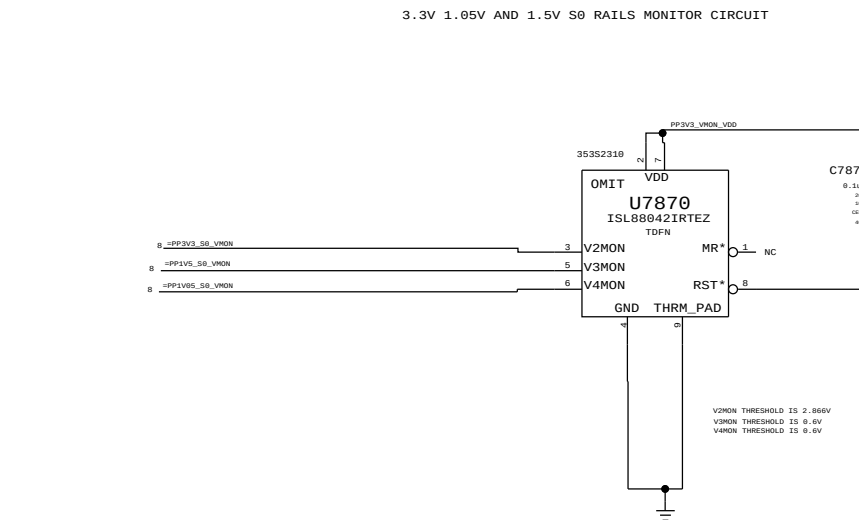
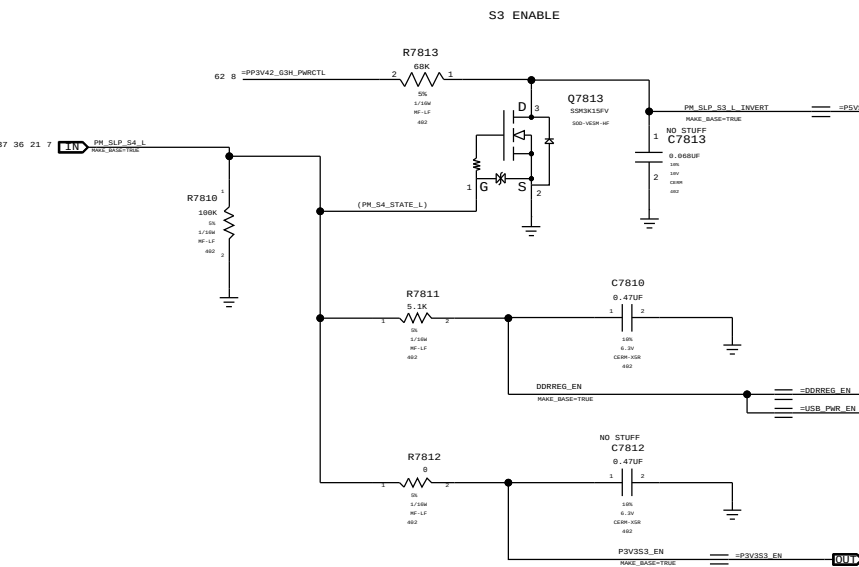
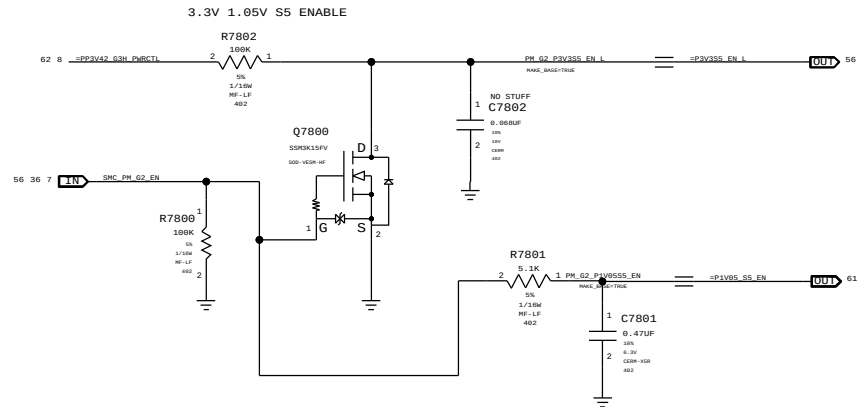


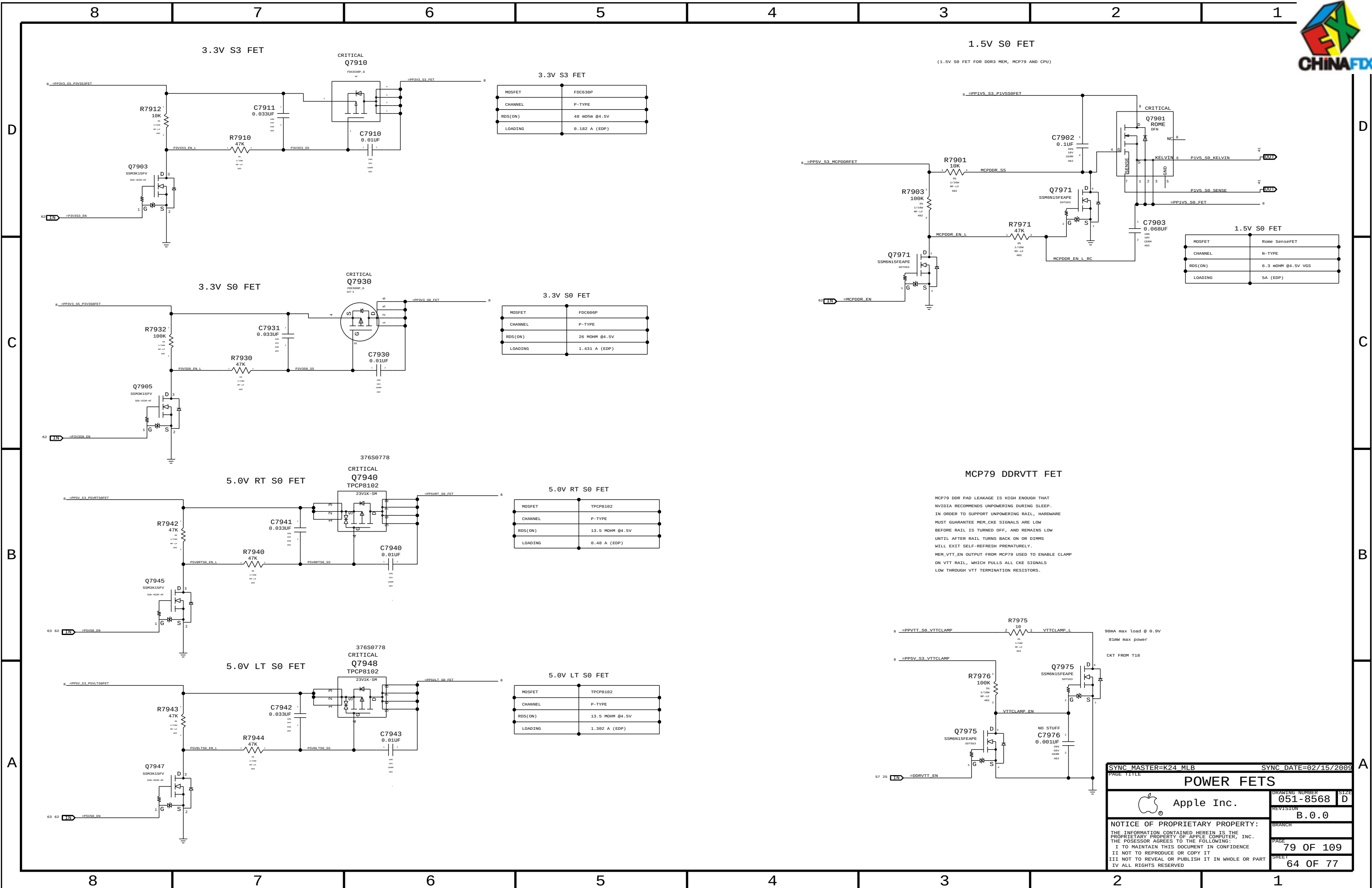
SYNC MASTER=K24 MLB		SYNC DATE=03/24/2006	
PAGE TITLE			
MISC POWER SUPPLIES			
 Apple Inc.		DRAWING NUMBER	051-8568
		REVISION	B.0.0
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Power Control Signals

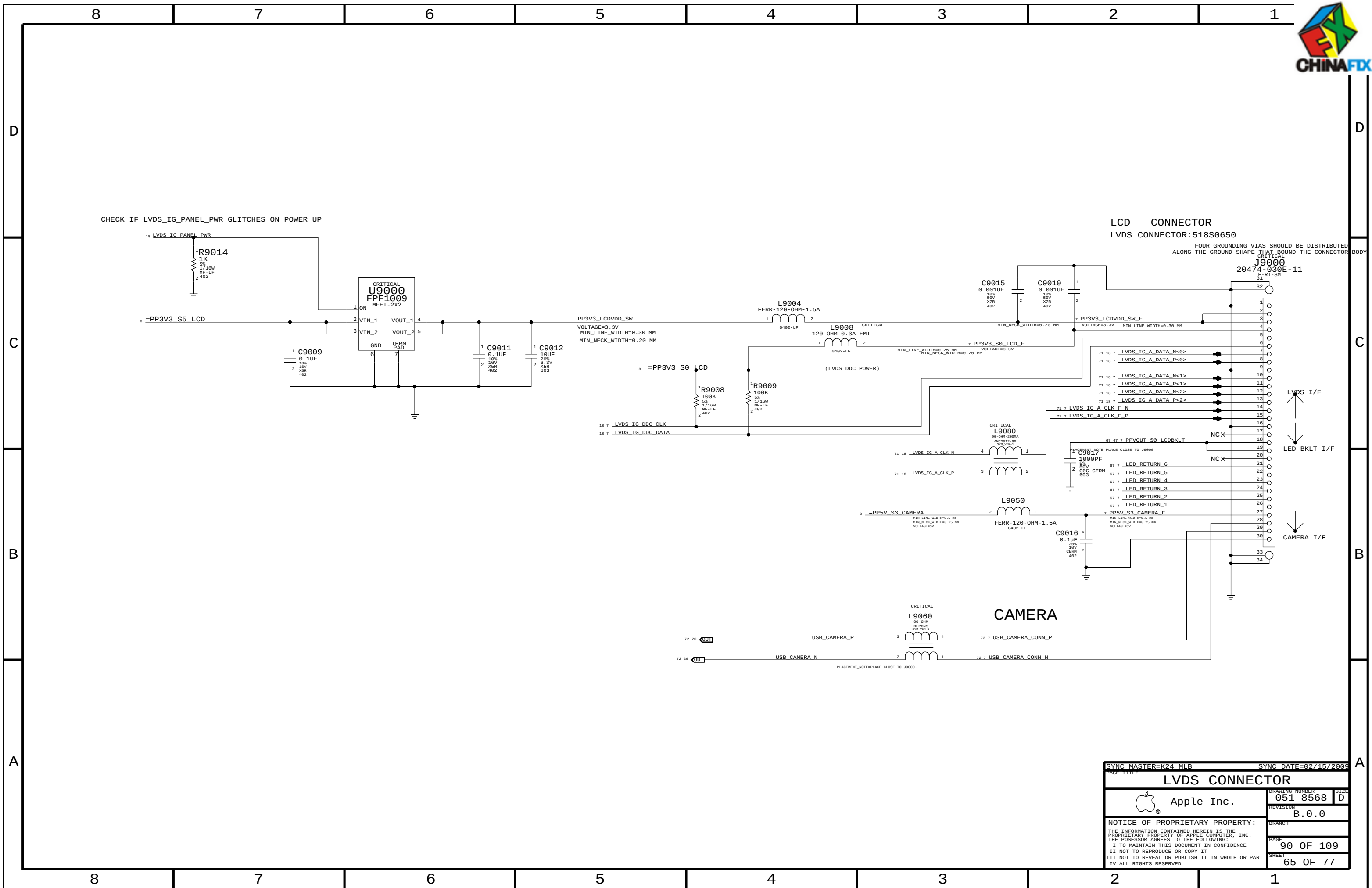
State	SMC_PM_G2_ENABLE	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	1	1	1
Sleep (S3)	1	1	0
Soft-Off (S5)	1	0	0
Battery Off (G3not)	0	0	0

3.3V_S0, 1.8V_S0 ENABLE
MCPDDR, CPUVTT, MCPCORES0 ENABLE
1.5V S0 AND 1.05V S0 ENABLE

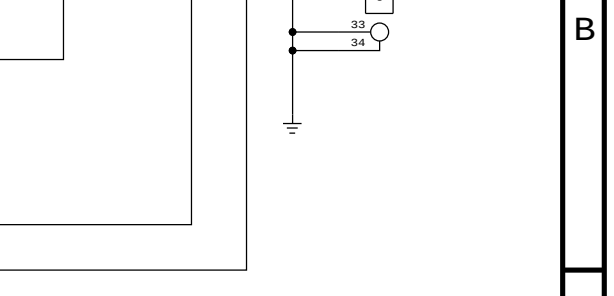
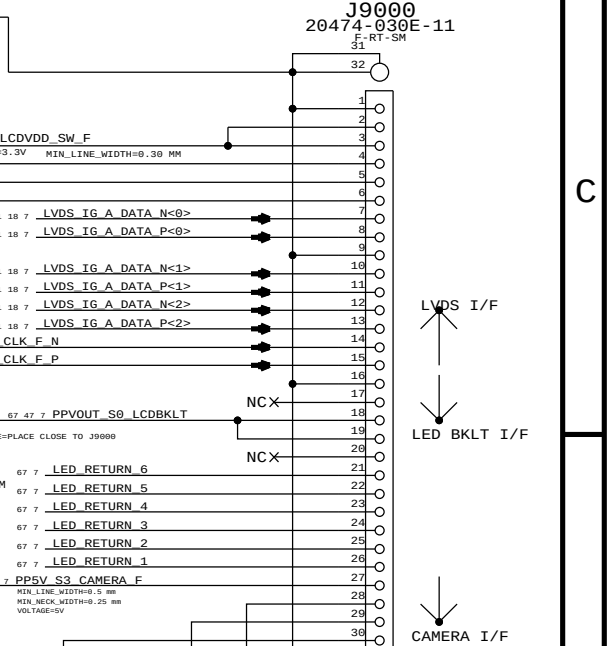




SYNC MASTER=K24 MLB		SYNC DATE=02/15/2009	
PAGE TITLE		POWER FETS	
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LCD CONNECTOR
LVDS CONNECTOR:518S0650
FOUR GROUNDING VIAS SHOULD BE DISTRIBUTED
ALONG THE GROUND SHAPE THAT BOUND THE CONNECTOR BODY

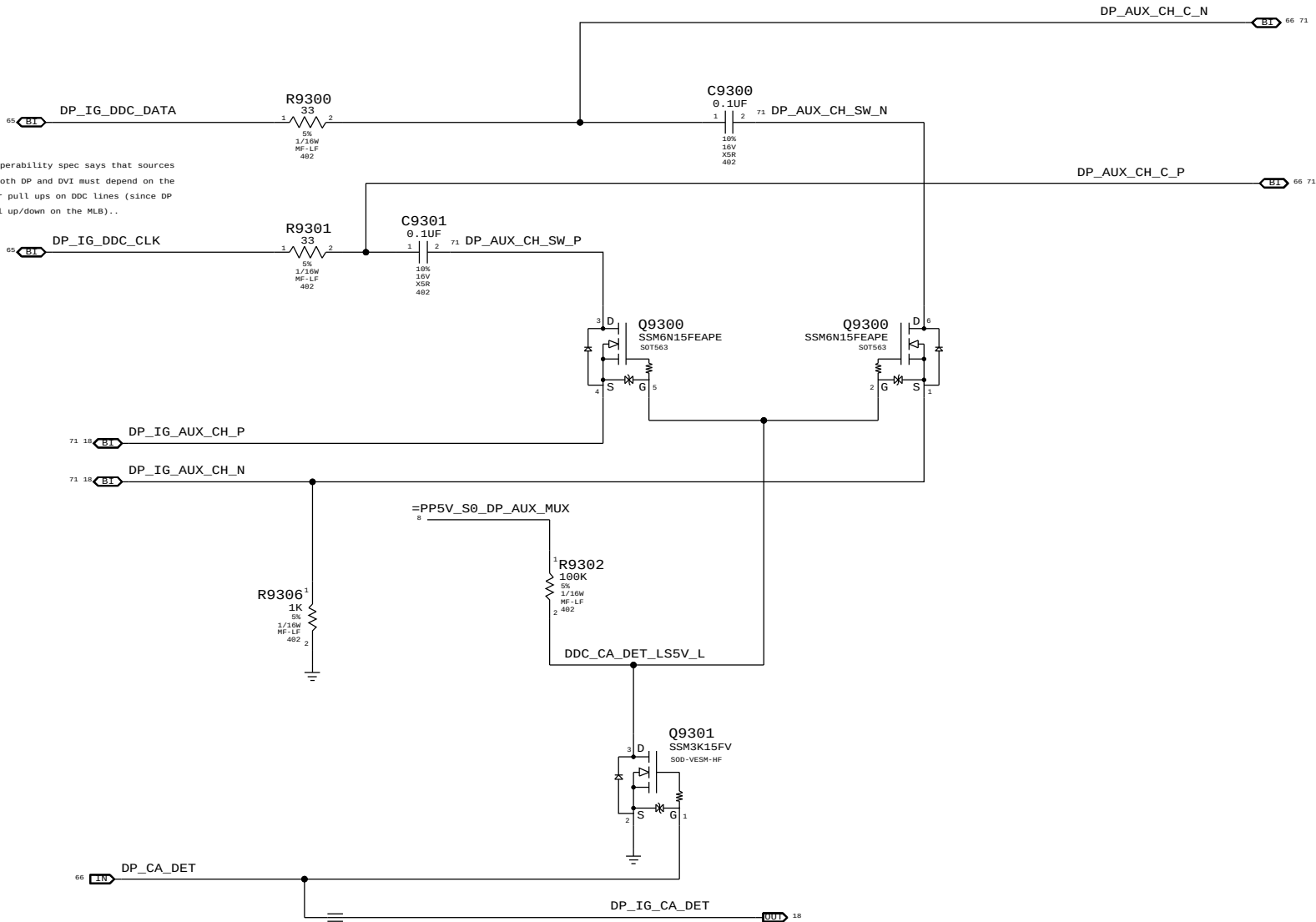


SYNC_MASTER=K24_MLB		SYNC_DATE=02/15/2009		A	
PAGE TITLE					
LVDS CONNECTOR					
 Apple Inc.		DRAWING NUMBER	051-8568	SIZE	D
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		BRANCH			
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18	=MCP_HDMI_TXC_P	---	DP_ML_P<3>	---	66	71
18	=MCP_HDMI_TXC_N	---	DP_ML_N<3>	---	66	71
18	=MCP_HDMI_TXD_P<0>	---	DP_ML_P<2>	---	66	71
18	=MCP_HDMI_TXD_N<0>	---	DP_ML_N<2>	---	66	71
18	=MCP_HDMI_TXD_P<1>	---	DP_ML_P<1>	---	66	71
18	=MCP_HDMI_TXD_N<1>	---	DP_ML_N<1>	---	66	71
18	=MCP_HDMI_TXD_P<2>	---	DP_ML_P<0>	---	66	71
18	=MCP_HDMI_TXD_N<2>	---	DP_ML_N<0>	---	66	71
18	=MCP_HDMI_HPD	---	DP_HPD	---	66	
18	=MCP_HDMI_DDC_CLK	---	DP_IG_DDC_CLK	---	65	
18	=MCP_HDMI_DDC_DATA	---	DP_IG_DDC_DATA	---	65	

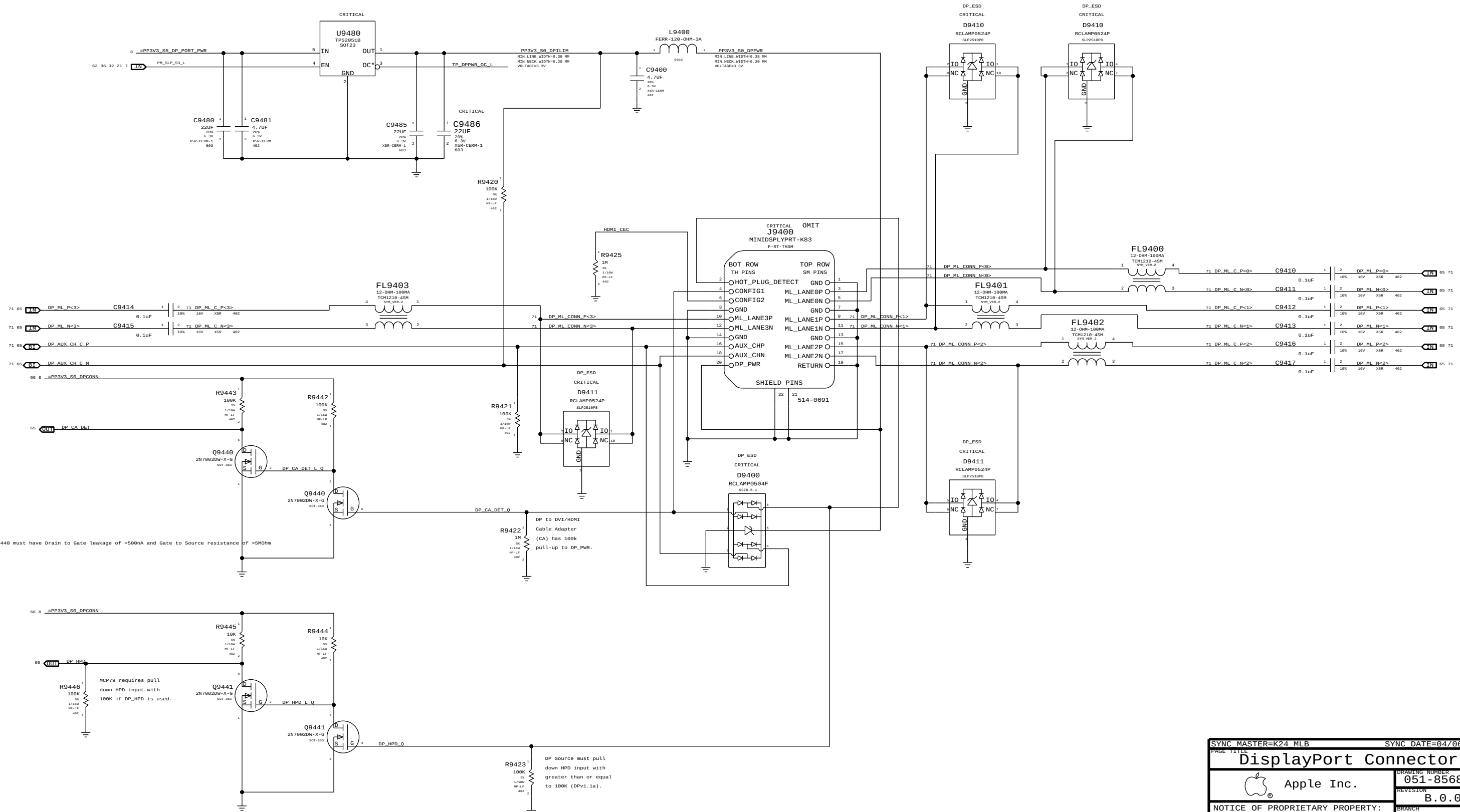
Display Port Interoperability spec says that sources or sinks which do both DP and DVI must depend on the external adapter for pull ups on DDC lines (since DP AUX CH has 100K pull up/down on the MLB)..

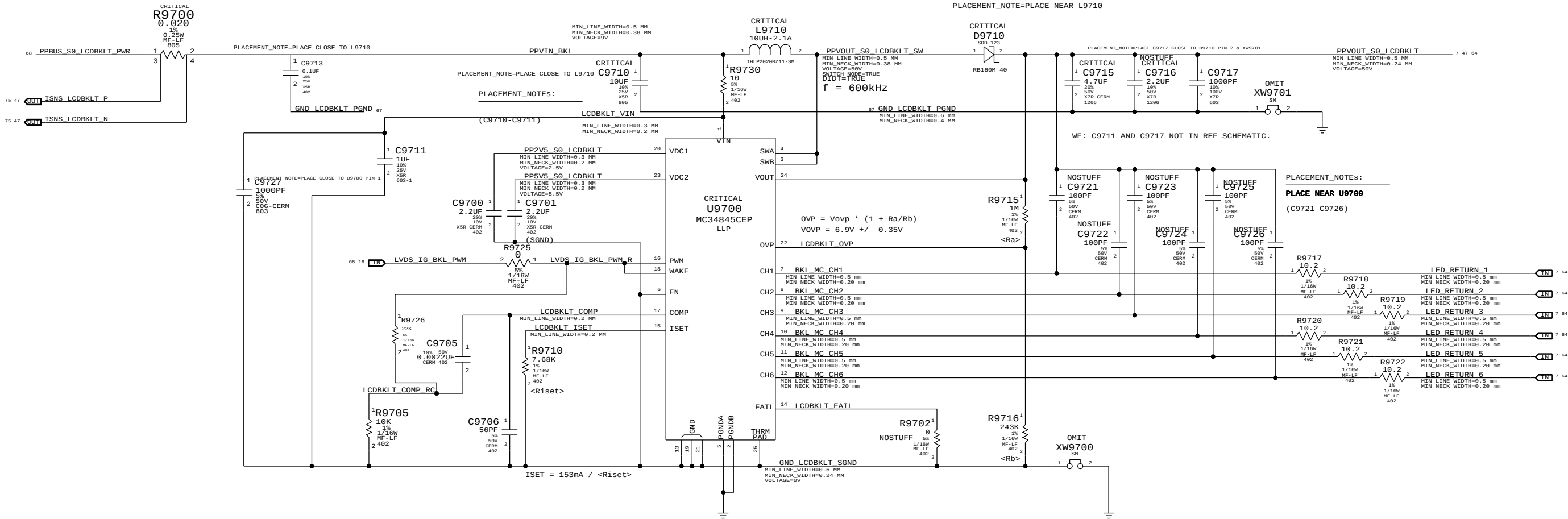


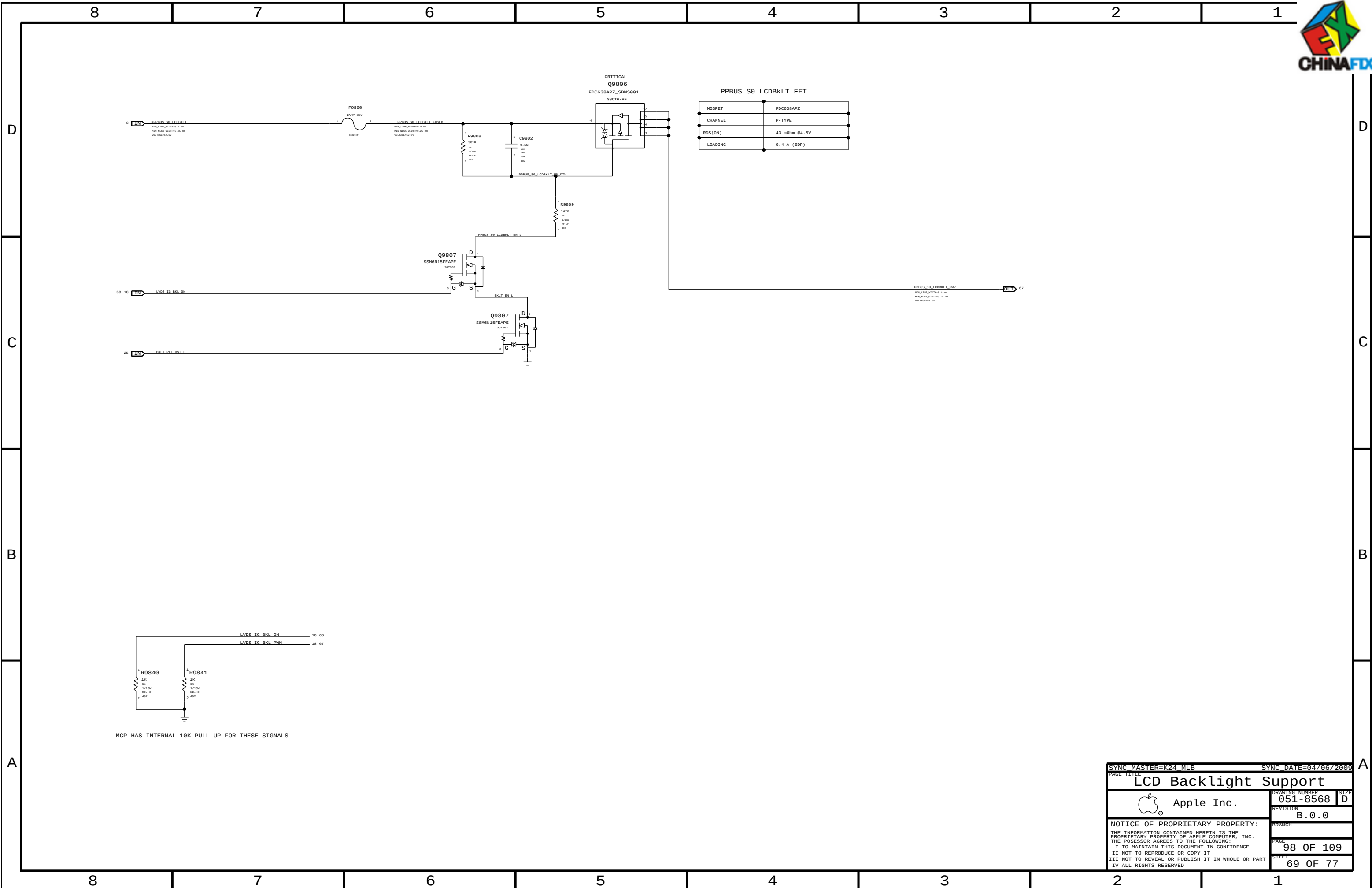
SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE			
DISPLAYPORT SUPPORT			
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		REVISION	B.0.0
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POR IS PLASTIC MINI DP CONNECTOR BUT METAL PART'S SCHEMATIC AND CAD SUMBOLS HAVE BEEN USED BEACUSE ITS LAND PATTERN CAN ACCOMODATE BOTH TYPES

Port Power Switch









FSB (Front-Side Bus) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
FSB_SBS	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
FSB_DSTB_SBS	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
FSB_DATA	*	=2x_DIELECTRIC	?	FSB_DATA	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_DSTB	*	=3x_DIELECTRIC	?	FSB_DSTB	TOP,BOTTOM	=5x_DIELECTRIC	?
FSB_ADDR	*	=STANDARD	?	FSB_ADDR	TOP,BOTTOM	=3x_DIELECTRIC	?
FSB_ADSTB	*	=2x_DIELECTRIC	?	FSB_ADSTB	TOP,BOTTOM	=4x_DIELECTRIC	?
FSB_1X	*	=STANDARD	?	FSB_1X	TOP,BOTTOM	=3x_DIELECTRIC	?

All 4x/2x/1x FSB signals with impedance requirements are 50-ohm single-ended.

FSB 4X signals / groups shown in signal table on right.

Signals within each 4x group should be matched within 5 ps of strobe.

DSTB# complementary pairs should be matched within 1 ps of each other, all DSTB#s matched to +/- 300 ps.

Spacing is 2x dielectric between DATA#, DINV# signals, with 3x dielectric spacing to the DSTB#s.

DSTB# complementary pairs are spaced normally and are NOT routed as differential pairs.

FSB 2X signals / groups shown in signal table on right.

Signals within each 2x group should be matched within 20 ps. ADTSB#s should be matched +/- 300 ps.

Spacing is 1x dielectric between ADDR#, REQ# signals, with 2x dielectric spacing to ADSTB#.

FSB 1X signals shown in signal table on right.

Signals within each 1x group should be matched to CPU clock, +/-1000 mls.

Design Guide recommends each strobe/signal group is routed on the same layer.

Intel Design Guide recommends FSB signals be routed only on internal layers.

NOTE: Intel Design Guide allows closer spacing if signal lengths can be shortened.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2

SOURCE: Santa Rosa Platform DG, Rev 1.5 (#22294), Sections 4.2 & 4.3

CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CPU_SBS	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

NOTE: 7 mil gap is for VCCSense pair, which Intel says to route with 7 mil spacing without specifying a target impedance.

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CPU_AGTL	*	=STANDARD	?	CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?
CPU_8MIL	*	8 MIL	?				
CPU_COMP	*	25 MIL	?				
CPU_STLREF	*	25 MIL	?				
CPU_TTP	*	=2:1_SPACING	?				
CPU_VCCSENSE	*	25 MIL	?				

Most CPU signals with impedance requirements are 55-ohm single-ended.

Some signals require 27.4-ohm single-ended impedance.

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2

SOURCE: Santa Rosa Platform DG, Rev 0.9 (#20517), Sections 4.4 & 5.8.2.4

MCP FSB COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MCP_SBS	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MCP_FSB_COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.4

FSB Clock Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CLK_FSB_180D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CLK_FSB	*	=3x_DIELECTRIC	?	CLK_FSB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP79 Interface DG (DG-03328-001_v01), Section 2.2.5

CPU / FSB Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
FSB_DATA_GROUPS	FSB_SBS	FSB_DATA	FSB_D_L<15..0>
FSB_DATA_GROUPS	FSB_SBS	FSB_DATA	FSB_DINV_L<0>
FSB_DSTB	FSB_DSTB_SBS	FSB_DSTB	FSB_DSTB_L_P<0>
FSB_DSTB	FSB_DSTB_SBS	FSB_DSTB	FSB_DSTB_L_N<0>
FSB_DATA_GROUP1	FSB_SBS	FSB_DATA	FSB_D_L<31..16>
FSB_DATA_GROUP1	FSB_SBS	FSB_DATA	FSB_DINV_L<1>
FSB_DSTB1	FSB_DSTB_SBS	FSB_DSTB	FSB_DSTB_L_P<1>
FSB_DSTB1	FSB_DSTB_SBS	FSB_DSTB	FSB_DSTB_L_N<1>
FSB_DATA_GROUP2	FSB_SBS	FSB_DATA	FSB_D_L<47..32>
FSB_DATA_GROUP2	FSB_SBS	FSB_DATA	FSB_DINV_L<2>
FSB_DSTB2	FSB_DSTB_SBS	FSB_DSTB	FSB_DSTB_L_P<2>
FSB_DSTB2	FSB_DSTB_SBS	FSB_DSTB	FSB_DSTB_L_N<2>
FSB_DATA_GROUP3	FSB_SBS	FSB_DATA	FSB_D_L<63..48>
FSB_DATA_GROUP3	FSB_SBS	FSB_DATA	FSB_DINV_L<3>
FSB_DSTB3	FSB_DSTB_SBS	FSB_DSTB	FSB_DSTB_L_P<3>
FSB_DSTB3	FSB_DSTB_SBS	FSB_DSTB	FSB_DSTB_L_N<3>
FSB_ADDR_GROUPS	FSB_SBS	FSB_ADDR	FSB_A_L<16..3>
FSB_ADDR_GROUPS	FSB_SBS	FSB_ADDR	FSB_REQ_L<4..0>
FSB_ADSTB	FSB_SBS	FSB_ADSTB	FSB_ADSTB_L<0>
FSB_ADDR_GROUP1	FSB_SBS	FSB_ADDR	FSB_A_L<35..17>
FSB_ADSTB1	FSB_SBS	FSB_ADSTB	FSB_ADSTB_L<1>
FSB_1X	FSB_SBS	FSB_1X	FSB_ADS_L
FSB_BREQB_L	FSB_SBS	FSB_1X	FSB_BREQB_L
FSB_BREQD_L	FSB_SBS	FSB_1X	FSB_BREQD_L
FSB_1X	FSB_SBS	FSB_1X	FSB_BNR_L
FSB_1X	FSB_SBS	FSB_1X	FSB_BPRT_L
FSB_1X	FSB_SBS	FSB_1X	FSB_DBSY_L
FSB_1X	FSB_SBS	FSB_1X	FSB_DEFER_L
FSB_1X	FSB_SBS	FSB_1X	FSB_DROY_L
FSB_1X	FSB_SBS	FSB_1X	FSB_HIT_L
FSB_1X	FSB_SBS	FSB_1X	FSB_HITM_L
FSB_1X	FSB_SBS	FSB_1X	FSB_LOCK_L
FSB_CPURST_L	FSB_SBS	FSB_1X	FSB_CPURST_L
FSB_1X	FSB_SBS	FSB_1X	FSB_RS_L<2..0>
FSB_1X	FSB_SBS	FSB_1X	FSB_TROY_L
CPU_ASYNC	CPU_SBS	CPU_AGTL	CPU_A20M_L
CPU_BSEL	CPU_SBS	CPU_AGTL	CPU_BSEL<2..0>
CPU_FERR_L	CPU_SBS	CPU_BMTL	CPU_FERR_L
CPU_ASYNC	CPU_SBS	CPU_AGTL	CPU_IOWM_L
CPU_INIT_L	CPU_SBS	CPU_AGTL	CPU_INIT_L
CPU_ASYNC_R	CPU_SBS	CPU_AGTL	CPU_INTR
CPU_ASYNC_R	CPU_SBS	CPU_AGTL	CPU_NMI
CPU_PROCHOT_L	CPU_SBS	CPU_AGTL	CPU_PROCHOT_L
CPU_PWRGD	CPU_SBS	CPU_AGTL	CPU_PWRGD
CPU_ASYNC	CPU_SBS	CPU_AGTL	CPU_SMT_L
CPU_ASYNC	CPU_SBS	CPU_AGTL	CPU_STECLK_L
PM_THRMSTRIP_L	CPU_SBS	CPU_BMTL	PM_THRMSTRIP_L
FSB_CPUVSLP_L	CPU_SBS	CPU_AGTL	FSB_CPUVSLP_L
CPU_FSBW_SS	CPU_SBS	CPU_AGTL	CPU_DPSLP_L
CPU_DPSRSTP_L	CPU_SBS	CPU_AGTL	CPU_DPSRSTP_L
CPU_ASYNC	CPU_SBS	CPU_AGTL	FSB_DPMR_L
MCP_CPU_COMP	MCP_SBS	MCP_FSB_COMP	MCP_BCLK_VML_COMP_VDD
MCP_CPU_COMP	MCP_SBS	MCP_FSB_COMP	MCP_BCLK_VML_COMP_GND
MCP_CPU_COMP	MCP_SBS	MCP_FSB_COMP	MCP_CPU_COMP_VCC
MCP_CPU_COMP	MCP_SBS	MCP_FSB_COMP	MCP_CPU_COMP_GND
FSB_CLK_CPU	CLK_FSB_180D	CLK_FSB	FSB_CLK_CPU_P
FSB_CLK_CPU	CLK_FSB_180D	CLK_FSB	FSB_CLK_CPU_N
FSB_CLK_TTP	CLK_FSB_180D	CLK_FSB	FSB_CLK_TTP_P
FSB_CLK_TTP	CLK_FSB_180D	CLK_FSB	FSB_CLK_TTP_N
FSB_CLK_MCP	CLK_FSB_180D	CLK_FSB	FSB_CLK_MCP_P
FSB_CLK_MCP	CLK_FSB_180D	CLK_FSB	FSB_CLK_MCP_N
(CPU_TERR_L)	CPU_SBS	CPU_BMTL	CPU_TERR_L
PM_DPSLPVR	CPU_SBS	CPU_AGTL	PM_DPSLPVR
(See above)	CPU_SBS	CPU_AGTL	IMVP_DPSLPVR
CPU_GTLREF	CPU_SBS	CPU_GTLREF	CPU_GTLREF
CPU_COMP	CPU_SBS	CPU_COMP	CPU_COMP<3>
CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP<2>
CPU_COMP	CPU_SBS	CPU_COMP	CPU_COMP<1>
CPU_COMP	CPU_27P4S	CPU_COMP	CPU_COMP<0>
XDP_TDI	CPU_SBS	CPU_TTP	XDP_TDI
XDP_TDO	CPU_SBS	CPU_TTP	XDP_TDO
XDP_TMS	CPU_SBS	CPU_TTP	XDP_TMS
XDP_TCK	CPU_SBS	CPU_TTP	XDP_TCK
XDP_TRST_L	CPU_SBS	CPU_TTP	XDP_TRST_L
XDP_BPM_L<4..0>	CPU_SBS	CPU_TTP	XDP_BPM_L<4..0>
XDP_BPM_L<5>	CPU_SBS	CPU_TTP	XDP_BPM_L<5>
(FSB_CPURST_L)	CPU_SBS	CPU_TTP	XDP_CPURST_L
CPU_VTD<6..0>	CPU_SBS	CPU_BMTL	CPU_VTD<6..0>
IMVP6_VTD<5..0>	CPU_SBS	CPU_BMTL	IMVP6_VTD<5..0>
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N
(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6_VSEN_P
(CPU_VCCSENSE)	CPU_27P4S	CPU_VCCSENSE	IMVP6_VSEN_N

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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_40S_VDD	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_760	*	=76_OHM_DIFF	=76_OHM_DIFF	=76_OHM_DIFF	=76_OHM_DIFF	=76_OHM_DIFF	=76_OHM_DIFF
MEM_760_VDD	*	=76_OHM_DIFF	=76_OHM_DIFF	=76_OHM_DIFF	=76_OHM_DIFF	=76_OHM_DIFF	=76_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MEM_CLK2MEM	*	=4:1_SPACING	?
MEM_CTRL2CTRL	*	=2:1_SPACING	?
MEM_CTRL2MEM	*	=2.5:1_SPACING	?
MEM_CMD2CMD	*	=1.5:1_SPACING	?
MEM_CMD2MEM	*	=3:1_SPACING	?
MEM_DATA2DATA	*	=1.5:1_SPACING	?
MEM_DATA2MEM	*	=3:1_SPACING	?
MEM_DQS2MEM	*	=3:1_SPACING	?
MEM_20THER	*	25 MIL	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2MEM
MEM_CLK	MEM_CTRL	*	MEM_CLK2MEM
MEM_CLK	MEM_CMD	*	MEM_CLK2MEM
MEM_CLK	MEM_DATA	*	MEM_CLK2MEM
MEM_CLK	MEM_DQS	*	MEM_CLK2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CTRL	MEM_CLK	*	MEM_CTRL2MEM
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL
MEM_CTRL	MEM_CMD	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DATA	*	MEM_CTRL2MEM
MEM_CTRL	MEM_DQS	*	MEM_CTRL2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DQS	MEM_CLK	*	MEM_DQS2MEM
MEM_DQS	MEM_CTRL	*	MEM_DQS2MEM
MEM_DQS	MEM_CMD	*	MEM_DQS2MEM
MEM_DQS	MEM_DATA	*	MEM_DQS2MEM
MEM_DQS	MEM_DQS	*	MEM_DQS2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CLK	*	MEM_CMD2MEM
MEM_CMD	MEM_CTRL	*	MEM_CMD2MEM
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_DATA	*	MEM_CMD2MEM
MEM_CMD	MEM_DQS	*	MEM_CMD2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_DATA	MEM_CLK	*	MEM_DATA2MEM
MEM_DATA	MEM_CTRL	*	MEM_DATA2MEM
MEM_DATA	MEM_CMD	*	MEM_DATA2MEM
MEM_DATA	MEM_DATA	*	MEM_DATA2DATA
MEM_DATA	MEM_DQS	*	MEM_DATA2MEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_DATA	*	*	MEM_20THER
MEM_DQS	*	*	MEM_20THER

Need to support MEM_*-style wildcards!

DDR2:

DQ signals should be matched within 20 ps of associated DQS pair.
DQS intra-pair matching should be within 1 ps, no inter-pair matching requirement.
All DQS pairs should be matched within 100 ps of clocks.
CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 140 ps.
A/BA/cmd signals should be matched within 75 ps, no CLK matching requirement.
All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate).
DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.

DDR3:

DQ signals should be matched within 5 ps of associated DQS pair.
DQS intra-pair matching should be within 1 ps, inter-pair matching should be within 180 ps
No DQS to clock matching requirement.
CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps.
A/BA/cmd signals should be matched within 5 ps of CLK pairs.
All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate).
DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.

SOURCE: MCP79 Interface DG (DG-93328-001_v00), Section 2.3

SOURCE: Santa Rosa Platform DG, Rev 1.0 (#21112), Section 6.2

MCP MEM COMP Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MCP_MEM_COMP	*	Y	7 MIL	7 MIL	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MCP_MEM_COMP	*	8 MIL	?

SOURCE: MCP79 Interface DG (DG-93328-001_v00), Section 2.3.4

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
MEM_A_CLK	MEM_760_VDD	MEM_CLK	MEM_A_CLK_P<5..0>	15 27
MEM_A_CLK	MEM_760_VDD	MEM_CLK	MEM_A_CLK_N<5..0>	15 27
MEM_A_CMT1	MEM_A0S_VDD	MEM_CTRL	MEM_A_CKE<3..0>	15 27
MEM_A_CMT1	MEM_A0S_VDD	MEM_CTRL	MEM_A_CS_L<3..0>	15 27
MEM_A_CMT1	MEM_A0S_VDD	MEM_CTRL	MEM_A_ODT<3..0>	15 27
MEM_A_CMD	MEM_A0S_VDD	MEM_CMD	MEM_A_A<14..0>	15 27
MEM_A_CMD	MEM_A0S_VDD	MEM_CMD	MEM_A_BA<2..0>	15 27
MEM_A_CMD	MEM_A0S_VDD	MEM_CMD	MEM_A_RAS_L	15 27
MEM_A_CMD	MEM_A0S_VDD	MEM_CMD	MEM_A_CAS_L	15 27
MEM_A_CMD	MEM_A0S_VDD	MEM_CMD	MEM_A_WE_L	15 27
MEM_A_DQ_BYTE0	MEM_A0S	MEM_DATA	MEM_A_DQ<7..0>	15 27
MEM_A_DQ_BYTE1	MEM_A0S	MEM_DATA	MEM_A_DQ<15..8>	15 27
MEM_A_DQ_BYTE2	MEM_A0S	MEM_DATA	MEM_A_DQ<23..16>	15 27
MEM_A_DQ_BYTE3	MEM_A0S	MEM_DATA	MEM_A_DQ<31..24>	15 27
MEM_A_DQ_BYTE4	MEM_A0S	MEM_DATA	MEM_A_DQ<39..32>	15 27
MEM_A_DQ_BYTE5	MEM_A0S	MEM_DATA	MEM_A_DQ<47..40>	15 27
MEM_A_DQ_BYTE6	MEM_A0S	MEM_DATA	MEM_A_DQ<55..48>	15 27
MEM_A_DQ_BYTE7	MEM_A0S	MEM_DATA	MEM_A_DQ<63..56>	15 27
MEM_A_DQ_BYTE8	MEM_A0S	MEM_DATA	MEM_A_DM<0>	15 27
MEM_A_DQ_BYTE1	MEM_A0S	MEM_DATA	MEM_A_DM<1>	15 27
MEM_A_DQ_BYTE2	MEM_A0S	MEM_DATA	MEM_A_DM<2>	15 27
MEM_A_DQ_BYTE3	MEM_A0S	MEM_DATA	MEM_A_DM<3>	15 27
MEM_A_DQ_BYTE4	MEM_A0S	MEM_DATA	MEM_A_DM<4>	15 27
MEM_A_DQ_BYTE5	MEM_A0S	MEM_DATA	MEM_A_DM<5>	15 27
MEM_A_DQ_BYTE6	MEM_A0S	MEM_DATA	MEM_A_DM<6>	15 27
MEM_A_DQ_BYTE7	MEM_A0S	MEM_DATA	MEM_A_DM<7>	15 27
MEM_A_DQS0	MEM_760	MEM_DQS	MEM_A_DQS_P<0>	15 27
MEM_A_DQS0	MEM_760	MEM_DQS	MEM_A_DQS_N<0>	15 27
MEM_A_DQS1	MEM_760	MEM_DQS	MEM_A_DQS_P<1>	15 27
MEM_A_DQS1	MEM_760	MEM_DQS	MEM_A_DQS_N<1>	15 27
MEM_A_DQS2	MEM_760	MEM_DQS	MEM_A_DQS_P<2>	15 27
MEM_A_DQS2	MEM_760	MEM_DQS	MEM_A_DQS_N<2>	15 27
MEM_A_DQS3	MEM_760	MEM_DQS	MEM_A_DQS_P<3>	15 27
MEM_A_DQS3	MEM_760	MEM_DQS	MEM_A_DQS_N<3>	15 27
MEM_A_DQS4	MEM_760	MEM_DQS	MEM_A_DQS_P<4>	15 27
MEM_A_DQS4	MEM_760	MEM_DQS	MEM_A_DQS_N<4>	15 27
MEM_A_DQS5	MEM_760	MEM_DQS	MEM_A_DQS_P<5>	15 27
MEM_A_DQS5	MEM_760	MEM_DQS	MEM_A_DQS_N<5>	15 27
MEM_A_DQS6	MEM_760	MEM_DQS	MEM_A_DQS_P<6>	15 27
MEM_A_DQS6	MEM_760	MEM_DQS	MEM_A_DQS_N<6>	15 27
MEM_A_DQS7	MEM_760	MEM_DQS	MEM_A_DQS_P<7>	15 27
MEM_A_DQS7	MEM_760	MEM_DQS	MEM_A_DQS_N<7>	15 27
MEM_B_CLK	MEM_760_VDD	MEM_CLK	MEM_B_CLK_P<5..0>	15 28
MEM_B_CLK	MEM_760_VDD	MEM_CLK	MEM_B_CLK_N<5..0>	15 28
MEM_B_CMT1	MEM_A0S_VDD	MEM_CTRL	MEM_B_CKE<3..0>	15 28
MEM_B_CMT1	MEM_A0S_VDD	MEM_CTRL	MEM_B_CS_L<3..0>	15 28
MEM_B_CMT1	MEM_A0S_VDD	MEM_CTRL	MEM_B_ODT<3..0>	15 28
MEM_B_CMD	MEM_A0S_VDD	MEM_CMD	MEM_B_A<14..0>	15 28
MEM_B_CMD	MEM_A0S_VDD	MEM_CMD	MEM_B_BA<2..0>	15 28
MEM_B_CMD	MEM_A0S_VDD	MEM_CMD	MEM_B_RAS_L	15 28
MEM_B_CMD	MEM_A0S_VDD	MEM_CMD	MEM_B_CAS_L	15 28
MEM_B_CMD	MEM_A0S_VDD	MEM_CMD	MEM_B_WE_L	15 28
MEM_B_DQ_BYTE0	MEM_A0S	MEM_DATA	MEM_B_DQ<7..0>	15 28
MEM_B_DQ_BYTE1	MEM_A0S	MEM_DATA	MEM_B_DQ<15..8>	15 28
MEM_B_DQ_BYTE2	MEM_A0S	MEM_DATA	MEM_B_DQ<23..16>	15 28
MEM_B_DQ_BYTE3	MEM_A0S	MEM_DATA	MEM_B_DQ<31..24>	15 28
MEM_B_DQ_BYTE4	MEM_A0S	MEM_DATA	MEM_B_DQ<39..32>	15 28
MEM_B_DQ_BYTE5	MEM_A0S	MEM_DATA	MEM_B_DQ<47..40>	15 28
MEM_B_DQ_BYTE6	MEM_A0S	MEM_DATA	MEM_B_DQ<55..48>	15 28
MEM_B_DQ_BYTE7	MEM_A0S	MEM_DATA	MEM_B_DQ<63..56>	15 28
MEM_B_DQ_BYTE8	MEM_A0S	MEM_DATA	MEM_B_DM<0>	15 28
MEM_B_DQ_BYTE1	MEM_A0S	MEM_DATA	MEM_B_DM<1>	15 28
MEM_B_DQ_BYTE2	MEM_A0S	MEM_DATA	MEM_B_DM<2>	15 28
MEM_B_DQ_BYTE3	MEM_A0S	MEM_DATA	MEM_B_DM<3>	15 28
MEM_B_DQ_BYTE4	MEM_A0S	MEM_DATA	MEM_B_DM<4>	15 28
MEM_B_DQ_BYTE5	MEM_A0S	MEM_DATA	MEM_B_DM<5>	15 28
MEM_B_DQ_BYTE6	MEM_A0S	MEM_DATA	MEM_B_DM<6>	15 28
MEM_B_DQ_BYTE7	MEM_A0S	MEM_DATA	MEM_B_DM<7>	15 28
MEM_B_DQS0	MEM_760	MEM_DQS	MEM_B_DQS_P<0>	15 28
MEM_B_DQS0	MEM_760	MEM_DQS	MEM_B_DQS_N<0>	15 28
MEM_B_DQS1	MEM_760	MEM_DQS	MEM_B_DQS_P<1>	15 28
MEM_B_DQS1	MEM_760	MEM_DQS	MEM_B_DQS_N<1>	15 28
MEM_B_DQS2	MEM_760	MEM_DQS	MEM_B_DQS_P<2>	15 28
MEM_B_DQS2	MEM_760	MEM_DQS	MEM_B_DQS_N<2>	15 28
MEM_B_DQS3	MEM_760	MEM_DQS	MEM_B_DQS_P<3>	15 28
MEM_B_DQS3	MEM_760	MEM_DQS	MEM_B_DQS_N<3>	15 28
MEM_B_DQS4	MEM_760	MEM_DQS	MEM_B_DQS_P<4>	15 28
MEM_B_DQS4	MEM_760	MEM_DQS	MEM_B_DQS_N<4>	15 28
MEM_B_DQS5	MEM_760	MEM_DQS	MEM_B_DQS_P<5>	15 28
MEM_B_DQS5	MEM_760	MEM_DQS	MEM_B_DQS_N<5>	15 28
MEM_B_DQS6	MEM_760	MEM_DQS	MEM_B_DQS_P<6>	15 28
MEM_B_DQS6	MEM_760	MEM_DQS	MEM_B_DQS_N<6>	15 28
MEM_B_DQS7	MEM_760	MEM_DQS	MEM_B_DQS_P<7>	15 28
MEM_B_DQS7	MEM_760	MEM_DQS	MEM_B_DQS_N<7>	15 28
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP_VDD	16
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP_GND	16

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PCI-Express

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCIE_980	*	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF
CLK_PCIE_1800	*	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCIE	*	=3X_DIELECTRIC	?
CLK_PCIE	*	20 MIL	?
MCP_PEX_COMP	*	8 MIL	?

SOURCE: MCP79 Interface D6 (DG-03328-001_v00), Section 2.4

Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
DP_1800	*	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF
LVDS_1800	*	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF
MCP_DV_COMP	*	?	20 MIL	20 MIL	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
DISPLAYPORT	*	=3X_DIELECTRIC	?
LVDS	*	=3X_DIELECTRIC	?

LVDS intra-pair matching should be 5 mils. Pairs should be within 100 mils of clock length.
DisplayPort/TMDS intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.
DisplayPort AUX CH intra-pair matching should be 5 ps. No relationship to other signals.
Max length of LVDS/DisplayPort/TMDS traces: 12 inches.
SOURCE: MCP79 Interface D6 (DG-03328-001_v00), Sections 2.5.3 & 2.5.4.

SATA Interface Constraints

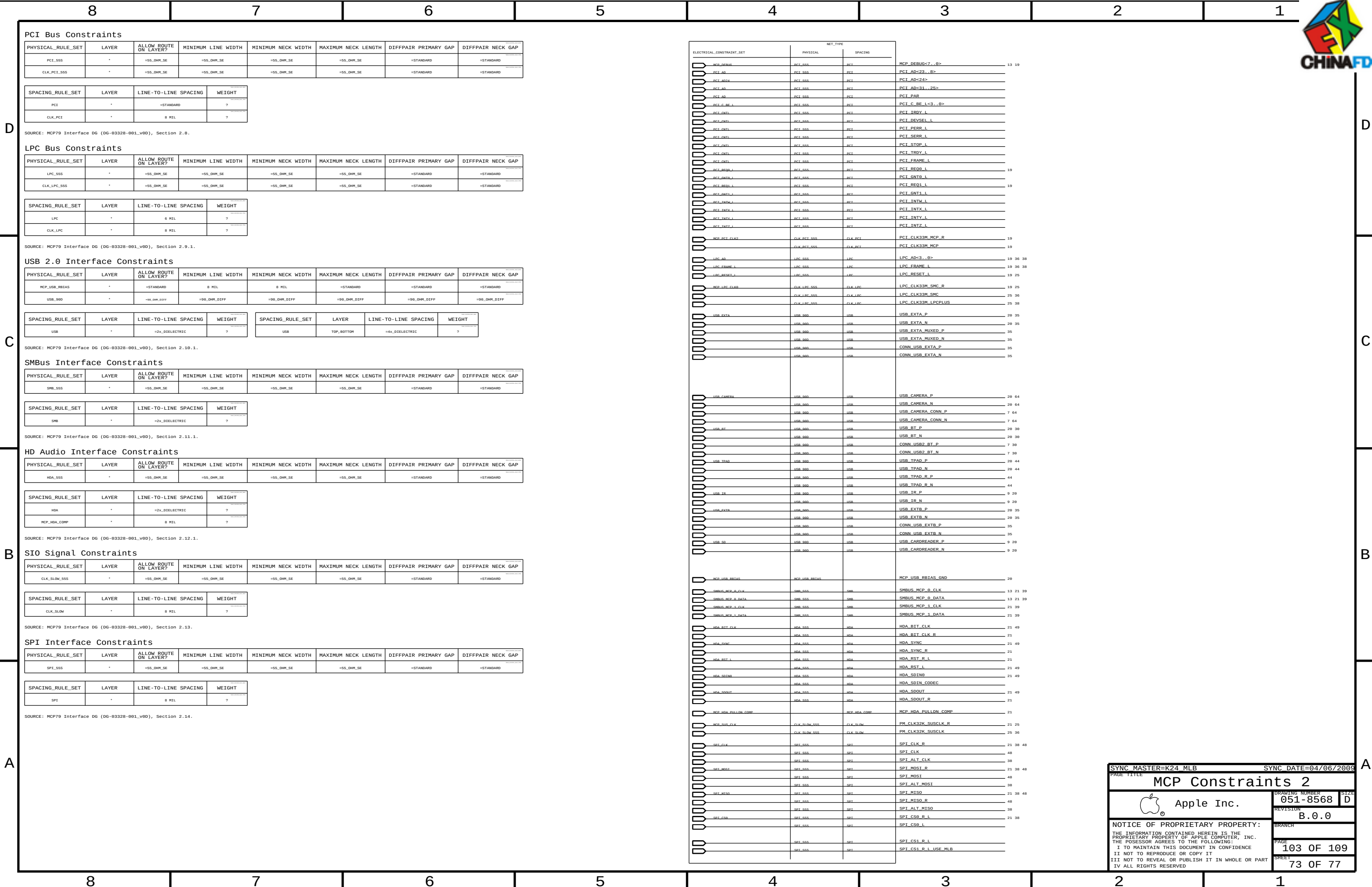
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SATA_1800	*	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF	=180_OHM_DIFF
SATA_980_HDD	*	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF	=98_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SATA	*	=4X_DIELECTRIC	?
SATA_TERM	*	8 MIL	?

SOURCE: MCP79 Interface D6 (DG-03328-001_v00), Section 2.7.1.

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	PCIE_980	PCIE	PCIE_MINI_R20_P
	PCIE_980	PCIE	PCIE_MINI_R20_N
	PCIE_MINI_R20	PCIE_980	PCIE_MINI_R20_C_P
	PCIE_980	PCIE	PCIE_MINI_R20_C_N
	PCIE_MINI_D2R	PCIE_980	PCIE_MINI_D2R_P
	PCIE_980	PCIE	PCIE_MINI_D2R_N
	PCIE_980	PCIE	PCIE_FW_R20_P
	PCIE_980	PCIE	PCIE_FW_R20_N
	PCIE_FW_R20	PCIE_980	PCIE_FW_R20_C_P
	PCIE_980	PCIE	PCIE_FW_R20_C_N
	PCIE_FW_D2R	PCIE_980	PCIE_FW_D2R_P
	PCIE_980	PCIE	PCIE_FW_D2R_N
	PCIE_980	PCIE	PCIE_FW_D2R_C_P
	PCIE_980	PCIE	PCIE_FW_D2R_C_N
	CLK_PCIE_1800	CLK_PCIE	PCIE_CLK100M_MINI_P
	CLK_PCIE_1800	CLK_PCIE	PCIE_CLK100M_MINI_N
	CLK_PCIE_1800	CLK_PCIE	PCIE_CLK100M_MINI_CONN_P
	CLK_PCIE_1800	CLK_PCIE	PCIE_CLK100M_MINI_CONN_N
	CLK_PCIE_1800	CLK_PCIE	PCIE_CLK100M_FC_P
	CLK_PCIE_1800	CLK_PCIE	PCIE_CLK100M_FC_N
	PCIE_980	PCIE	CONN_PCIE_MINI_R20_P
	PCIE_980	PCIE	CONN_PCIE_MINI_R20_N
	PCIE_980	PCIE	CONN_PCIE_MINI_D2R_P
	PCIE_980	PCIE	CONN_PCIE_MINI_D2R_N
		MCP_PEX_COMP	MCP_PEX_CLK_COMP
	DP_1800	DISPLAYPORT	TMDS_IG_TXC_P
	DP_1800	DISPLAYPORT	TMDS_IG_TXC_N
	TMDS_IG_TXD	DP_1800	TMDS_IG_TXD_P<2..8>
	TMDS_IG_TXD	DP_1800	TMDS_IG_TXD_N<2..8>
	DP_1800	DISPLAYPORT	DP_ML_P<3..8>
	DP_1800	DISPLAYPORT	DP_ML_C_P<3..8>
	DP_1800	DISPLAYPORT	DP_ML_N<3..8>
	DP_1800	DISPLAYPORT	DP_ML_C_N<3..8>
	DP_1800	DISPLAYPORT	DP_IG_AUX_CH_P
	DP_1800	DISPLAYPORT	DP_IG_AUX_CH_N
	DP_1800	DISPLAYPORT	DP_AUX_CH_SW_P
	DP_1800	DISPLAYPORT	DP_AUX_CH_SW_N
	DP_1800	DISPLAYPORT	DP_AUX_CH_C_P
	DP_1800	DISPLAYPORT	DP_AUX_CH_C_N
	MCP_HDMI_VPROBE	MCP_HDMI_VPROBE	MCP_HDMI_RSET
	LVDS_1800	LVDS	LVDS_IG_A_CLK_P
	LVDS_1800	LVDS	LVDS_IG_A_CLK_F_P
	LVDS_IG_A_CLK	LVDS_1800	LVDS_IG_A_CLK_N
	LVDS_1800	LVDS	LVDS_IG_A_CLK_F_N
	LVDS_IG_A_DATA	LVDS_1800	LVDS_IG_A_DATA_P<2..8>
	LVDS_IG_A_DATA	LVDS_1800	LVDS_IG_A_DATA_N<2..8>
	DP_1800	DISPLAYPORT	DP_ML_CONN_P<3..8>
	DP_1800	DISPLAYPORT	DP_ML_CONN_N<3..8>
	MCP_IFPAB_VPROBE	MCP_IFPAB_VPROBE	MCP_IFPAB_RSET
	SATA_980_HDD	SATA	SATA_HDD_R2D_C_P
	SATA_980_HDD	SATA	SATA_HDD_R2D_C_N
	SATA_980_HDD	SATA	SATA_HDD_R2D_P
	SATA_980_HDD	SATA	SATA_HDD_R2D_N
	SATA_980_HDD	SATA	SATA_HDD_R2D_UF_P
	SATA_980_HDD	SATA	SATA_HDD_R2D_UF_N
	SATA_980_HDD	SATA	SATA_HDD_D2R_P
	SATA_980_HDD	SATA	SATA_HDD_D2R_N
	SATA_980_HDD	SATA	SATA_HDD_D2R_C_P
	SATA_980_HDD	SATA	SATA_HDD_D2R_C_N
	SATA_980_HDD	SATA	SATA_HDD_D2R_UF_P
	SATA_980_HDD	SATA	SATA_HDD_D2R_UF_N
	SATA_1800	SATA	SATA_ODD_R2D_C_P
	SATA_1800	SATA	SATA_ODD_R2D_C_N
	SATA_1800	SATA	SATA_ODD_R2D_P
	SATA_1800	SATA	SATA_ODD_R2D_N
	SATA_1800	SATA	SATA_ODD_R2D_UF_P
	SATA_1800	SATA	SATA_ODD_R2D_UF_N
	SATA_1800	SATA	SATA_ODD_D2R_P
	SATA_1800	SATA	SATA_ODD_D2R_N
	SATA_1800	SATA	SATA_ODD_D2R_C_P
	SATA_1800	SATA	SATA_ODD_D2R_C_N
	SATA_1800	SATA	SATA_ODD_D2R_UF_P
	SATA_1800	SATA	SATA_ODD_D2R_UF_N
		SATA_TERM	MCP_SATA_TERM

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PCI Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
PCI	*	=STANDARD	?
CLK_PCI	*	8 MIL	?

SOURCE: MCP79 Interface D6 (DG-03328-001_v00), Section 2.8.

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SOURCE: MCP79 Interface D6 (DG-03328-001_v00), Section 2.9.1.

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MCP_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_900	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: MCP79 Interface D6 (DG-03328-001_v00), Section 2.10.1.

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

SOURCE: MCP79 Interface D6 (DG-03328-001_v00), Section 2.11.1.

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?
MCP_HDA_COMP	*	8 MIL	?

SOURCE: MCP79 Interface D6 (DG-03328-001_v00), Section 2.12.1.

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
CLK_SLOW_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
CLK_SLOW	*	8 MIL	?

SOURCE: MCP79 Interface D6 (DG-03328-001_v00), Section 2.13.

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SPI	*	8 MIL	?

SOURCE: MCP79 Interface D6 (DG-03328-001_v00), Section 2.14.

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
MCP_DEBUG	PCT_55S	PCT	MCP_DEBUG<7..8> 13 19
PCT_A0	PCT_55S	PCT	PCT_AD<23..8>
PCT_A0A	PCT_55S	PCT	PCT_AD<24>
PCT_A0	PCT_55S	PCT	PCT_AD<31..25>
PCT_A0	PCT_55S	PCT	PCT_PAR
PCT_C_BE_L	PCT_55S	PCT	PCT_C_BE_L<3..8>
PCT_CTRL	PCT_55S	PCT	PCT_TRDY_L
PCT_CTRL	PCT_55S	PCT	PCT_DEVSEL_L
PCT_CTRL	PCT_55S	PCT	PCT_PERR_L
PCT_CTRL	PCT_55S	PCT	PCT_SERR_L
PCT_CTRL	PCT_55S	PCT	PCT_STOP_L
PCT_CTRL	PCT_55S	PCT	PCT_TRDY_L
PCT_CTRL	PCT_55S	PCT	PCT_FRAME_L
PCT_CTRL	PCT_55S	PCT	PCT_RE00_L
PCT_CTRL	PCT_55S	PCT	PCT_GNT0_L
PCT_CTRL	PCT_55S	PCT	PCT_REQ1_L
PCT_CTRL	PCT_55S	PCT	PCT_GNT1_L
PCT_CTRL	PCT_55S	PCT	PCT_INT0_L
PCT_CTRL	PCT_55S	PCT	PCT_INTX_L
PCT_CTRL	PCT_55S	PCT	PCT_INTX_L
PCT_CTRL	PCT_55S	PCT	PCT_INTY_L
PCT_CTRL	PCT_55S	PCT	PCT_INTZ_L
MCP_PCT_CLK2	CLK_PCT_55S	CLK_PCT	PCT_CLK33M_MCP_R 19
	CLK_PCT_55S	CLK_PCT	PCT_CLK33M_MCP 19
LPC_A0	LPC_55S	LPC	LPC_AD<3..8> 19 36 38
LPC_FRAME_L	LPC_55S	LPC	LPC_FRAME_L 19 36 38
LPC_RESET_L	LPC_55S	LPC	LPC_RESET_L 19 25
MCP_LPC_CLK0	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC_R 19 25
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_SMC 25 36
	CLK_LPC_55S	CLK_LPC	LPC_CLK33M_LPCPLUS 25 38
USB_EXT0	USB_900	USB	USB_EXT0_P 26 35
	USB_900	USB	USB_EXT0_N 26 35
	USB_900	USB	USB_EXT0_MUXED_P 35
	USB_900	USB	USB_EXT0_MUXED_N 35
	USB_900	USB	CONN_USB_EXT0_P 35
	USB_900	USB	CONN_USB_EXT0_N 35
USB_CAMERA	USB_900	USB	USB_CAMERA_P 26 64
	USB_900	USB	USB_CAMERA_N 26 64
	USB_900	USB	USB_CAMERA_CONN_P 7 64
	USB_900	USB	USB_CAMERA_CONN_N 7 64
USB_BT	USB_900	USB	USB_BT_P 26 38
	USB_900	USB	USB_BT_N 26 38
	USB_900	USB	CONN_USB2_BT_P 7 38
	USB_900	USB	CONN_USB2_BT_N 7 38
USB_TP00	USB_900	USB	USB_TP00_P 26 44
	USB_900	USB	USB_TP00_N 26 44
	USB_900	USB	USB_TP00_R_P 44
	USB_900	USB	USB_TP00_R_N 44
USB_TR	USB_900	USB	USB_TR_P 9 26
	USB_900	USB	USB_TR_N 9 26
USB_EXT0	USB_900	USB	USB_EXT0_P 26 35
	USB_900	USB	USB_EXT0_N 26 35
	USB_900	USB	CONN_USB_EXT0_P 35
	USB_900	USB	CONN_USB_EXT0_N 35
USB_S0	USB_900	USB	USB_CARDREADER_P 9 26
	USB_900	USB	USB_CARDREADER_N 9 26
MCP_USB_RBIAS	MCP_USB_RBIAS		MCP_USB_RBIAS_GND 26
SMBUS_MCP_0_CLK	SMB_55S	SMB	SMBUS_MCP_0_CLK 13 21 39
SMBUS_MCP_0_DATA	SMB_55S	SMB	SMBUS_MCP_0_DATA 13 21 39
SMBUS_MCP_1_CLK	SMB_55S	SMB	SMBUS_MCP_1_CLK 21 39
SMBUS_MCP_1_DATA	SMB_55S	SMB	SMBUS_MCP_1_DATA 21 39
HDA_BT_CLK	HDA_55S	HDA	HDA_BT_CLK 21 49
	HDA_55S	HDA	HDA_BT_CLK_R 21
HDA_SYNC	HDA_55S	HDA	HDA_SYNC 21 49
	HDA_55S	HDA	HDA_SYNC_R 21
HDA_RST_L	HDA_55S	HDA	HDA_RST_L 21 49
	HDA_55S	HDA	HDA_RST_L 21
HDA_SDIN0	HDA_55S	HDA	HDA_SDIN0 21 49
	HDA_55S	HDA	HDA_SDIN_CODEC 21 49
HDA_SDOUT	HDA_55S	HDA	HDA_SDOUT 21 49
	HDA_55S	HDA	HDA_SDOUT_R 21
MCP_HDA_PULLUP_COMP		MCP_HDA_COMP	MCP_HDA_PULLUP_COMP 21
MCP_SIO_CLK	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK_R 21 25
	CLK_SLOW_55S	CLK_SLOW	PM_CLK32K_SUSCLK 25 36
SPT_CLK	SPT_55S	SPT	SPI_CLK_R 21 38 48
	SPT_55S	SPT	SPI_CLK 48
	SPT_55S	SPT	SPI_ALT_CLK 38
SPT_MOSI	SPT_55S	SPT	SPI_MOSI_R 21 38 48
	SPT_55S	SPT	SPI_MOSI 48
	SPT_55S	SPT	SPI_ALT_MOSI 38
SPT_MISO	SPT_55S	SPT	SPI_MISO 21 38 48
	SPT_55S	SPT	SPI_MISO_R 48
	SPT_55S	SPT	SPI_ALT_MISO 38
SPT_CS0	SPT_55S	SPT	SPI_CS0_R_L 21 38
	SPT_55S	SPT	SPI_CS0_L
	SPT_55S	SPT	SPI_CS1_R_L
	SPT_55S	SPT	SPI_CS1_R_L_USE_MLB

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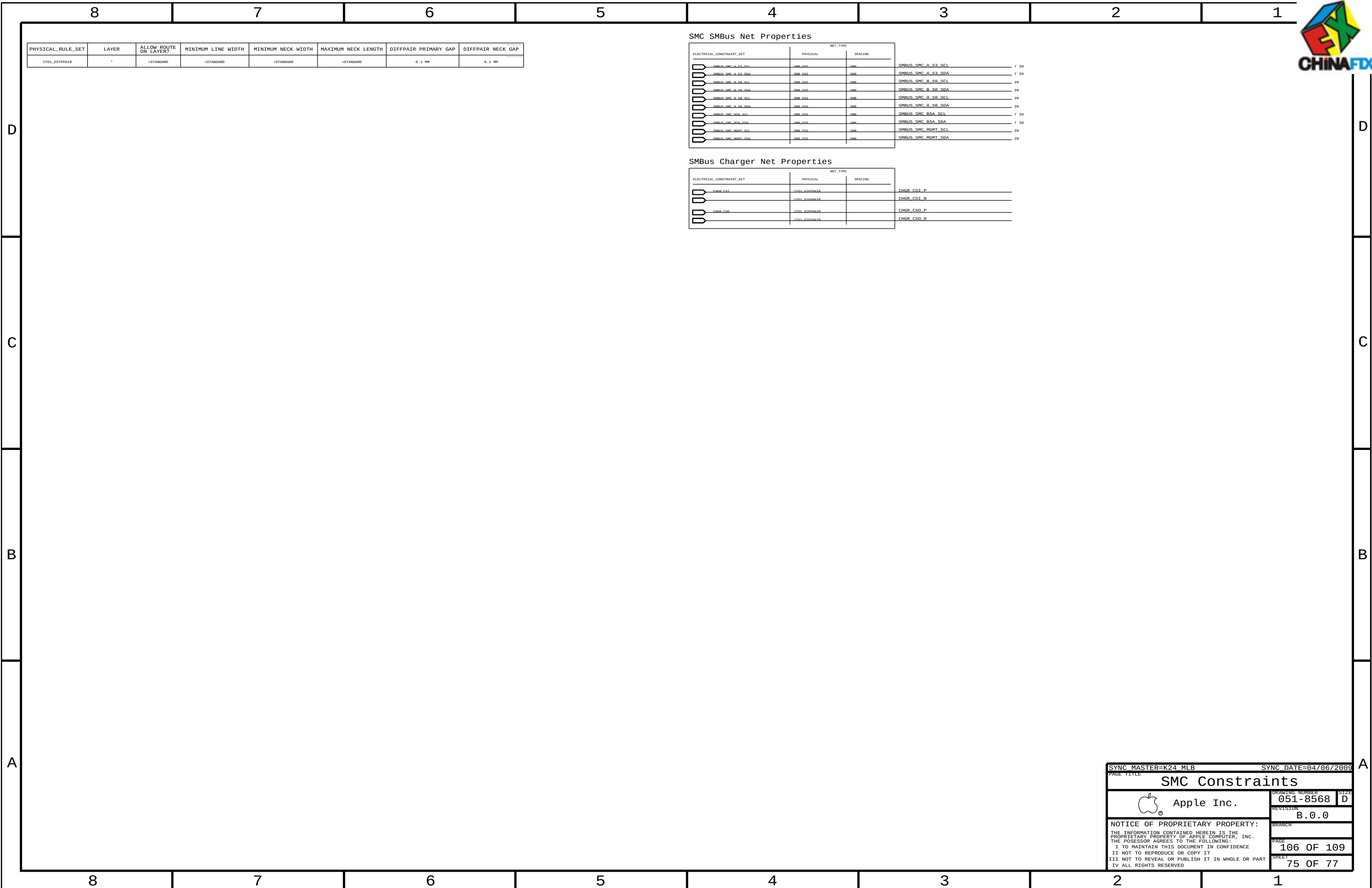
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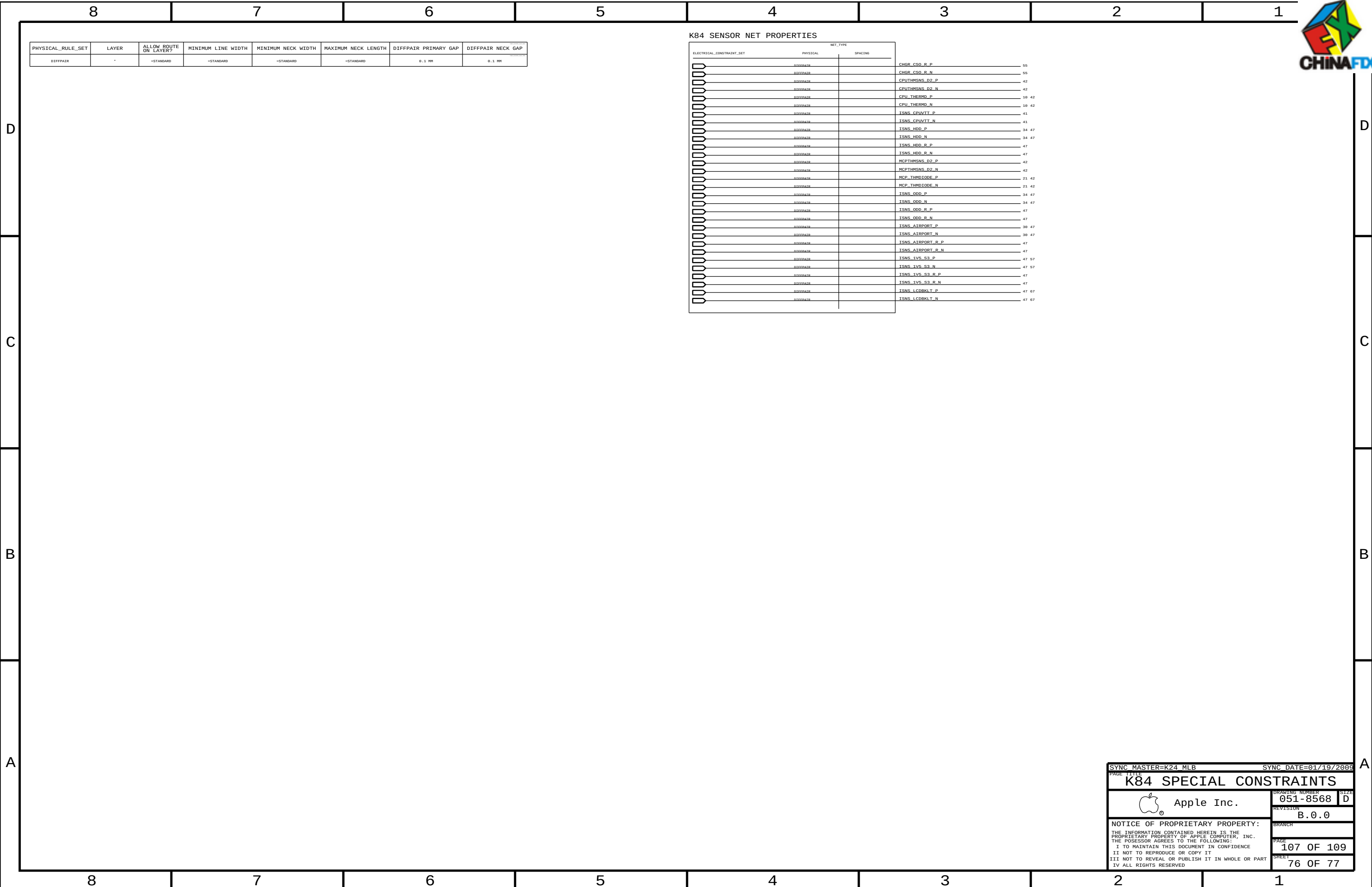
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8	7	6	5	4	3	2	1
K84 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS							
BOARD LAYERS				BOARD AREAS		BOARD UNITS (ALL OF MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM				NO. TYPE, BGA_P1MM		MM	15.5.1
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHM_SE	0.100MM	30 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	12.7 MM	=DEFAULT	=DEFAULT
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP, BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP, BOTTOM	Y	0.115 MM	0.115 MM			
50_OHM_SE	*	Y	0.076 MM	0.076 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP, BOTTOM	Y	0.165 MM	0.100 MM			
40_OHM_SE	*	Y	0.126 MM	0.100 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP, BOTTOM	Y	0.310 MM	0.310 MM			
27P4_OHM_SE	*	Y	0.222 MM	0.222 MM	=STANDARD	=STANDARD	=STANDARD
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
70_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.151 MM	0.100 MM	=STANDARD	0.224 MM	0.224 MM
70_OHM_DIFF	TOP, BOTTOM	Y	0.185 MM	0.100 MM		0.200 MM	0.200 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.095 MM	0.095 MM		0.234 MM	0.234 MM
90_OHM_DIFF	TOP, BOTTOM	Y	0.112 MM	0.112 MM		0.220 MM	0.220 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.075 MM	0.075 MM		0.244 MM	0.244 MM
100_OHM_DIFF	TOP, BOTTOM	Y	0.091 MM	0.091 MM		0.230 MM	0.230 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHM_DIFF_HDD	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHM_DIFF_HDD	ISL3, ISL4, ISL9, ISL10	Y	0.083 MM	0.083 MM		0.400 MM	0.400 MM
100_OHM_DIFF_HDD	TOP, BOTTOM	Y	0.095 MM	0.095 MM		0.400 MM	0.400 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHM_DIFF	ISL3, ISL4, ISL9, ISL10	Y	0.075 MM	0.075 MM		0.330 MM	0.330 MM
110_OHM_DIFF	TOP, BOTTOM	Y	0.077 MM	0.077 MM		0.330 MM	0.330 MM
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?
BGA_P3MM	*	=DEFAULT	?
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1.5:1_SPACING	*	0.15 MM	?
2:1_SPACING	*	0.2 MM	?
2.5:1_SPACING	*	0.25 MM	?
3:1_SPACING	*	0.3 MM	?
4:1_SPACING	*	0.4 MM	?
SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	TOP, BOTTOM	0.140 MM	?
3X_DIELECTRIC	TOP, BOTTOM	0.210 MM	?
4X_DIELECTRIC	TOP, BOTTOM	0.280 MM	?
5X_DIELECTRIC	TOP, BOTTOM	0.350 MM	?
2X_DIELECTRIC	*	0.126 MM	?
3X_DIELECTRIC	*	0.180 MM	?
4X_DIELECTRIC	*	0.252 MM	?
5X_DIELECTRIC	*	0.315 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA_P1MM	BGA_P1MM
MEM_CLK	*	BGA_P1MM	BGA_P2MM
CLK_FSB	*	BGA_P1MM	BGA_P2MM
CLK_LPC	*	BGA_P1MM	BGA_P2MM
CLK_PC1	*	BGA_P1MM	BGA_P2MM
CLK_PCIE	*	BGA_P1MM	BGA_P2MM
CLK_SLOW	*	BGA_P1MM	BGA_P2MM
FSB_DSTB	FSB_DSTB	BGA_P1MM	BGA_P3MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_485	BGA_P1MM	STANDARD
MEM_485_VDD	BGA_P1MM	STANDARD

SYNC MASTER=K24 MLB

SYNC DATE=01/19/2009

PAGE TITLE

K84 RULE DEFINITIONS

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